

FOOD DREAM

Planning Phase

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22/2/2025



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Introduction

I have been hired to create the Food Dream LTD mobile application (Food Dream App) to replace Food Dream LTD's old paper system and change the way customers shop from in-store to online. The application will have to meet standards in functionality, usability, user experience and performance above or near other applications on the market. The goal for the project is to make a seamless application for customers to easily browse, buy and interact with the company's products. Store managers should be able to change products on the Food Dream App storefront for customers to see. There should also be a support page for customers to ask for help from our support team and an analytics tracker for customers viewing and purchasing products to see which of the products are performing the best. For this to happen, I will need to plan out the systems that will need to be added to the application, conduct interviews to gain further knowledge of the business, schedule all the tasks and requirements needed and design wireframes and diagrams to show the business & myself the idea of how the application will be laid out.

Interpretation of the Project Idea

Ordering System:

- **Description:** System that allows customers to browse, view/interact and choose our products they'd like to purchase at the checkout.
- **Communication:**
 - * **Product Catalog System**
 - * **Analytics System**
- **Features:**
 - * **Basket:** Basket for customers to add or remove multiple items to manage or save later for purchasing at the checkout.
 - * **Order Summary:** Before moving to the checkout system to purchase the order, the customers will be able to view a summary of their order which will

contain a list of items, total price of the items, quantity, individual prices of items, stock status, and names of items they have put in their basket.

- * **Error Handling:** In case adding, removing or managing (increasing quantity e.g.) items to the basket doesn't work an error message should appear with solutions or an ability to retry adding, removing or managing (increasing quantity e.g.) to the basket. The error should be a simple, clear and understandable message for the customer and should have some solutions for them to try and fix their given problem; The system should also allow the user to retry adding to the basket. In case stock statuses cannot be updated - an error message should appear to try and help the customer resolve the problem with solutions given or retry loading the status.
- * **Real-Time Updates:** Stock statuses inside the product information on the order summary & basket page will be updated in real time via WebSocket integration to give an accurate status for customers and stop products being bought when out-of-stock.

Checkout System:

- **Description:** System for customers to pay for their items within their basket, handle the transaction and give the customer a receipt of their transaction.
- **Communication:**
 - * **Ordering System**
 - * **Inventory Management System**
 - * **Notification System**
 - * **Customer Account System**
 - * **Analytics System**
- **Features:**
 - * **Payment Methods:** Support for multiple payment methods such as credit/debit cards, Google Pay and PayPal. This allows for more accessibility and ease for customers to pay for their items.

- * **Receipt:** Gives a receipt/summary of the order and payment made by the customer, should show the basket of items they ordered, the total amount before and after tax, date, name, address, and the payment they used.
- * **Form:** A form for needed customer details (email, address, name and payment method) for trans actioning the order, updates on order, and delivery of the order. This form will also have validation for the input data such as the address and payment method/details.
- * **Error Handling:** Incase the payment fails to transact the system should allow the customer to retry the payment or reenter their details if necessary. The error should be a simple, clear and understandable message for the customer and should have some solutions for them to try and fix their given problem.
- * **SQL Database for Data Storage:** This system utilizes a SQL database to store data of the transactions made by customers. This allows the application to keep track of transactions so that customers or employees can view them in case any intervention is needed such as a refund, it also enables the company to gain analytics on stock, products purchased and transactions.

Inventory Management System (IMS):

- **Description:** The IMS is used to monitor and manage the inventory of items as customers purchase items, items are lost, items expire or are stolen and if new stock is added to the inventory. For the application, however, this system should only concentrate on updating stock as customers purchase items in their baskets. This helps automate the inventory management process, saving time on updating them manually allowing the company to focus on other important matters. And it helps give real-time updates on items ensuring that customers can always see the true availability of products, preventing the sale of out-of-stock items and enhancing customer satisfaction.
- **Communication:**

- * **Product Catalog System**
- * **Ordering System**
- * **Product Management System**
- **Features:**
 - * **Real-Time Stock Updates:** When customers buy items at the checkout, the IMS should automatically update the inventory stock in real time, ensuring the stock of products is accurate for customers and the system and that products aren't oversold when they aren't in stock.
 - * **SQL Database for Data Storage:** This system utilizes a SQL database to store data of the inventory and its history. This allows the application to display the stock levels of products and keep track of the inventory, it also enables the company to gain analytics on stock and transactions.
 - * **WebSocket Integration:** Adds a constant connection between the backend and frontend so that real-time updates are pushed to the frontend whenever changes are made in the backend. This ensures that stock information and other relevant data are always up to date with no need to manually update them or risk products being sold out of stock.
 - * **Error Handling:** Error handling should be integrated in case of errors such as network disconnects, duplicate data, failure to update data, etc. This feature of the system should display error messages with solutions, queue requests until network reconnects, and maintain last known data to prevent errors in the frontend (until real-time updates can continue) as this can help with keeping a seamless user experience and prevent disruptions and keep the application from breaking.

Notification System:

- **Description:** Alerts the customer of any information, events, and updates relevant for them to know. This informs customers of, for example, order confirmations of order they have made in app. These notifications will be presented to customers both in app (in app notifications via toast) and out of

app (via email) to ensure that customers are aware of important actions or events without needing to actively check the app.

- **Communication:**
 - * **Customer Support System**
 - * **Customer Account System**
 - * **Ordering System**
 - * **Product Catalog System**
 - * **Product Management System**
- **Features:**
 - * **In-app Notifications:** In-app notifications will occur usually when an error occurs or an event has happened such as when an order is successfully transacted, customers will receive an in-app notification from toast confirming that the order has been made.
 - * **Outer-app Notifications:** When an order is successfully transacted, customers will receive an outer-app notification via email confirming that the order has been made it will also include relevant information such as an order ID, basket of items, payment method and total amount paid, confirming that the order has been made.
 - * **Error Handling:** In case a notification cannot be sent/load to the application an error message should appear describing the problem and giving solutions to fix the problem or a button should appear to retry sending the notification.

Customer Account System (CAS):

- **Description:** The CAS allows for customers to create, manage and update accounts they have made (or are just about to make) in the app. These accounts will store sensitive information such as order history which enables customers to track their purchases in app, names, emails, passwords, addresses and payment methods (information inside the order history section). They are also able to update/recover some of these details within the customer account

system's features. This system will also provide safety mechanisms to ensure customers' accounts are secure such as password recovery, email authentication, and encryption of sensitive data.

- **Communication:**

- * **Notification System**
- * **Ordering System**
- * **Checkout System**
- * **Product Catalog System**
- * **Customer Support System**

- **Features:**

- * **Account Registration:** Customers can seamlessly sign up to create an account by inputting information such as email, name, and password. And also agreeing to our terms of service and policies. They will also have to verify their email with a code sent to their email they signed up with in order for the account to be created.
- * **Guests:** Customers have the option to shop as guests until they decide to log in or create an account.
- * **Login and Authentication:** Customers can securely log in to their accounts with their registered credentials (email and password) and verify the two-factor authentication with a code sent to their email. There should also be an option for customers at the login screen to recover their passwords where they'd have to verify the account is there's via an email code before they enter the password recovery page.
- * **Manage Account Information:** Customers have the ability to change their account information such as name, email address and password.
- * **Account Deletion:** Customers have the option to delete their data from our database as they have the right to do so under the GDPR regulation.

- * **Sessions:** Customers who have logged in will remain logged in a session if the user leaves the app (inactive not a full exit) or doesn't interact with the application within a certain time the session will be closed and a message will appear prompting the customer to log in again unless they manually log out or interact again with the app before the session time closes.
- * **Order History/Tracking:** Customers who have made orders on their account will be able to see a detailed history of them inside their account page. Order ID's, Dates, Items, Payment Method and Total will be included.
- * **SQL Database for Data Storage:** All account information related to customers' details will be encrypted and stored on the database as it's essential for other systems/features to function such as the login feature.
- * **Error Handling:** If there's any failure for the customer to register or log in due to invalid credentials or an authentication issue, customers will be alerted with a clear error message and solutions to try and fix the problem, customers will be given the ability to retry logging in, registering or being sent a new verification code. If the order history cannot be loaded an error message will appear with solutions to try fix the problem and a button to retry loading the orders.

Product Catalog System (PCS):

- **Description:** The PCS displays all products available to customers in a categorized format, enabling a user-friendly view of the catalog. It ensures that customers can seamlessly browse and navigate through the catalog to find what they need. Each product in the catalog will include essential details such as a product name, image, stock status and price. For more information, customers can click on a product to be transferred to its product page, where additional details will be displayed such as country of origin, product description, barcode, ingredients, etc.
- **Communication:**
 - * **Ordering System**

- * **Customer Account System**
 - * **Notification System**
 - * **Inventory Management System**
 - * **Analytics System**
-
- **Features:**
 - * **Categorized Catalog:** Products in the catalog will each have categories for easier navigation and browsing for customers (*Subcategories might be added to improve navigation and searching if enough time to implement*).
 - * **Searching:** Search bar will be added for users to specifically look for the item they'd like to purchase. Search suggestions will be added to help customers with spelling and help them find the items much quicker. Searches can be specific with user inputs, or they can use filters already added by the system to sort and search products with a certain tag such as category.
 - * **Dedicated Product Page:** Customers can click on a specific product to get additional information about that product. The system should transfer them to a dedicated page of the product selected and present the additional product information on the page.
 - * **Products Catalog Page:** This page will list all products available in the catalog. Each product displayed will have details attached to them comprising of a product name, image, stock status and price for a clear and concise page for the customer.
 - * **Error Handling:** If a connection is lost to the page an in-app notification will alert the customer that the page is not loading and try the solution given to fix the problem. If a product fails to load an in-app notification will alert the customer of this problem and to try again later if no solutions can be found. A button will be implemented to retry loading to page if this happens.

- * **Real-Time Updates:** Stock statuses inside the product information on the catalog page will be updated in real time via WebSocket integration to give an accurate status for customers and stop products being bought when out-of-stock.
- * **SQL Database for Data Storage:** All product information will be stored on the SQL database rather than being stored locally (apart from product images), for a more compact application.

Customer Support System (CSS):

- **Description:** The CSS is used as a primary means of communication with Food Dream LTD and the customer. If a customer has an issue such as an accidental order or unsatisfied order, they will request a refund or ask questions with our team via email and this page.
- **Communication:**
 - * **Notification System**
 - * **Customer Account System**
- **Features:**
 - * **Email Support:** Customers can ask for assistance from our team on any issue they require to be resolved, whether that's a refund, unsatisfactory products, deletion of data, and general questions they'd like answered.
 - * **Error Handling:** In case an email a customer would like to send to us cannot be sent, an in-app notification will appear with a clear message and solutions to solve the problem they may have.

Analytics and Monitoring System (AMS):

- **Description:** The AMS is a simple system that monitors customers viewing of products and purchases of products to give the company an idea of what the most popular products bought are. This data will be added to the database with the columns "views" and "sales" made for the products table.
- **Communication:**
 - * **Customer Account System**

- * **Product Catalog System**
- * **Checkout System**
- **Features:**
 - * **Product View Tracking:** This feature monitors customer viewing of products in the application, logging data (views) for each product viewed by a customer, this will help the company understand what the customers are interested in and what their most popular products are based on customer viewings.
 - * **Product Sales Tracking:** This feature monitors sales of products made by customers when an order is made. The system will log data (sales) for each product in the customer order/purchase. This helps the company gain valuable information on their products and customers' interests.
 - * **SQL Database for Data Storage:** All data made by the customers interactions will be logged into the database under the columns “views” and “sales” which will be added to each product in the product table.
 - * **Error Handling:** In case of a failure to add the data to the database another attempt to add the data will be made once the problem is handled if the problem persists then no data will be added, and an error will be logged.

Product Management System

- **Description:** This system is intended to allow managers of the store to manage the product catalog of the application. The manager is able to log into the application with an admin account and is allowed to add, edit and delete products on the application with updates to the Inventory Management System.
- **Communication:**
 - * **Inventory Management System**
- **Features:**
 - * **View Admin Product Catalog:** This page is basically the same as the customers version of the page but is different as there will be an option to add items at the top of the page and each product displayed will have an option to edit or delete for the admin.

- * **Add New Product:** This page allows admins to add a new item to the database and catalog. They will have to add details into a form fields and all inputs will be validated before being sent to the database.
- * **Edit Product:** Admins on this page can edit existing products within the catalog. Edits will be made on a form similar to the add a new product page and all inputs will be validated before being sent to the database.
- * **Delete Product:** There is no page for the delete product feature as this does not need to add new data, it only needs a prompt to make sure the admin is correctly deleting the correct product. The prompt will ask the admin if they are sure they want to delete said product and if so, they have to enter that product name before submission to make sure they are correctly deleting the product.
- * **Error Handling:** Validation of input fields from the forms will ensure that all information is formatted correctly, and no vulnerabilities can occur such as SQL injection. Most fields will be required to let the admin know that they need to input something into the field. If certain processes such as deletion of an item fails, a retry method will try to delete it again within a certain time frame and if unable to a notification will occur to show that the deletion process has failed.

Requirements

Functional

- **Ordering System - High Priority (Required)**
 - Customers should be able to browse, view and interact with available products.
 - Customers should be able to add, remove and manage (e.g. change quantity) items in their basket or in catalog. Customers can access the basket at any time.
 - Order/Basket summary should be displayed when viewing the basket. It should contain information such as total price, stock status, item names & quantities and individual item prices.
 - Error handling for the basket and stock updates in case of issues that may occur, error messages will have clear solutions and retry options for the issues to be resolved.
- **Checkout System - High Priority (Required)**

- Support for Google Pay, PayPal and credit/debit card payments.
 - Checkout form for customers to input necessary information for transactions (name, address, payment information).
 - An order receipt will be made after the transaction summarizing their order and information used to transact the order.
 - A secure option to retry payment in case of failure or error.
 - Input cleaning and validation of the customer form.
 - Needs at least one item in the basket to checkout
-
- **Inventory Management System - High Priority (Required)**
 - Updates to stock after customer orders are transacted.
 - Stock level data kept in the SQL database.
 - Error handling for failed inventory updates utilizing retry options and queued requests for e.g. disconnects
-
- **Product Catalog System - High Priority (Required)**
 - Products in the catalog should be in a categorized format.
 - Products should each have a dedicated page with information tied to that product for the customer to click on and view.
 - Each product on the product catalog page should contain information such as stock status, name, brand and price to be displayed for the customer.
 - Clear error messages and retry options for handling issues that may occur with e.g. loading products on the page.
-
- **Login System - High Priority (Required)**
 - In case the customer account system is not implemented this will have to be a substitute for the admins to log in. (*Admin accounts will be added via the database not the register page/system.*)
 - Secure login with admin email.
 - Customers can continue as guests.
-
- **Product Management System - High Priority (Required)**
 - Options like deletion, edit, addition of products should work on the admin product catalog page.
 - Forms for inputting and submitting data of the product.

- Logging in with an admin account should direct the admin to the admin catalog page not the customer page.
- Clear error messages and retry options for handling issues that may occur with e.g. deleting products.
- **Notification System – Medium Priority (Optional)**
 - In-app notifications via toast for certain events such as order confirmations and errors.
 - Out-of-app notifications for order confirmation/summary.
 - Error message and retry option for failed notifications.
- **Customer Account System – Medium Priority (Optional)**
 - Ability to register/create accounts with verification via email or customers can continue as guests until they decide to create one.
 - Secure login with verification via email.
 - Password recovery via email.
 - Customers order history with their order/transaction data for them to view.
 - Delete the customer account/data, customers can delete their account and data.
 - Change account information, customers can change their name, email and password on the manage account page.
 - Encryption of sensitive data such as name, password, address, etc. Which is added to the SQL database.
- **Customer Support System - Low Priority (Time-Dependent)**
 - Ability for customers to send support requests via the customer support page which will send an email to the company inbox via mailer send API.
 - Error handling and ability to retry sending emails if they fail to be sent.
- **Analytics and Monitoring System - Low Priority (Time-Dependent)**
 - The system should track customers who view a particular product and log the view data in the SQL database.
 - Sales should be tracked after a customer has made a purchase of products, each individual product with quantity should be logged in the sale column of the database. Each product will have a sale column.

- Attempts at retry requests should be made if errors occur when saving/logging the data in the SQL database, if the error persists however the error should be logged.

Non-Functional

- **Usability**

- All systems & features in the application should be a seamless, user-friendly experience for customers and should be quick to understand and use. *(I will get into more about usability in the HCI section of the plan.)*

- **Scalability**

- All systems in the application should be able to be easily expanded for an influx of more users and products to the application.
- The source code of the application should be clearly documented with comments and adhering to industry standard formatting expansion of the application.

- **Maintainability**

- The source code of the application should be clearly documented with adherence to industry standard formatting (code comments, camelCase, etc.) for easy debugging, testing and updates.

- **Accessibility**

- Provide features in the design for customers with disabilities, features such as e.g. read aloud for dyslexia, blind or difficult vision. *(More information on accessibility will be in the HCI section of the plan.)*

- **Data Privacy**

- The systems in the application particularly the SQL database, customer account system and the analytics and monitoring system, should adhere to the GDPR regulation as these systems and the application use sensitive data taken from our customers with permission.

- Customers should be able to manage their data, e.g. deletion of all data in our database, as this is a requirement as part of the GDPR regulation which the company must follow.

Aims

The aims of this application are to make systems that are seamless and efficient for customers, managers and Food Dream LTD to use such features as, purchasing products, editing products, tracking customer sales, etc. *(All requirements are layed out in the interpretation section of the plan and functional section.)* The old Food Dream LTD paper system is inefficient because it takes way too long to input the information and hard store find the paper documents compared to a more automated system, this means that the aims of Food Dream LTD cannot be met without a change to the system. A new mobile online, database-driven system will be implemented to change their older system to more efficiently meet the aims of the company.

Research

Interview Questions

Deadline?	5th May 2024.
What is the application and why does it need to be developed?	The application is a mobile-based platform designed to replace our outdated paper-based system. It will allow customers to browse, purchase, and interact with products online, while also enabling store managers to efficiently manage product inventory and catalog digitally. The goal is to streamline operations, improve customer experience, and provide better data management for the company.
What goals should this mobile application reach.	The application should provide a seamless, user-friendly experience for customers, including browsing products, managing their basket, checking out, and making secure payments. It should support multiple payment methods like Google Pay, PayPal, and credit/debit cards. Additionally, the app should provide

	inventory updates, ensure secure transactions, and support order management. The app should perform better than or at least equal to competitors in terms of functionality, usability, and performance.
What are some of the general requirements of the application being developed.	The application should allow customers to create and manage accounts, view and interact with products, add, remove, or update items in their basket, and complete the checkout process. Customers should be able to review order summaries, track orders, and receive support. Managers should have the ability to log in securely, access and admin portal, and manage the product catalog (e.g., adding, editing, deleting products).
What should happen if an error occurs in the application.	When an error occurs, the application should display a clear error message with potential solutions and provide retry options where applicable. For example, if a payment fails, the customer should have the option to retry using a different method or reattempt the transaction after resolving the issue.
What else should the customer be notified of other than an error?	Customers should receive an in-app notification when their order is placed successfully, as well as an email confirmation with a summary of their order. Notifications should also be sent for shipping updates, stock availability, or if there are issues with the order.
What details should the customer put into the registration form for their customer account.	The registration form should require the customer to provide their email address, full name, and a strong password. A secure verification code should be sent to the customer's email to confirm the registration. This will ensure secure account creation and enable users to log in after account creation. Optional: Address details for checkout can be added after account creation.

Is customer account creation necessary for shopping on the app.	No, customers can proceed as guests, but creating an account will provide added benefits such as order history tracking, secure access to personal details, and streamlined future purchases.
Can customers delete their accounts.	Yes, customers will have the ability to delete their accounts and personal data through the app. This is compliant with GDPR regulations, ensuring customers can manage and erase their personal information if they choose.
Will there be any analytics that will need to be tracked and saved.	Yes, the app will need to track customer behavior, such as product views, and log sales data after each purchase. This information will be stored in the database to help the company analyze popular products and customer interests for business intelligence and marketing purposes.
What payments will be available for customers to use to pay for their shopping.	The app will support multiple payment methods, including credit/debit cards, Google Pay, and PayPal. Customers will have a secure and flexible range of options for completing their transactions.
Any logos, fonts, etc. To be used in the application.	The font can be chosen by the developer, with a recommendation to keep it simple and legible. A logo will be provided by the company to be used in the application for brand consistency.
Does an item need to be in the basket before a customer can proceed to the checkout?	Yes, a customer must have at least one item in their basket before they can proceed to the checkout process and finalize the purchase.
Any color scheme you would like used in the application.	The color scheme should be white, red, and black. This aligns with the brand identity and provides a clear and appealing user interface.
Is the basket always able to be viewed when shopping.	Yes, there should always be a visible basket icon that users can click at any time to view their cart, ensuring easy access to the items they've added.
Can customers modify their order after checkout?	No, once the customer has completed the checkout process and the order has been

	processed, modifications to the order (such as changing the quantity or adding/removing items) will not be allowed. However, customers will be able to track the status of their order and request cancellations or adjustments through customer support if necessary.
What kind of payment validation will be in place for security?	Payment validation will be handled through secure payment gateways, including Google Pay, PayPal, and credit/debit cards. The system will perform basic validation for the format of payment details, ensuring that the customer's information is valid before proceeding to the payment processor. Additionally, sensitive data such as credit card information will be securely encrypted before it is transmitted.
What should the checkout page display?	The checkout page should display a summary of the items in the customer's basket, including item names, quantities, prices, and the total price of the order. It should also show the billing address and payment method fields, ensuring the customer can review and confirm the information before finalizing the purchase. A confirmation message will appear upon successful payment completion.
How will the application handle customer support queries?	Customers will be able to send support requests through a dedicated customer support page within the app. These requests will be forwarded to the company's support team via an email notification system. If an email fails to send, the system will attempt to resend the message automatically or offer the user an option to retry sending the support request.
Will the application be GDPR-compliant?	Yes, the application will be fully GDPR-compliant. This includes providing customers with the ability to manage their personal data, including deleting accounts and erasing personal information from the database upon request. The system will also ensure that all sensitive data is stored

	and processed in accordance with GDPR regulations, with clear consent mechanisms for collecting and using customer data.
What happens if a product is discontinued or removed from the catalog?	If a product is discontinued or removed from the catalog, the system will notify customers who have the product in their basket and prevent the customer from purchasing the product. The product will also be removed from the storefront. However, customers who have already purchased the product will be informed via email about its discontinuation, and any backorders will be handled appropriately (e.g., offering a similar product or issuing a refund).
How will customer data be handled during account deletion?	Upon request, customers can delete their accounts and all associated data from the system. The system will ensure that all personal data (name, email, order history, etc.) is fully erased from the database. This will be done in compliance with GDPR regulations, ensuring that the customer's right to be forgotten is respected.
Any design preferences?	It's up to the developer to decide.
Will the customers be able to access the application offline.	It's up to the developer to decide.

Current System

Inventory Log			
Product Name			
Product Price			
Product Location			
Barcode Number			
Expiry Date			
Product Stock on Hand			
Time			
Date			
Manager Name			

Order Log			
Product Name	Product Price	Quantity	Barcode Number
Order ID			
Customer Name			
Customer Address			
Number of Products			
Order Total			
Date			
Time			

Return Sheet	
Reason for Return	
Order ID	
Customer Name	
Customer Address	
Date	
Time	

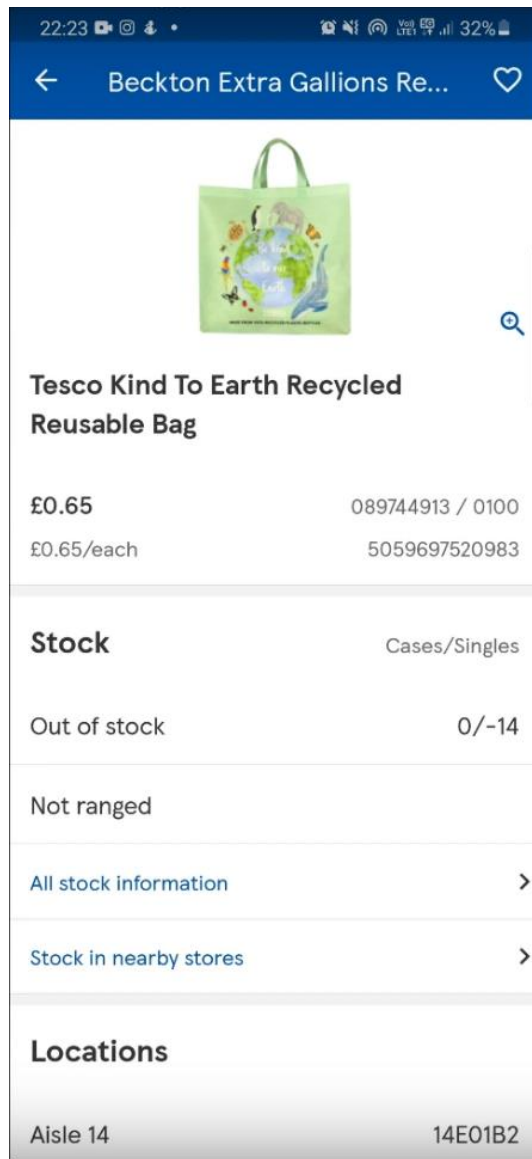


Similar Applications

Inform (Tesco)

A similar system in the industry, inform is used to search and view products on a specific store. I as the developer feel that some components of this app could be implemented into

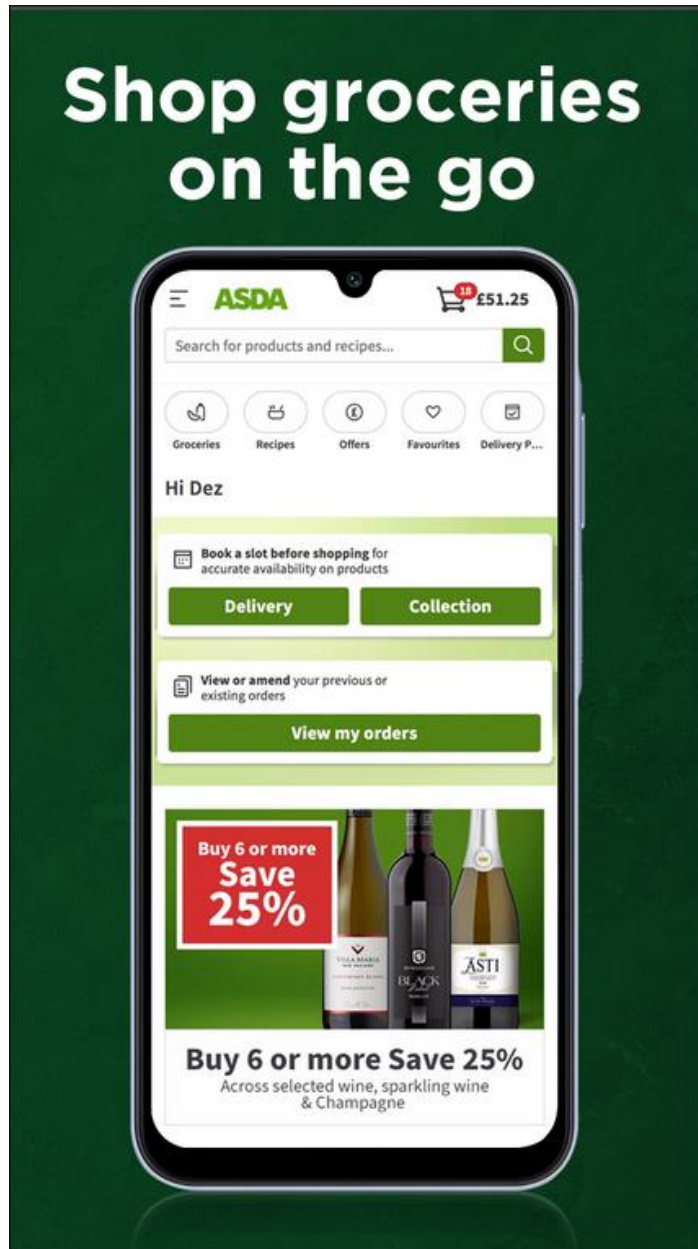
the Food Dream LTD application such as the layout of the product page and the ability to scan product barcode to find an item on the app as this could make the application more seamless and easy to use as this would the customer more options to find items in the app for example.



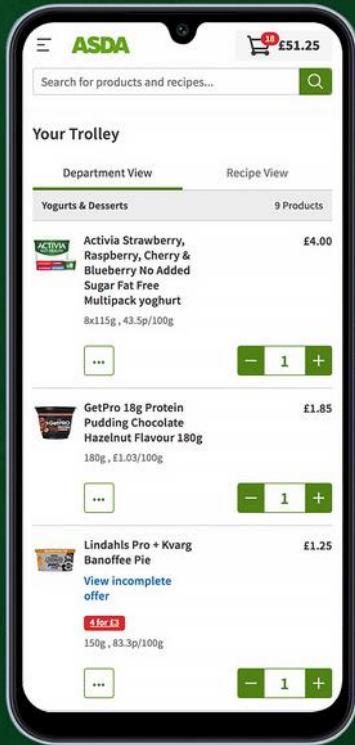
Asda Groceries Mobile

With this application I found that the design of the home page is excellent as all the most frequently used pages are immediately shown making it much easier for the user to navigate through the application. The design of the basket page, checkout page and sorting/filtering system is also excellent as the pages immediately seem clear to use, I

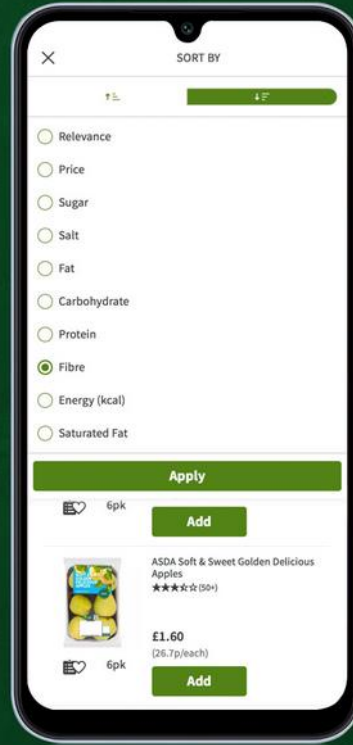
think elements of this application should also be added to the Food Dream LTD application.



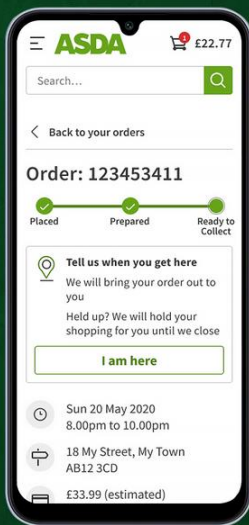
Amend your order easily



Nutritional Filters

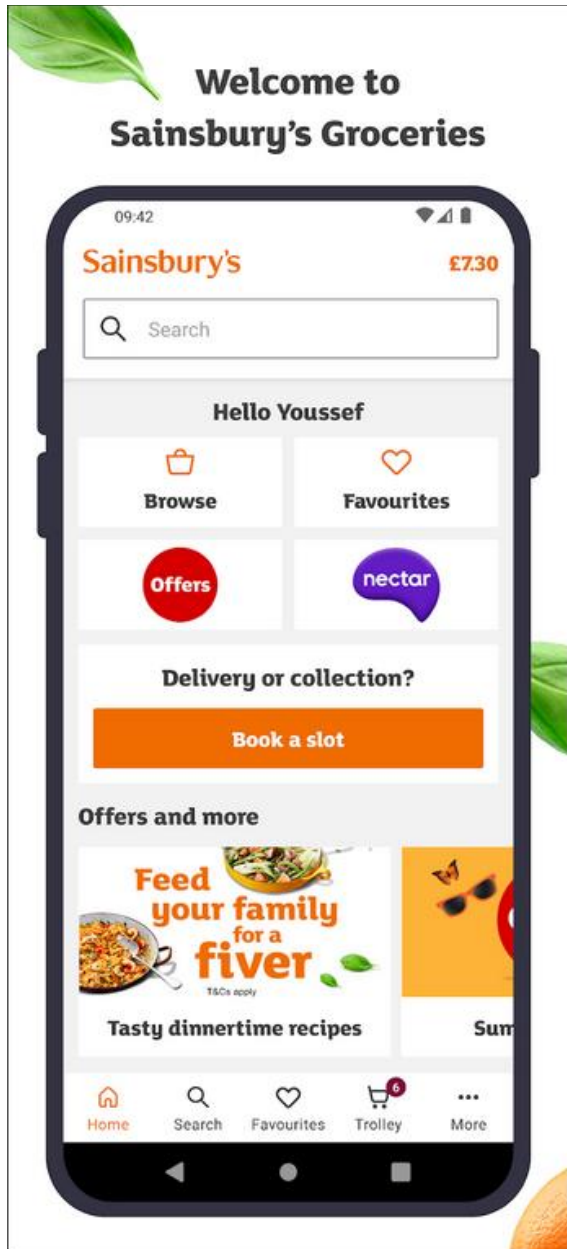


Get faster grocery collection



Sainsbury Groceries

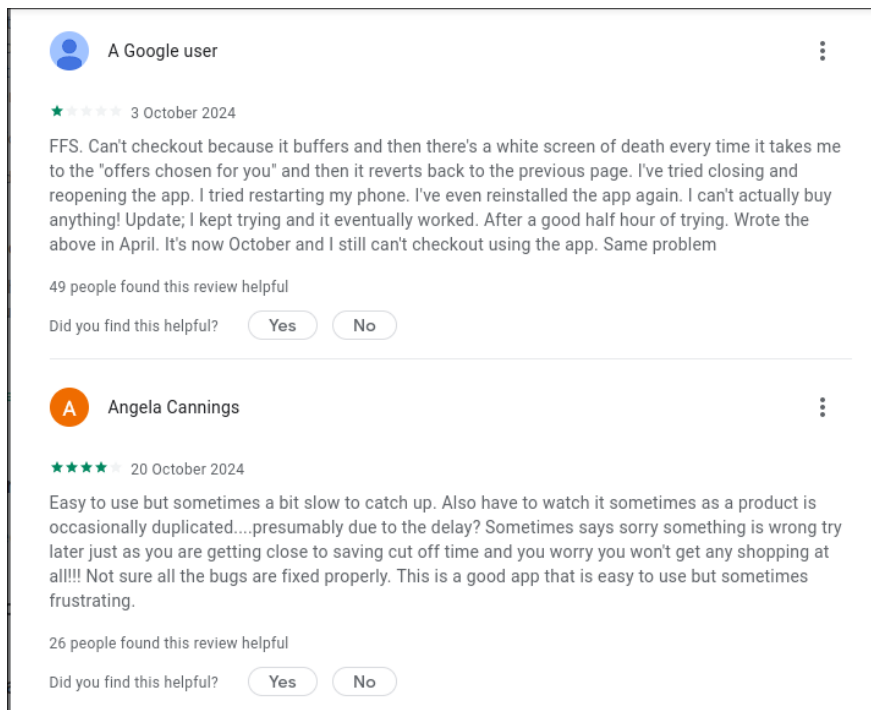
I found that I also like the navigation bar design in this system as it is also very clear and displays the most common/frequent pages and tools for the customer to use seamlessly.



Reviews

I've also found that some customers have given crucial feedback to this application that can I use to make sure the application performs much better. The customer states that alot of the applications systems perform poorly such as freezing and crashes. The customer support was also poor as they did not give a proper solution to their problem. As the developer I see that the performance aspect of the application is important for the customer's experience, and I will make sure to reach the requirements layed out in the document as it is crucial for good publicity.

Tesco



Asda

**Kit Kat**

★☆☆☆☆ 24 January 2025

I don't know what has happened to this app but it's horrendous. It's a nightmare to use and crashes all the time. I don't understand how a billion pound company can have such a cheap looking, difficult to use app. It crashes all the time. if you click on a item and then go back, it takes you to the top of the page even if you were all the way at the end. If I do a click and collect, it puts the nearest shop at my old address even though I've changed it in my settings. The constant suggestions.

50 people found this review helpful

Did you find this helpful?

**Alexis**

★☆☆☆☆ 20 January 2025

Been glitchy for several months now, If you don't book a slot before you add items to your basket, it clears your basket when you book, even though you are not needing to change which shop, which it claims you need to every time. Then you try to save your trolley, it doesn't save it. It's slow and you have to scroll through lots of irrelevant items when searching. If you add items to your basket and log in on another device, they won't be there. Previous orders regularly disappear. Nightmare!

35 people found this review helpful

Did you find this helpful?

**Dan Layton**

★☆☆☆☆ 26 January 2025

Can you please stop tinkering with the app? Every new iteration is worse than the last. Glitchy, slow, and with dreadful navigation. No one is asking for a super-sophisticated, super-shiny, bells-and-whistles experience - just something that actually WORKS. It was fine about 18 months ago, so can we go back to how it was, please?

30 people found this review helpful

Did you find this helpful?

Sainsbury



Cat Lawson

★★★★☆

12 December 2024

The app keeps reloading when you are scrolling through items. I was sent an incorrect item and it seems really difficult to check if it was a replacement or a mistake. I've clicked on the order and then the e-receipt. The e-receipt doesn't load so I can't view it. Getting help seems convoluted and awkward. It feels like it is deliberately difficult to check if you should get a refund.

15 people found this review helpful

Did you find this helpful?

Yes

No



MattMoo 44

★★★☆☆

11 December 2024

Recent updates have made it unusable. The search and favourites functions are broken, keeps refreshing and showing the same few results over and over. Amending an order often causes a freeze where the existing order is not loaded in the basket but the discard changes button appears, have to force close the app to get out of that. Quite often coupons and nectar won't redeem. Customer service just say 'use the website'. That is not a solution. The app needs to be fixed.

12 people found this review helpful

Did you find this helpful?

Yes

No



Alison Grant

★★★★★

11 October 2024

Barely ever works. Usually can't even load My Delivery Slot information on the main page. Today it managed that but I can't amend my order at all "Oops. Something went wrong" message. My Internet connection is fine. On the days it does work enough to try to amend my order it will never load my previous orders page, just spins. Sometimes it just crashes on checkout. I usually try once to see how it's behaving today but 99% of the time I give up and use the web page.

12 people found this review helpful

Did you find this helpful?

Yes

No



helen wilson

★★★☆☆

9 February 2025

Latest update has broken what was a good app. Basket/current orders won't load on first attempt, have to close and reopen every time. So annoying when you want to quickly amend an order. I'm using a pixel 7a phone. UPDATE FEB 2025 - Still crashes EVERY TIME I open the app. I then have to close it and re open, then it works. I've reported this previously and nothing changes. Please sort this out it's beyond frustrating.

1 person found this review helpful

Did you find this helpful?

Yes

No

User Groups

Users

Customers

Relationship	Interacts with products from the application.
Key Goals	- Buy their shopping on the application. - View products they may be interested in on the application.
Characteristics	- Basic knowledge of technology. Able to operate phone (install and navigate applications) 20-60 in age (varies a lot in age), and in a fair financial state.
Tasks	- Register for the application. - Login into the application - Enter the (customer) product catalog page and view products. - Add chosen product to the basket. - Enter basket and click on checkout. - Choose payment method, enter details and pay.
Constraints	- Easy to read fonts, large font sizes. - Simplistic layout for customers to interact with necessary features easily.
Technical Environment	- Fairly new phones (fair knowledge of the phone used). - Up to date android version.
Physical Environment	- Indoor and outdoor environments. - On the move, sitting down, etc.

Managers/Admins

Relationship	Administers the product catalog system.
Key Goals	- Manage the products inside the catalog page. - Can add new products to the catalog page.
Characteristics	30-50s in age. - Basic knowledge of technology. Able to operate phone (install and navigate applications)
Tasks	- Log into the admin account. - Enter the admin catalog page. - Add new products to the catalog page. - Delete existing products from the catalog page. - Edit existing products from the catalog page.
Constraints	- Simplistic layout for admins to interact with necessary features easily. - Easily readable fonts.

	• Protections to stop admins from accidentally changing the wrong product.
Technical Environment	• Fairly new work phones (fair knowledge of the phone used). • Up to date android version.
Physical Environment	• Indoor environments. • On the move, sitting down, etc.

Stakeholders

Food Dream LTD (Client)

Relationship	• Owners of the application (Food Dream App).
Key Goals	• Working with the developer to create the application to improve their customer satisfaction and sales.
Characteristics	• Basic knowledge of technology. • Good communication skills. • Great knowledge of the grocery market, working of their business and their customers.
Tasks	• Meetings with the developer to explain how they'd like the app to function, feel and perform. • End user requirements they need for the application. • Any other needs or issues that they have with the application.
Constraints	• Client may miss a requirement due to basic knowledge of technology.
Technical Environment	• Phone, laptop and desktop for communication between the client and developer. • Windows, Linux, macOS, IOS, Android. • Teams or discord for messages and calls if needed.
Physical Environment	• Indoor environment in an office setting. • A round table for face-to-face meetings.

Developer

Relationship	• Plan, design, develop, test, deploy and evaluate (Agile Development Method) the application for Food Dream LTD.
Key Goals	• Gather & analyze the requirements and workings of Food Dream LTD and implement it into the application. • Create a project plan, implement and test the application. • Document the testing and evaluate the application. • Create a user guide for the application.
Characteristics	• Great knowledge of technology

	No prior knowledge of the requirements and the workings of Food Dream LTD.
Tasks	- Analyse the requirements, workings of the company and the users. - Document system features and requirements needed for the user's tasks and clients end user requirements. - Create a project schedule for developer tasks. - Create UML diagrams to help give a clear and envisioned idea of the inner workings of the system, in turn helping the implementation process. - Implement the systems with the prior schedule and diagrams made. - Test the implemented systems functionality, security, performance and usability. - Document the test and evaluate the systems implemented to see if they meet all the requirements set. - Create a user guide for the users.
Constraints	- Research into the inner workings of Food Dream LTD and its users is required. - A major amount of time will need to be allocated for implementing the system. - Designing a simple system for users to use may take a lot of time to create as well.
Technical Environment	- Lenovo IdeaPad Gaming 3 laptop (32GB DDR4 RAM, AMD Ryzen 5 5600H, RTX 3050ti Graphics Card, 64-bit operating system). - Arch Linux x86_64 - Hyprrland (Window Manager) - Built-in keyboard and display (15" 1920x1080, 144hz Display) - Logitech G Pro Wireless Mouse.
Physical Environment	- Office based setting. - Home based setting. - On the move with laptop.

User Scenarios

Customers

Register to the application

The customer should open the Food Dream application which should show the login page first on load. The customer should click on the register link on the page to be transferred and enter the registration page. Once on the page the user should enter their email address, name and password into the appropriate form boxes, they must also agree to Food Dream LTD terms and conditions to proceed with registration and verify they're with a code sent to them via email.

Login to the application

The customer will move back to the login page after registering (the system will automatically do this). And should enter their newly registered details into the login page form (email, password) and should get a two-factor authentication email code sent to them to be entered in for logging in.

Enter the product catalog page and view products.

The customer, after logging in, should be redirected to the product catalog page and the customer will be to interact with the products on the page; The customer can also search for products on the catalog page.

Add chosen product to the basket.

When the customer finds a product, they'd like to purchase it, they can add it to the basket by clicking on the add to basket link on the individual product page or on the product box in the catalog page.

Enter basket and click on checkout.

When the customer is finished with their shopping and is ready to checkout. The customer can click on the basket icon on the catalog page to be redirected to the basket page with an order summary of their basket. Once they're happy with the basket the customer can click the checkout button and move to the checkout page.

Choose payment method, enter details and pay.

Once the customer has clicked on the checkout button and redirected to the checkout page. The customer should choose a payment method of 3 (Google Pay, Credit/Debit

Cards and PayPal) enter their details into the appropriate form boxes and hit pay to transact their shopping.

Managers/Admins

Log into the admin account.

The admin will enter the login page after loading the Food Dream application. The admin should enter their admin account details into the login page form (work email, password) to log in.

Enter the admin catalog page.

After login the system will redirect the admin to the admin catalog page. The admin can view & interact with the products on the catalog page.

Add new products to the catalog page.

At the top of the admin catalog page there is a button the admin can tap to enter the add new product page. The admin once on the new product page can enter the new product details into the appropriate fields on the new product page form. Once the details have been added to the form the admin can tap the submit button at the bottom of the page and confirm it, if they're sure they would like to add the new product.

Delete existing products from the catalog page.

When the admin hovers over a product in the admin catalog page two options will appear to edit or delete the product. Once the admin clicks on the delete item button the admin is shown a confirm notification to choose whether they confirm or not that they would like to delete the item.

Edit existing products from the catalog page.

When admin hovers over a product in the admin catalog page two options will appear to edit or delete the product. Once the admin clicks on the edit item button the admin is redirected to the edit product page. The admin once on the edit product page can enter the new product details into the appropriate fields on the edit product page form. Once the details have been added to the form the admin can tap the submit button at the bottom of the page and confirm it, if they're sure they would like to edit the product and add the new details.

Task Analysis

Customer

Register to the application

- Step 1. Turn on device.
- Step 2. Open the Food Dream Application.
- Step 3. Register a new account for the Food Dream Application.
- Step 3.1. Verify email to proceed with registration.

Login to the application

- Step 1. (After registering the system will move the customer to login page) The customer should login to their newly registered account.
- Step 1.1. Verify two-factor authentication via email to proceed with login.

Enter the product catalog page and view products.

- Step 1. (After login system will redirect customer to the customer catalog page) The customer is able view & interact with the products on the catalog page.
- Step 2. Search for a particular product they'd like to purchase.

Add chosen product to the basket.

- Step 1. Customer will add the product to the basket via entering the individual product page and clicking on the add to basket button.

Enter basket and click on checkout.

- Step 1. The customer will click on the basket icon and enter the basket page.
- Step 2. The customer should click on the checkout button to proceed to the checkout page.

Choose payment method, enter details and pay.

- Step 1. The customer should choose their payment method and enter their details into the checkout form
- Step 2. Then hit the pay button to pay for their items.

Managers/Admins

Log into the admin account.

- Step 1. The admin should load the Food Dream application on their work phone
- Step 2. Enter the login page and input their admin account details into the appropriate fields.

Enter the admin catalog page.

- Step 1. (After login system will redirect the admin to the admin catalog page) The admin is able view & interact with the products on the catalog page.

Add new products to the catalog page.

- Step 1. At the top of the admin catalog page there is a button the admin can tap to enter the add new product page.
- Step 2. The admin once on the new product page can enter the new product details into the appropriate fields on the new product page form.
- Step 3. Once the details have been added to the form the admin can tap the submit button at the bottom of the page and confirm it, if they're sure they would like to add the new product.

Delete existing products from the catalog page.

- Step 1. When admin hovers over a product in the admin catalog page two options will appear to edit or delete the product.
- Step 2. Once the admin clicks on the delete item button the admin is shown a confirm notification to choose whether they confirm or not that they would like to delete the item.

Edit existing products from the catalog page.

- Step 1. When admin hovers over a product in the admin catalog page two options will appear to edit or delete the product.
- Step 2. Once the admin clicks on the edit item button the admin is redirected to the edit product page.
- Step 3. The admin once on the edit product page can enter the new product details into the appropriate fields on the edit product page form.
- Step 4. Once the details have been added to the form the admin can tap the submit button at the bottom of the page and confirm it, if they're sure they would like to edit the product and add the new details.

Assumptions

- Users using the application will have access to an android phone and Wi-Fi or roaming.
- The android version is fairly new or up to date.
- Users have an understanding of the shopping experience and checkout experience. And basic knowledge of how to operate a phone and many applications.
- Customers have an email they can use to sign up with.
- Managers are given their admin account details.
- All changes to products by the managers are all predetermined by the company.

Constraints

- Time scale is quite tight (21st April) to implement and test the application so some features may be partially developed or undeveloped.
- Customers will have different devices with different screen types and resolutions.
- Some users may have little knowledge of how to use applications or a device and may need a guide to understand how to use the application.
- The developer will have to spend time researching and planning the inner workings of Food Dream LTD and requirements they need for the application.
- Wi-Fi/roaming will have to be used to access our application.

Security Requirements

SQL Injection

SQL injection is a form of attack that focuses on performing unauthorized SQL statements within the database. The statements could edit, add, delete and list/view the sensitive data held by the database. Protection against this is crucial in the protection of sensitive data and the functionality of the application. Prepared SQL statements is a good method to stop this as it cleans user inputs to stop the possibility of any malicious statements being sent to the database; if there is any malicious inputs entered by the user the input data will not be sent, and the user will have to re-enter a valid input to proceed.

The Data Protection Act & GDPR (General Data Protection Regulation)

The data protection act is the UK implementation of GDPR otherwise known as general data protection regulation. The act makes everyone who holds personal data responsible for using the data fairly, lawfully and transparently only used in specific or explicit purposes in a way that is necessarily relevant to the purpose. The data needs to be accurate and not held for longer than the necessary period. Not abiding by this act can affect a company via unlimited fines, bad publication and to an even higher degree imprisonment.

GDPR is closely like the UK's data protection act with differences to where the law is applied for example, the UK's act only applies in the UK and GDPR only applies to its EU members. The systems will need to abide by this act and regulation to avoid any prosecution by the UK government or EU.

Encryption

Encrypting customers' passwords is a secure method of protecting the authentication of a customer's account. However, for these passwords to be strong & secure they need to follow a certain policy or rules in the creation of them such as meeting a certain length and using multiple types of characters to make the password in plain text more complex. If the policy is not met however, the password is rejected until the password is strong enough to be encrypted.

After the creation of a strong password a further step will be needed to obscure the password from unauthorized individuals. Encryption, specifically an algorithm to hash the password into ciphertext making it extremely difficult to decrypt and see the true password. A particular algorithm I will make use of is bcrypt as it is particularly more secure than SHA algorithms which is more vulnerable to brute force and rainbow tables.

Performance Requirements

Server-side

- **Response Time:** 1-2 seconds (minimum); This should ensure seamless and quick responses to customer requests, giving them a smooth user experience. The time for loading pages, updates and processes should also be in a short time frame to minimize any frustration from the customer and avoid any lag or delay.
- **Database Query Timeframe:** Queries such as updates on stock, product prices, names, passwords, emails. Should be performed within a time frame of under 1

second (minimum) to ensure a seamless user experience. However, queries that contain more complex data or higher amounts of data should be performed within a time frame of 3 seconds (minimum) to allocate for the more complex data being retrieved.

- Error Handling Timeframe: In case of certain errors such as failed database queries and network errors which will affect the processing of data, a timeframe for a retry method should be fairly short to keep our customers from being agitated, under 5 seconds (minimum).

Client-side

- Operating System: Android 6.0 (Marshmallow) (minimum); as this is the oldest version of android that the application will be suited/compatible with and allow the developer to use android's new features.
- RAM: 2GB (minimum), 3GB-4GB (recommended).
- Storage: 50MB of free space (minimum); this application is small to moderate in size so it shouldn't need more than 50MB of free space to be installed.
- Network: 2MB/s (minimum) 10MB/s (recommended); real-time updates from the WebSocket's and updates from the database will rely on a stable network connection from the customer to the server to process the updates to the application.

Device Recommendation

I'd recommend the Samsung Galaxy A04s for a customer on a budget or a company buying new work devices, as the device meets all the requirements stated for a low price of £169.99.

Samsung Galaxy A04s
(Copper, 4GB RAM,
64GB Storage)

[Visit the Samsung Store](#)

4.4 ★★★★★ ✓ 1,135 ratings

Amazon's Choice

£169⁹⁹

Pay with Instalments by Barclays. Subject
to financial status & eligibility.

Credit broker: Amazon EU S.a.r.l, Lender:
Barclays. [Learn more](#)



Roll over image to zoom in



Brand	Samsung
Operating system	Android 12.0
RAM memory installed size	4 GB
CPU model	Snapdragon
CPU speed	2 GHz
Memory storage capacity	64 GB
Screen size	6.5 Inches
Resolution	720 x 1600
Refresh rate	90 Hz
Model name	Galaxy A04

For a more expensive option for the customer, I'd recommend the Google Pixel 6a, as it will provide a smoother experience as it has an increase in RAM and Cellular technology compared to the budget phone.



Roll over image to zoom in

Installation options available

Brand Google

Operating system Android 12.0

RAM memory installed size 6 GB

CPU model Snapdragon

Memory storage capacity 128 GB

Screen size 6.1 Inches

Resolution 1080 x 2340

Model name Pixel 6a

Wireless carrier Unlocked

Cellular technology 5G

[^ See less](#)

Google Pixel 6a – Unlocked Android 5G Smartphone with 12 megapixel camera and 24-hour battery – Charcoal - 6.1 Inches

[Visit the Google Store](#)

4.4 ★★★★★ 4,009 ratings

-15% £339¹⁵

RRP: £399.00

prime

FREE Returns

Savings Experience 60 days of Audible free [Terms](#)

Pay with Instalments by Barclays. Subject to financial status & eligibility.

Credit broker: Amazon EU S.a.r.l, Lender: Barclays. [Learn more](#)

Colour: **Charcoal**

Roll over image to zoom in

Resources

Asset Log

File Name	File Type	File Ext	File Size	File Res	File Origin	Description	Example
Logo	Image	PNG	124KB	500x500	AI-generated.	Logo for the application.	
Product Image	Image	PNG	237KB	360x360	https://pngtree.com/freepng/fast-food-big-ham-burger_8648590.html - free	Product image for the catalog page.	

					images of food.		
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Software

System Environment

- Operating System: Arch Linux x86_64.
- Browser: Mozilla Firefox.
- Search Engine: Google.

Planning

- Documentation: Microsoft Word.
- Project Plan: Trello.
- Wireframes: mockups.com
- Diagrams: draw.io.

Development

- IDE's: PHPStorm, Android Studio, Neovim.
- Documentation: Microsoft Word, GitHub, Trello.
- Database: phpMyAdmin.
- Research: GitHub, Microsoft Word, Stack Overflow, YouTube, Kotlin In-Depth (Book), ChatGPT.

Hardware

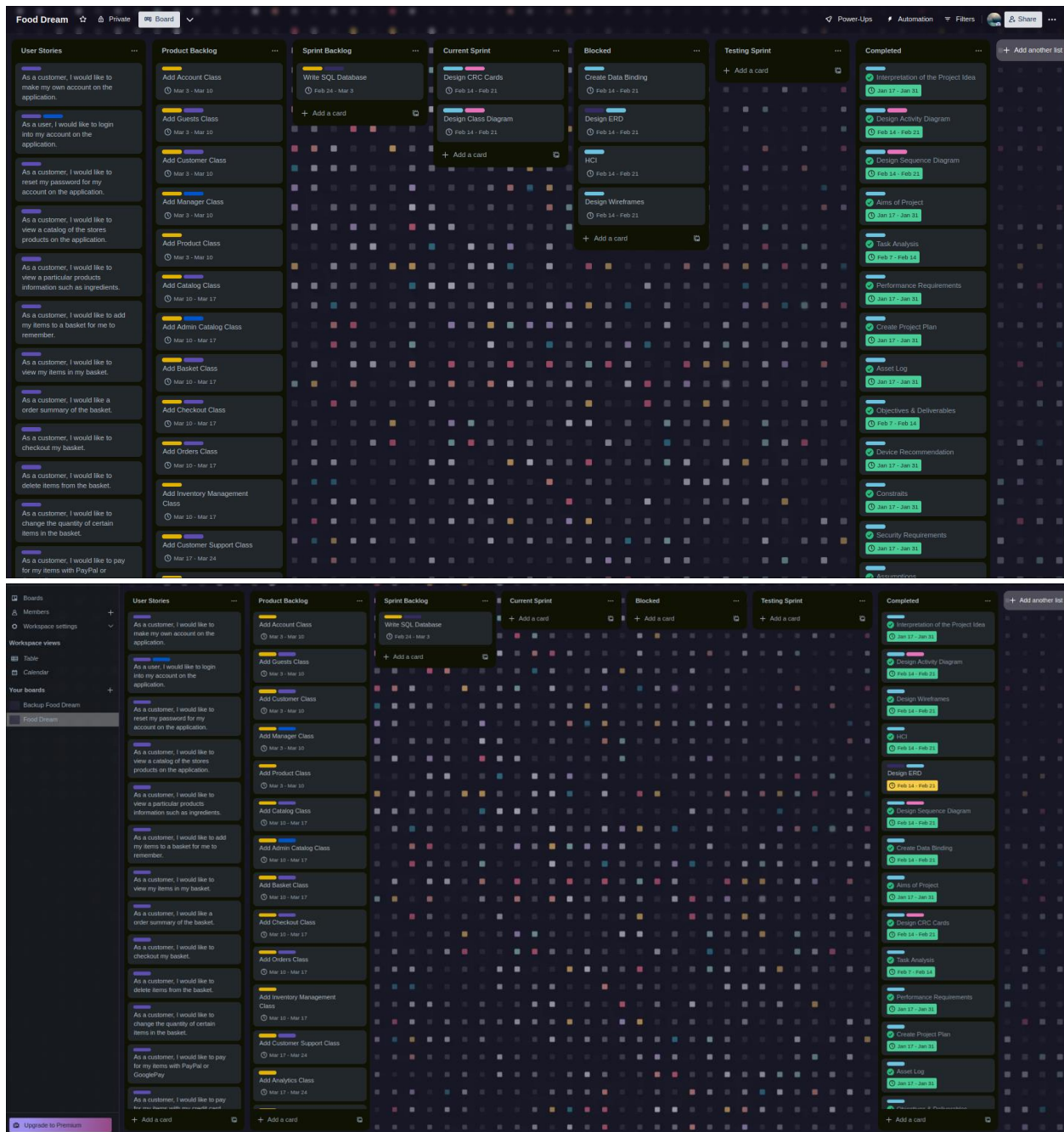
- Lenovo IdeaPad Gaming 3 Laptop.
- 32GB DDR4 RAM.
- AMD Ryzen 5 5600H.
- RTX 3050ti Graphics Card.
- Arch Linux x86_64.
- Hyprland (Window Manager).
- Wayland (Composer).
- Built-in keyboard and display (15" 1920x1080, 144hz).
- Logitech G Pro Wireless Mouse.

Project Timeline

This project timeline lays out the required tasks for the whole project and schedules them for completion based on certain deadlines. Tasks are mutable/subject to change if necessary or creation of new tasks may occur during the timeline. Tasks will be

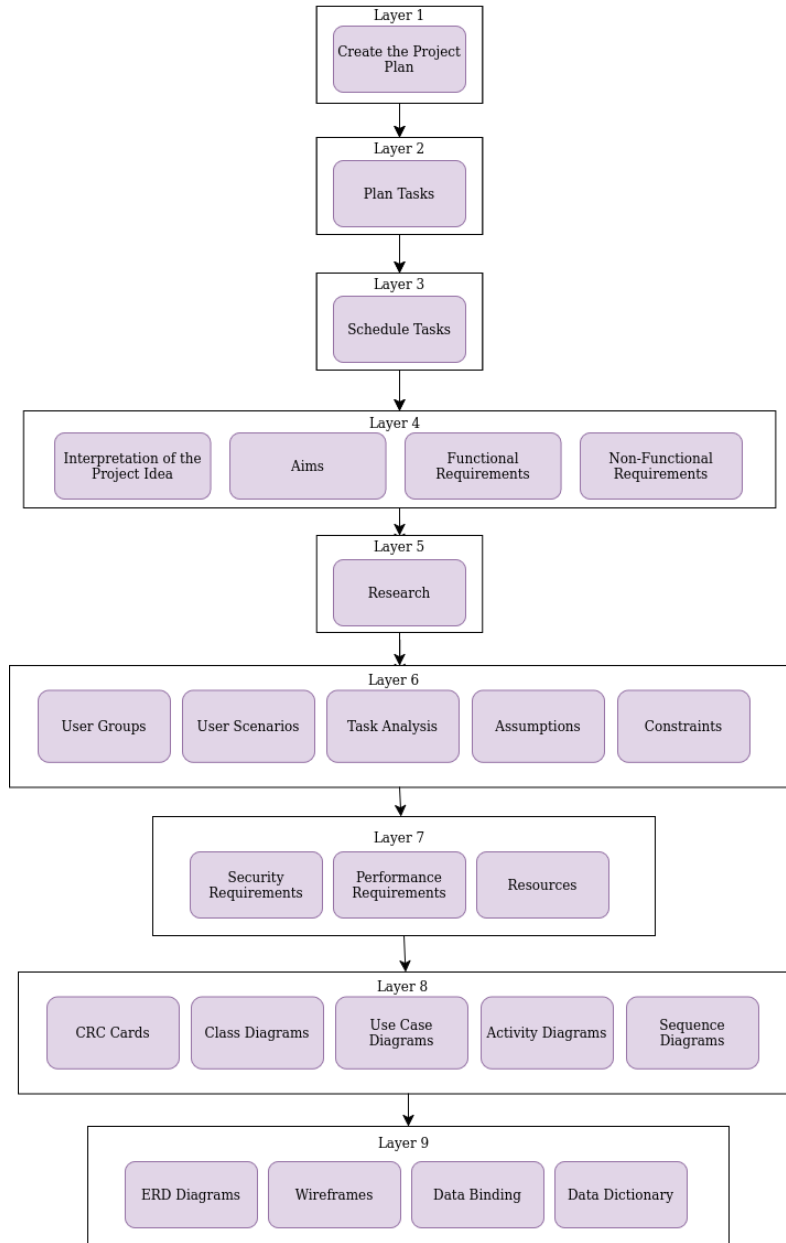
documented and ordered in Trello with each task given a group and priority/layer. User stories are used in the creation of these tasks and placed into a product backlog, they are then ordered and grouped and placed into the sprint backlog until the task is ready to be carried out in the current sprint. Upon completion of the task in the current sprint, the task is moved into the completed block if the task does not require testing, however if the task does require testing, they are pushed into the testing sprint until the task is fully tested which then will finally be moved to the completed block.

Trello was completed as of 17th February.



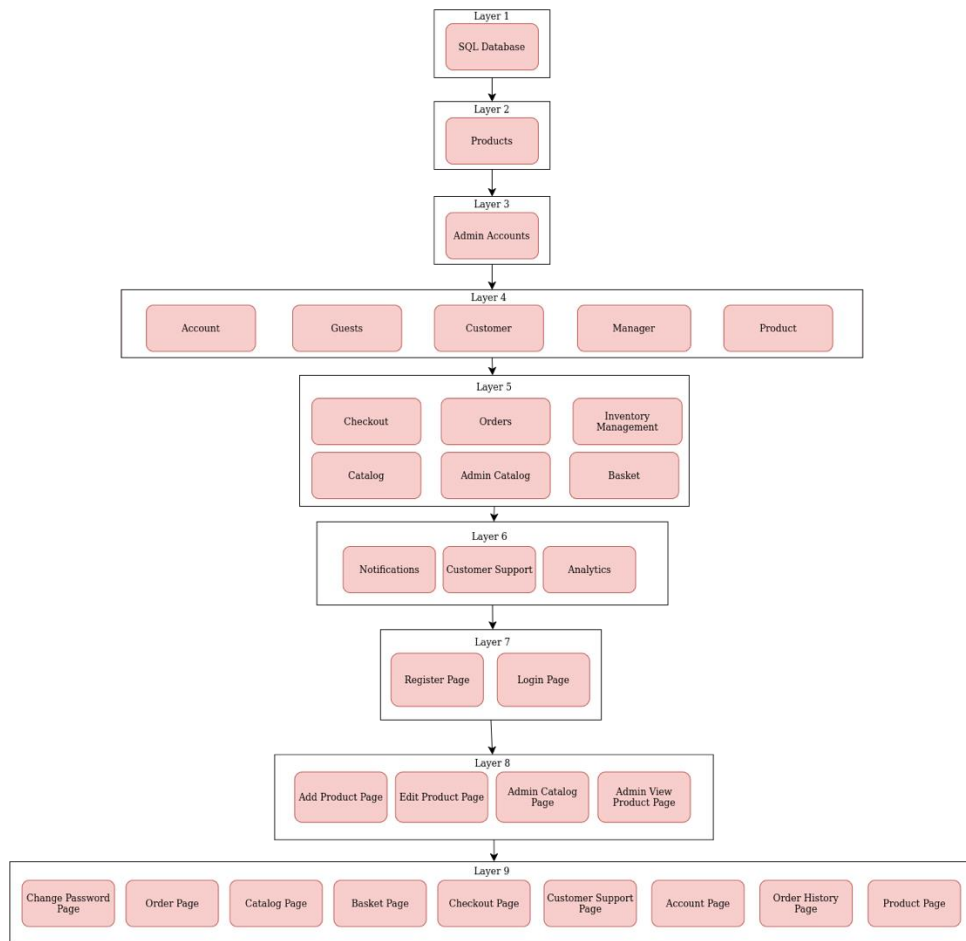
Planning Tasks

This diagram shows the flow of the completion tasks. Each task is layered and those that are higher are completed first before any of the tasks in the layer below e.g. tasks on layer 1 must be completed before tasks on layer 2 are completed.



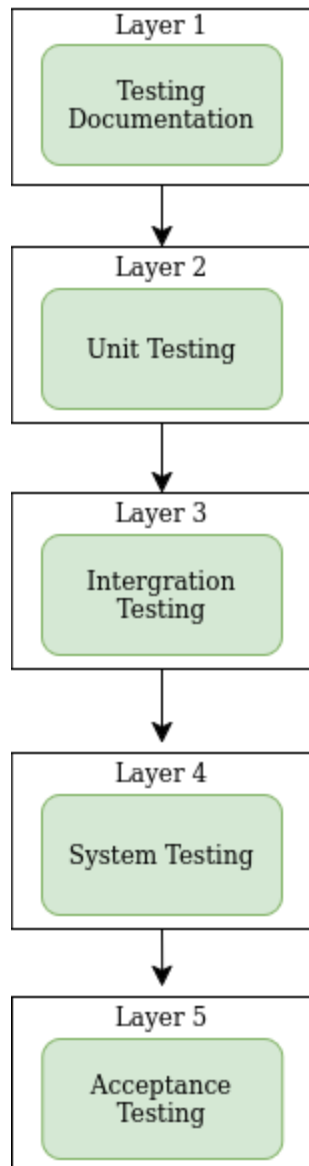
Implementation Planning

All planning tasks must be completed before the implementation can be started. Again, each task is layered and those that are higher are completed first before any of the tasks in the layer below e.g. tasks on layer 1 must be completed before tasks on layer 2 are completed.



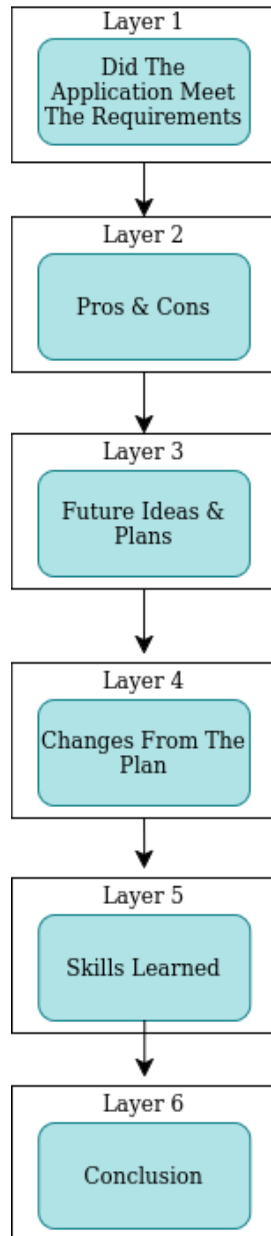
Test Planning

All implementation tasks must be completed before the testing can be started. Again, each task is layered and those that are higher are completed first before any of the tasks in the layer below e.g. tasks on layer 1 must be completed before tasks on layer 2 are completed.



Evaluation

All testing tasks must be completed before the evaluation can be started. Again, each task is layered and those that are higher are completed first before any of the tasks in the layer below e.g. tasks on layer 1 must be completed before tasks on layer 2 are completed.



Schedules

	Planning	Implementation	Testing	Evaluation
Week 1	Project Plan, Brief and Analysis			

Week 2	Project Plan, Brief and Analysis			
Week 3	Project Timeline			
Week 4	Project Diagrams			
Week 5 Planning Phase Due	Transition Week	Write SQL Database based on the ERD model and Data Dictionary		
Week 6		Create High Priority Models		
Week 7		Create High Priority Models		
Week 8		Create Other Models (Layer 6)		
Week 9		Create Starting Pages & Admin Pages		
Week 10		Create Other Pages (Layer 9)		
Week 11		Transition Week	Start Project Testing Following the Test Plan	
Week 12			Unit Test, Intergration Testing, System Testing and Acceptance Testing	
Week 13 Implementation and Testing Phase Due		Transition Week	Transition Week	Start Project Evaluation
Week 14				Finish Evaluation

Week 15 Evaluation Phase Due				Transition Week
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Objectives & Deliverables

These are objectives and outcomes of each project phase; Planning, Implementation, Testing and Evaluation.

Planning Phase

- **Project Scheduling**
 - Create a project timeline, based on:
 - Task Priority
 - Deadlines
 - Schedule tasks within the project timeline.
- **Project Idea Analysis**
 - Interpretation of the project idea.
 - Lay out functional and non-functional requirements.
 - Aims of the project.
- **Project Research**
 - Define User Groups
 - Define User Scenarios
 - Layout Task Analysis
 - Define Assumptions
 - Define Constraints
 - Conduct Interview with Client
 - Research Similar Systems
- **Additional Project Requirements**
 - Define Performance Requirements
 - Define Security Requirements
 - Define Device Requirements
- **Resources**
 - Create asset log
 - Define software used
 - Define hardware used
- **Design Phase**
 - Create use case diagram
 - Create activity diagram
 - Create class diagram

- Create sequence diagram
- Create ERD diagram
- Create data dictionary
- Create wireframes
- Create data binding diagram
- Create CRC cards

Implementation Phase

- **SQL Database**
 - Code SQL for the database based on the ERD and data dictionary diagram.
- **Admin & Product Data**
 - Add the admin & product data to the database.
- **Creation of High Priority Models**
 - Create Account Model.
 - Create Guests Model.
 - Create Customer Model.
 - Create Manager Model.
 - Create Product Model.
 - Create Catalog Model.
 - Create Admin Catalog Model.
 - Create Basket Model.
 - Create Checkout Model.
 - Create Orders Model.
 - Create Inventory Management Model.
- **Creation of Other Priority Models**
 - Create Notification Model.
 - Create Customer Support Model.
 - Create Analytics Model.
- **Creation of UX Pages**
 - Create a login page based on the created login page wireframe.
 - Create a subpage for the register page based on the created register page wireframe.
 - Create a subpage for the forgot password page based on the created forgot password page wireframe.
 - Create Admin Product Catalog Page based on the create wireframe.
 - Create an Admin Individual Product Page.
 - Create add product page.
 - Create edit product page.

- Create Customer Catalog Page
- Create Customer Individual Product Page
- Create Order Page
- Create Checkout Page
- Create Basket Page
- Create Customer Support Page
- Create Account Page
- Create Order History Page

Testing Phase

- **Testing Documentation**
 - Create unit testing documentation.
 - Create integration testing documentation.
 - Create system testing documentation based on the functional requirements of the project.
 - Create acceptance testing documentation based on the business and user requirements of the project.
- **Perform Tests**
 - Carry out unit tests.
 - Implement changes if needed after carrying out tests then test again.
 - Carry out integration tests.
 - Implement changes if needed after carrying out tests then test again.
 - Carry out system tests.
 - Implement changes if needed after carrying out tests then test again.
 - Carry out acceptance tests
 - Implement changes if needed after carrying out tests then test again.

Evaluation

- **Evaluation Documentation**
 - Has the application met the requirements?
 - Pros & Cons of Decisions Made
 - Future ideas and plans.
 - Changes from the original plan
 - Skills learned from this project
 - Conclusion of the whole project.

Main Objectives

- **Planning Phase**
 - Project Scheduling
 - Project Idea Analysis
 - Project Research
 - Additional Project Requirements
 - Resources
 - Design Phase
- **Implementation Phase**
 - SQL Database
 - Admin & Product Data
 - Creation of High Priority Models
 - Creation of Other Priority Models
 - Creation of UX Pages
- **Testing Phase**
 - Testing Documentation
 - Perform Tests
- **Evaluation**
 - Evaluation Documentation

Business Model

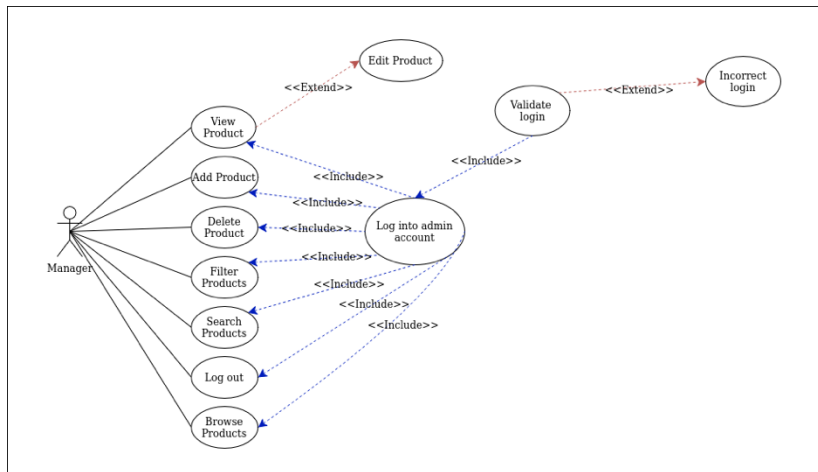
These UML diagrams were created using draw.io software.

Use Case Diagram

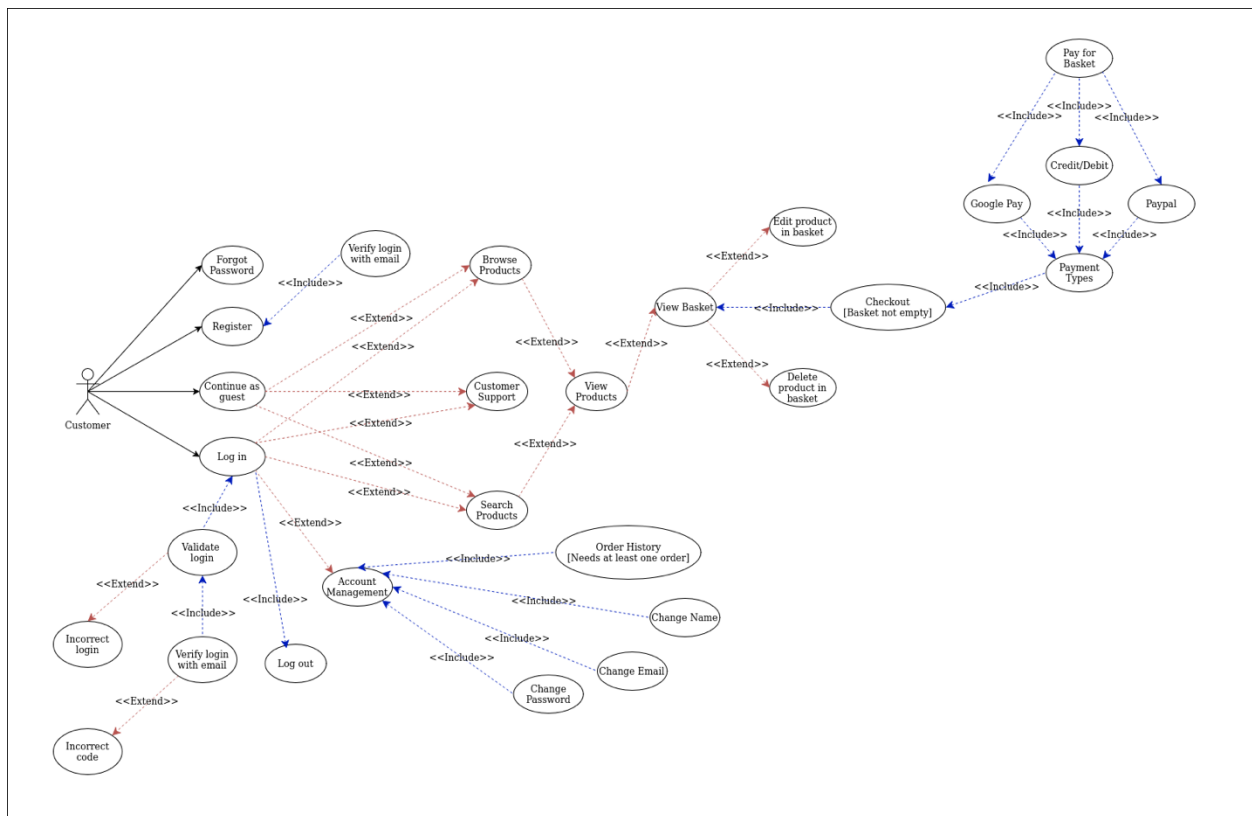
This is a high-level look at the application as a whole. It focuses on the use cases of the customers and managers when using the application. The managers are responsible for managing the products in the storefront/catalog for the customers to interact with. They will have an admin account given to them to log in with as well.

The customer can interact with systems such as creating accounts, logging to accounts, editing account, viewing products, adding, editing and removing products from their basket, view their basket and checking out with their basket.

Manager



Customer

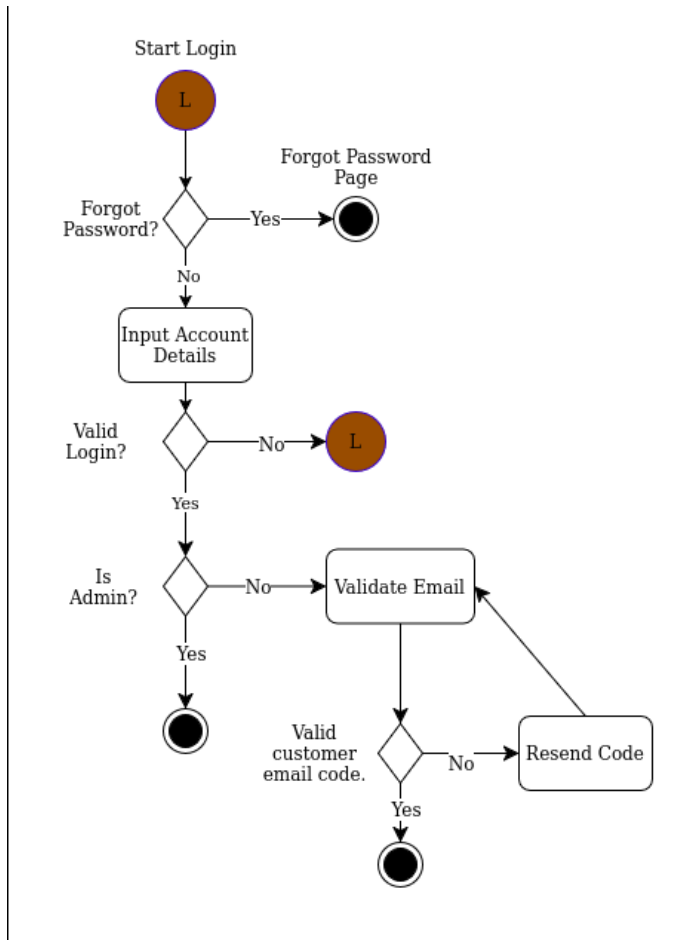


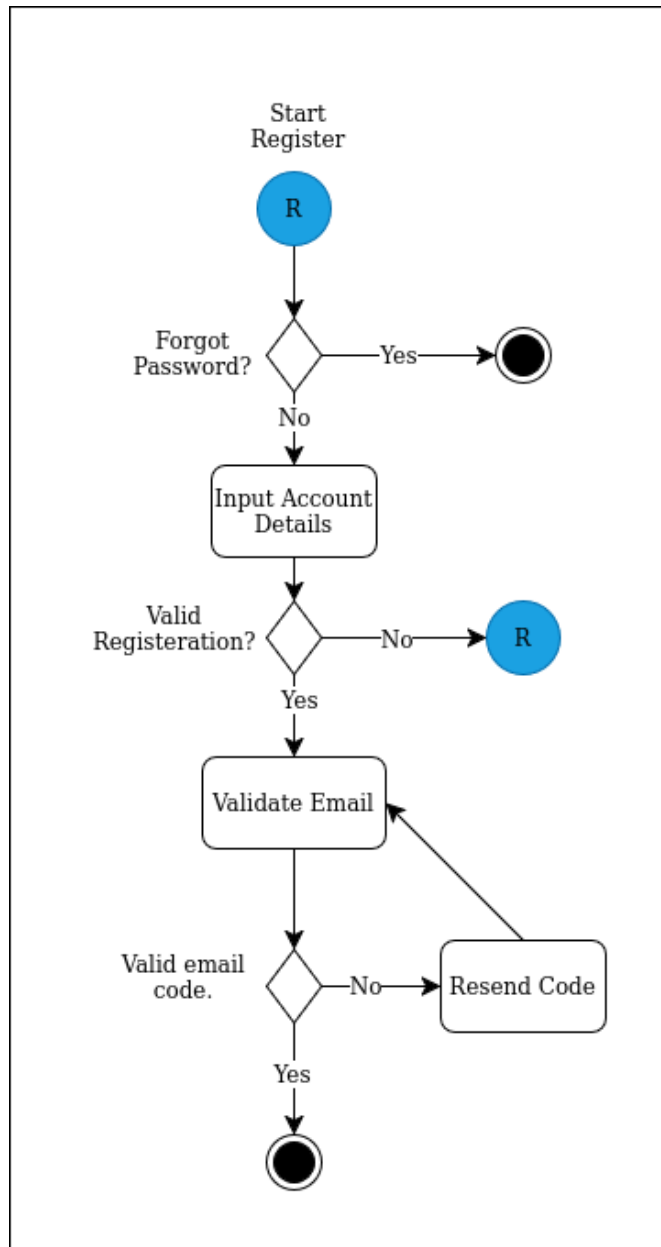
Activity Diagram

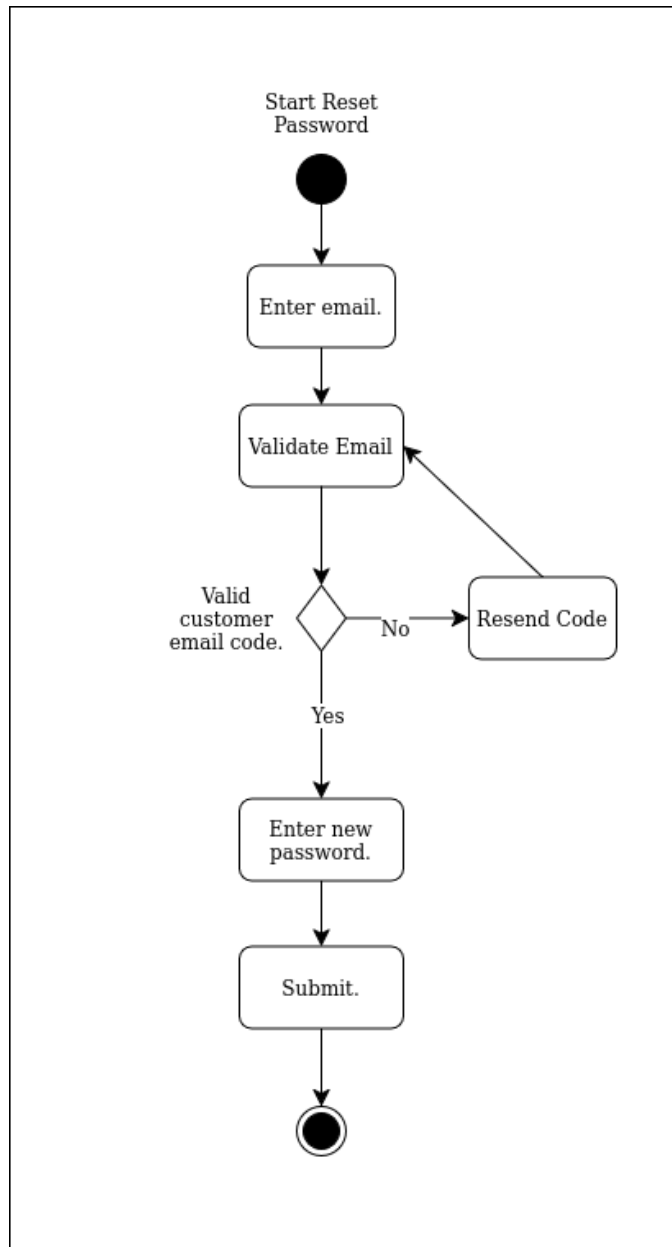
These activity diagrams depict the procedure of the system and individual systems for the customer and manager based on conditions and events that happen throughout the process of the system or systems.

Diagrams in svg files will be supported for a better view.

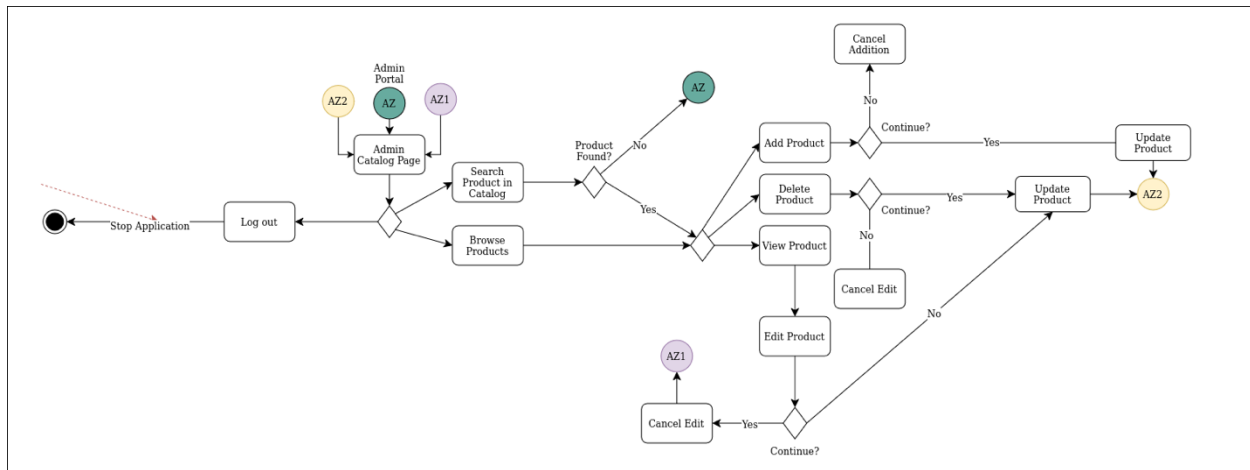
Login System



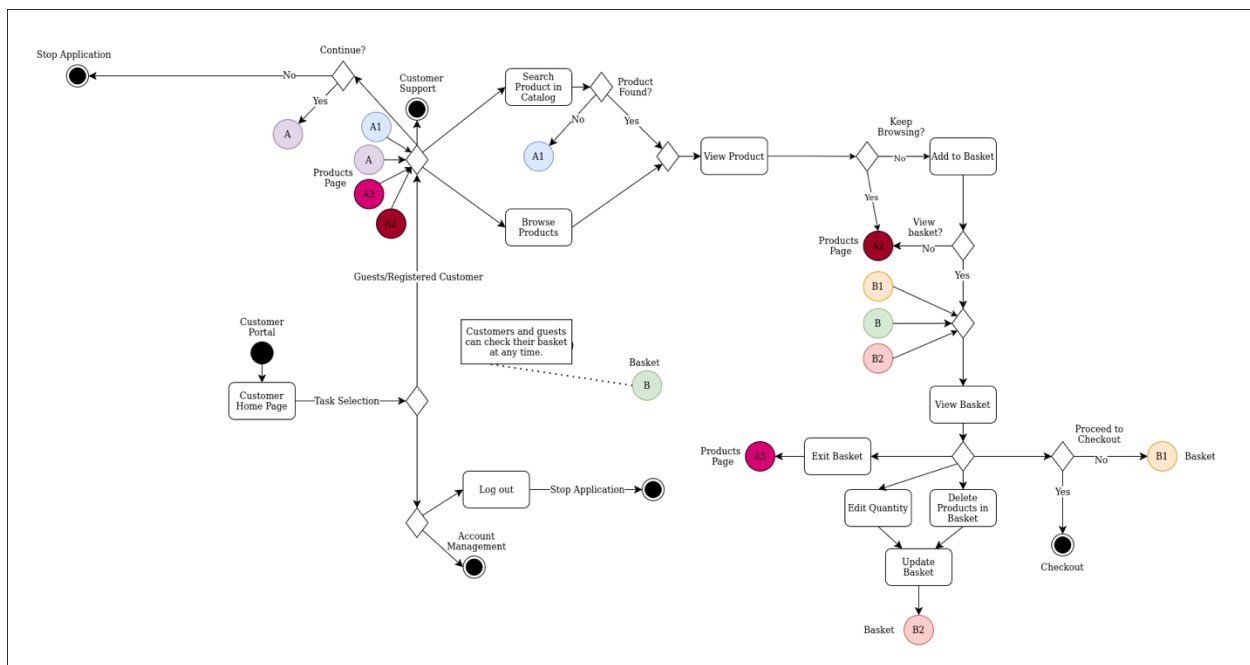
Register System

Forgot Password System

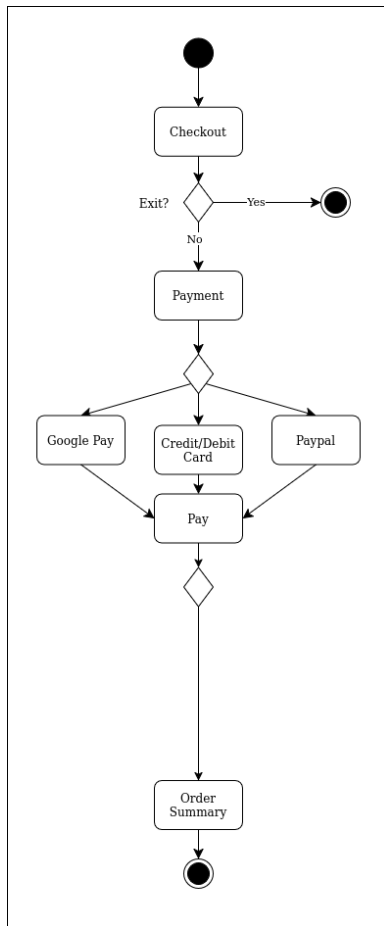
Product Management System



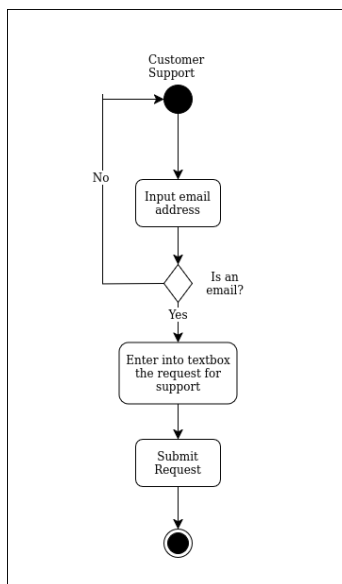
Ordering System



Checkout System



Customer Support System

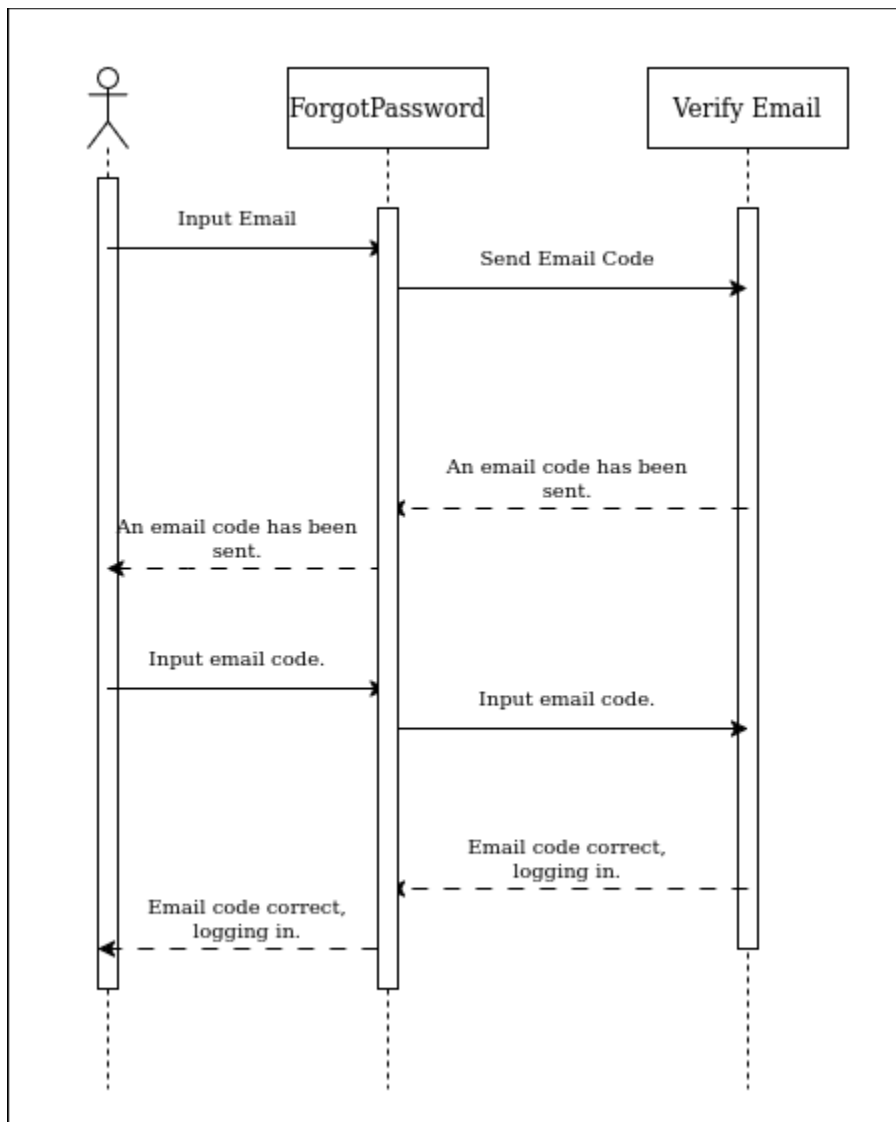


Sequence Diagrams

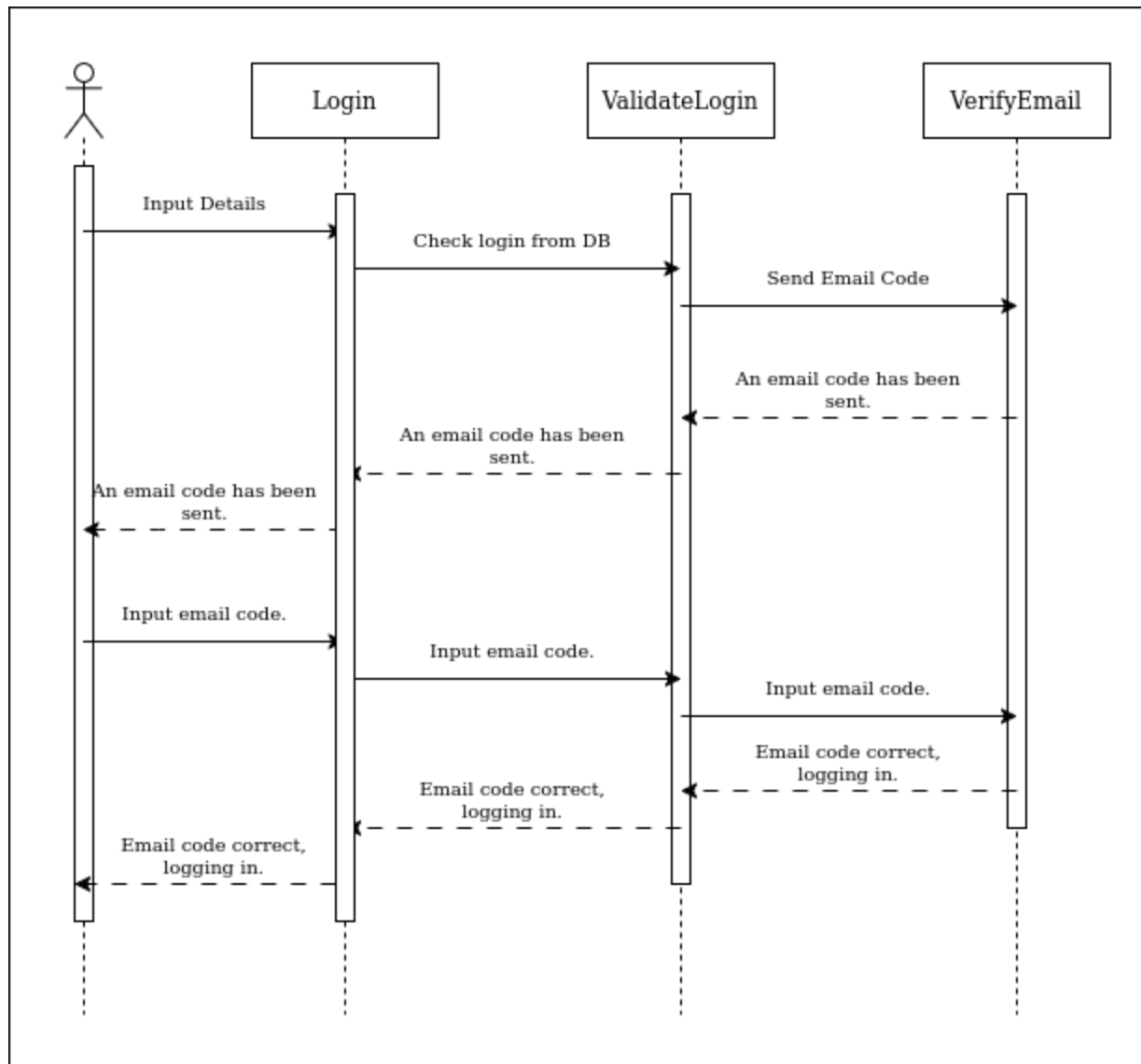
These sequence diagrams show how objects within the system interact with one another. Each diagram represents a specific use case, depicting user interactions with the system. These diagrams are supposed to give a simplified view of the system/systems within the application.

All Users

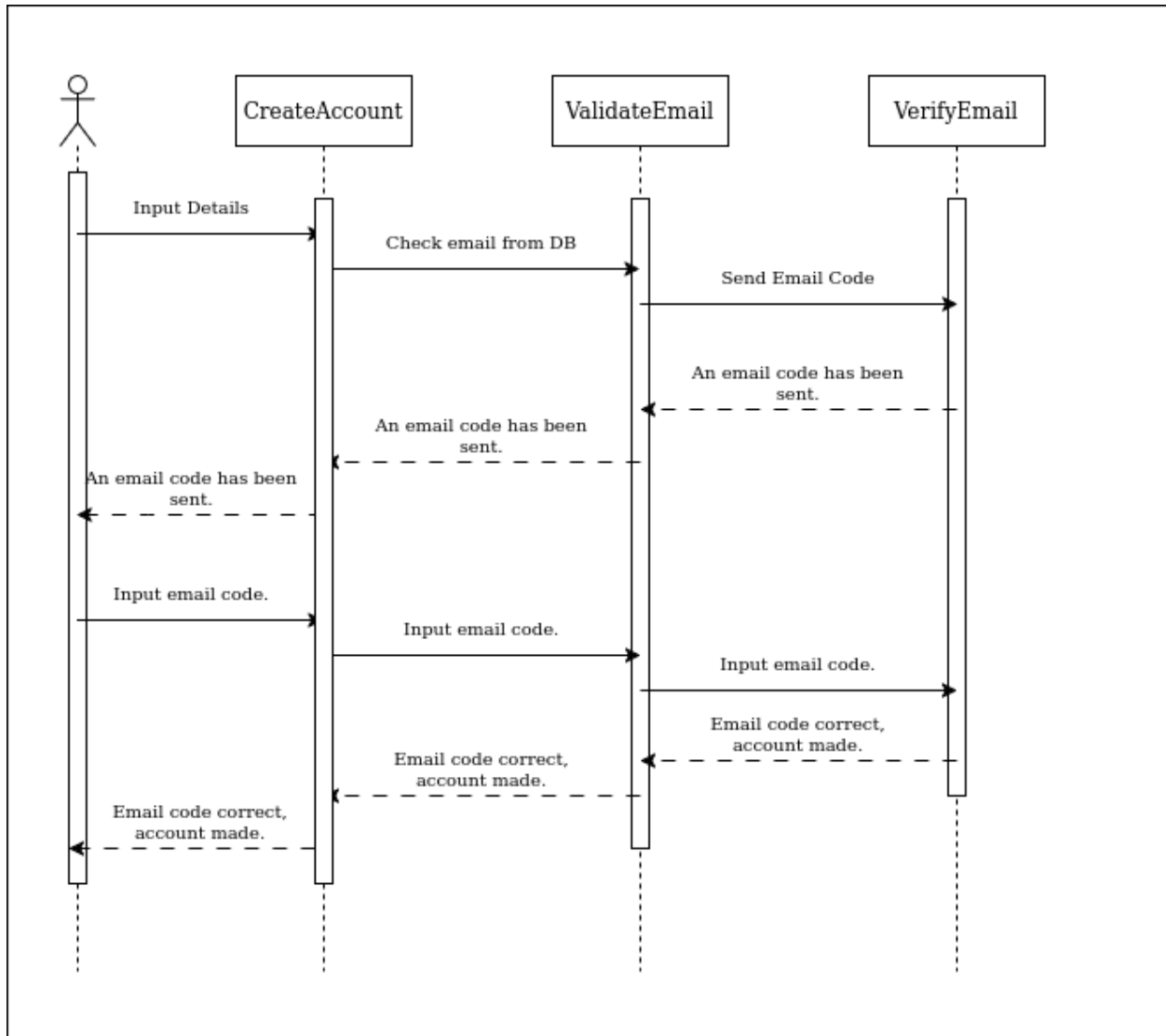
Forgot Password:



Login:

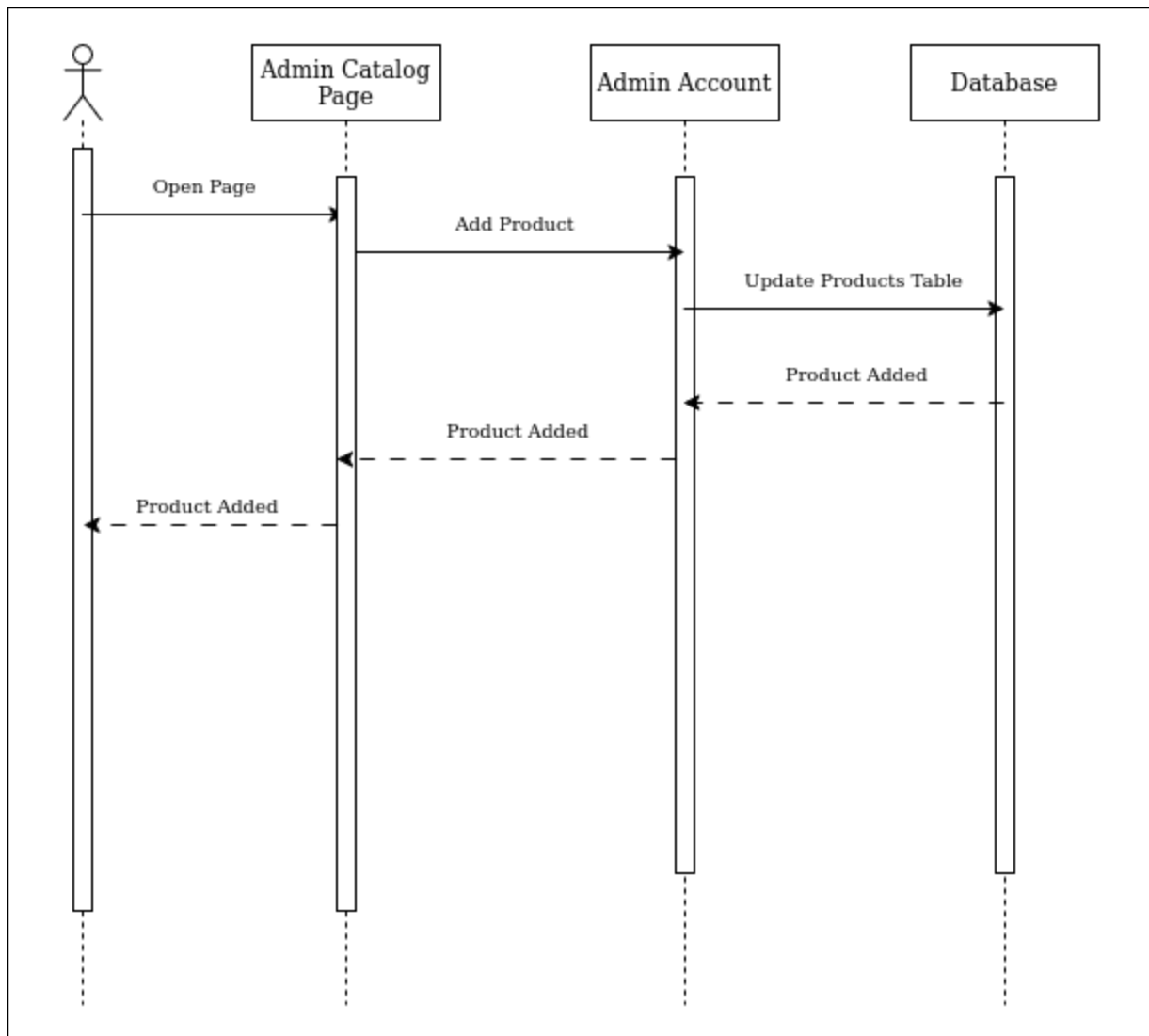


Create Account:

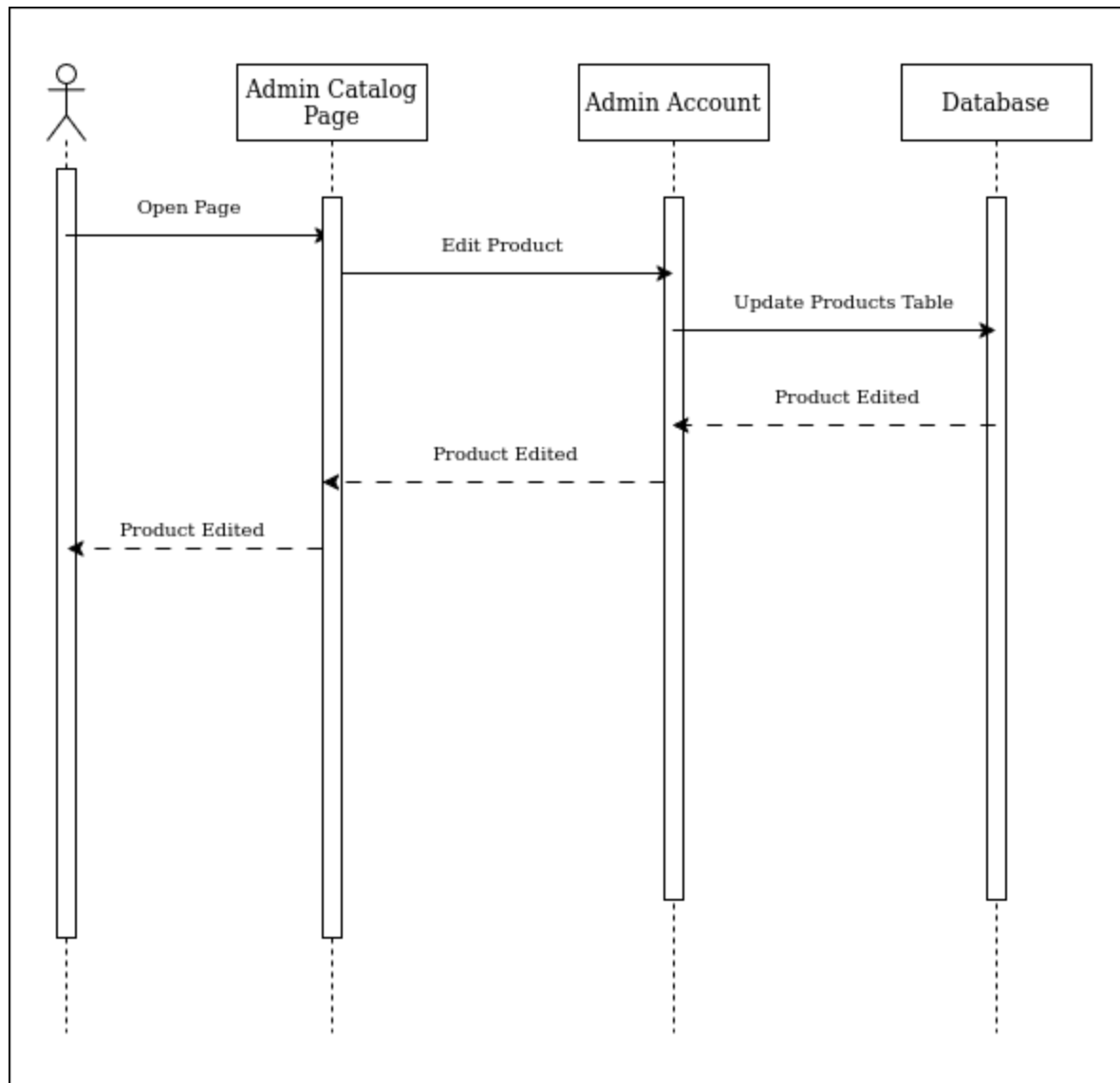


Admin

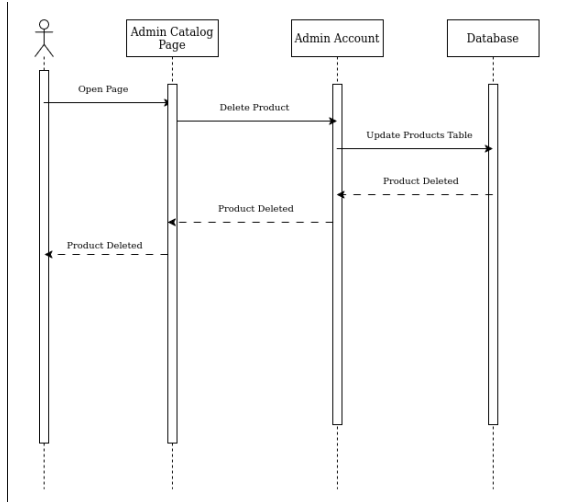
Add Product:

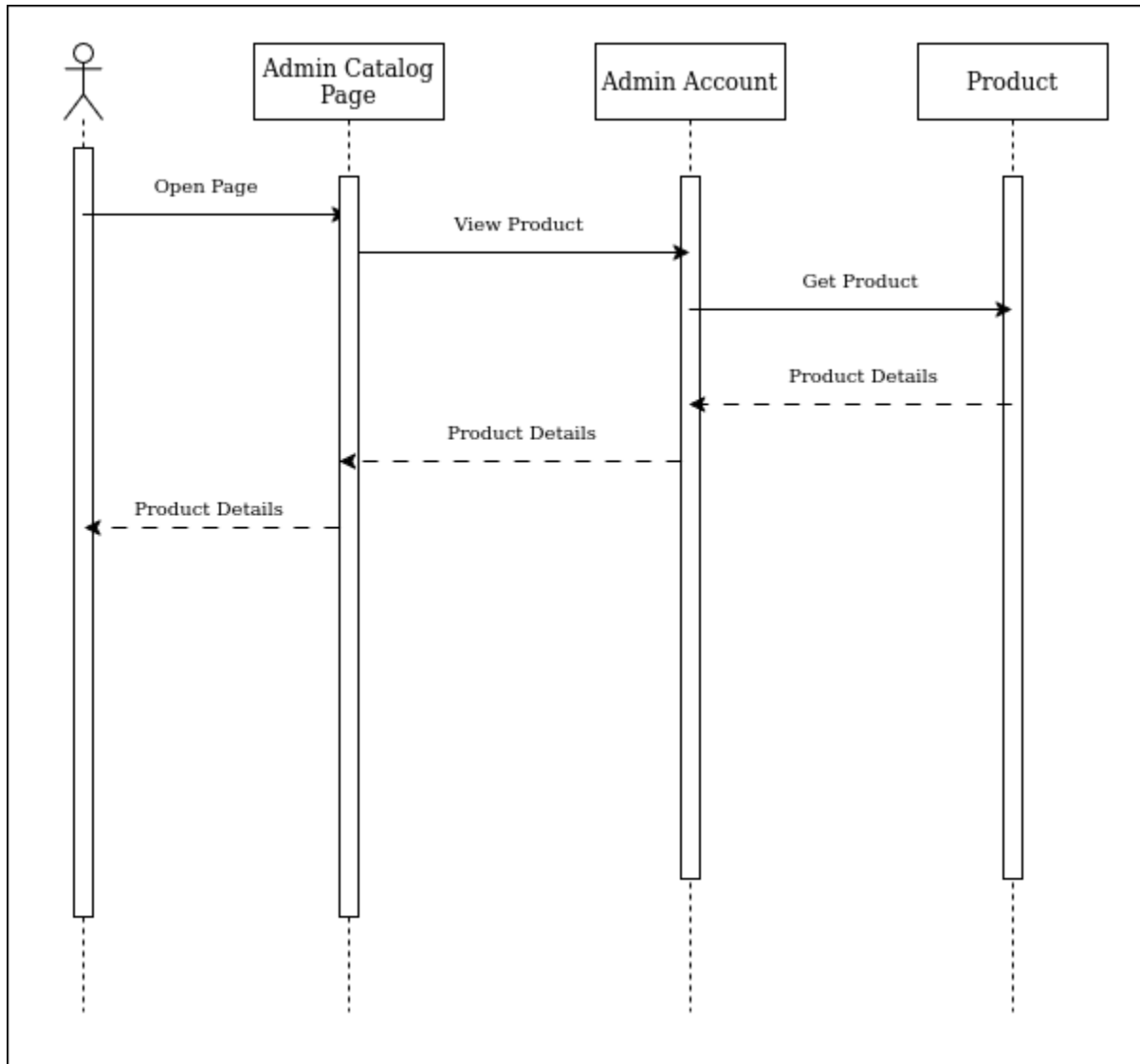


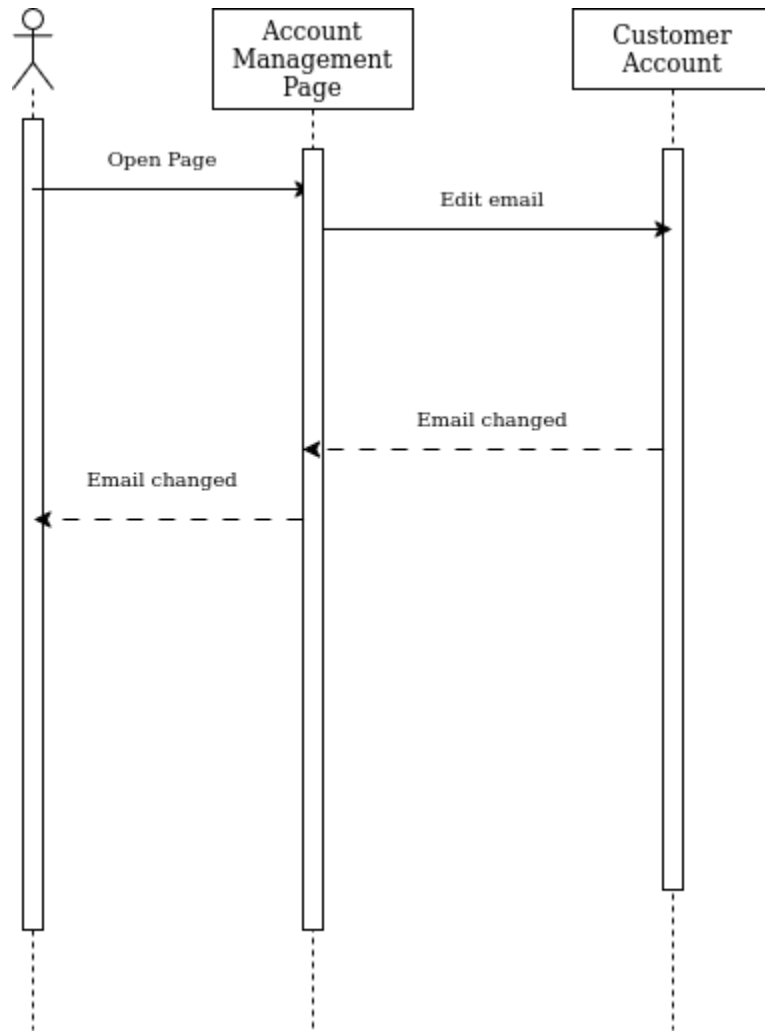
Edit Product:

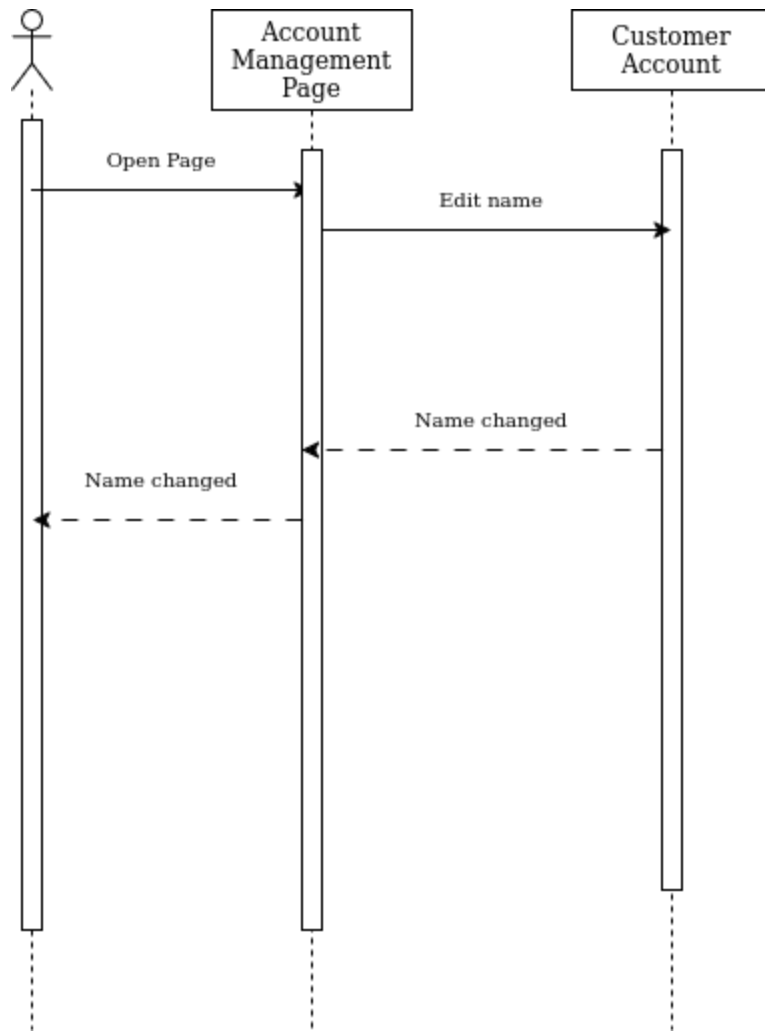


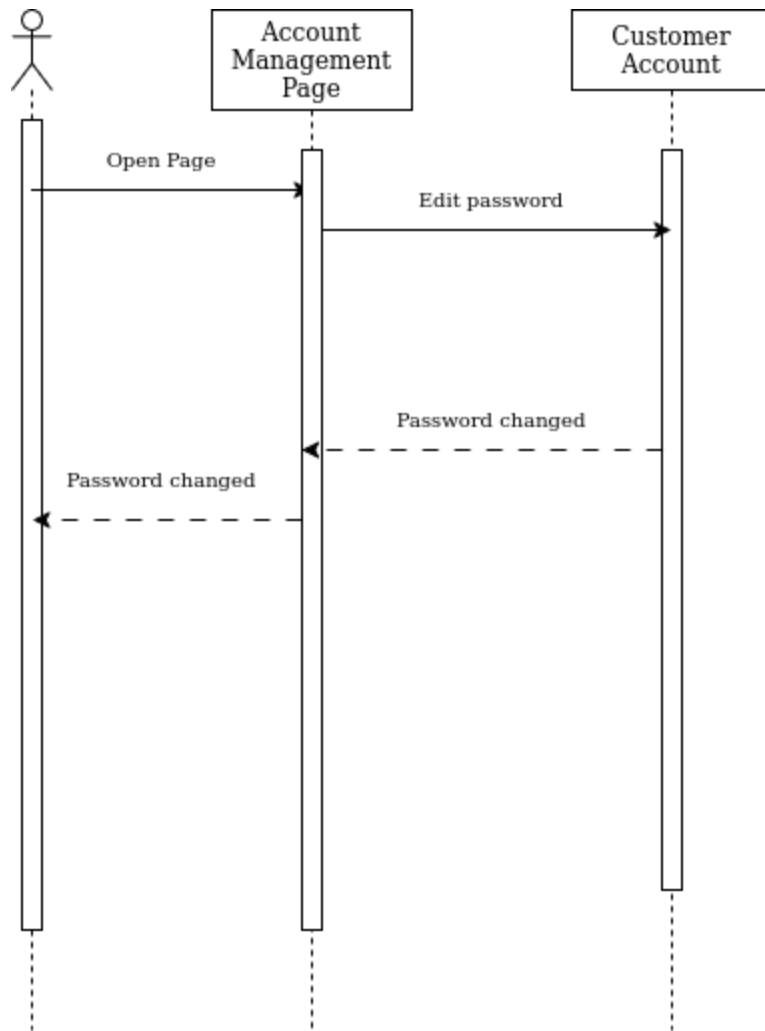
Delete Product:



View Product:

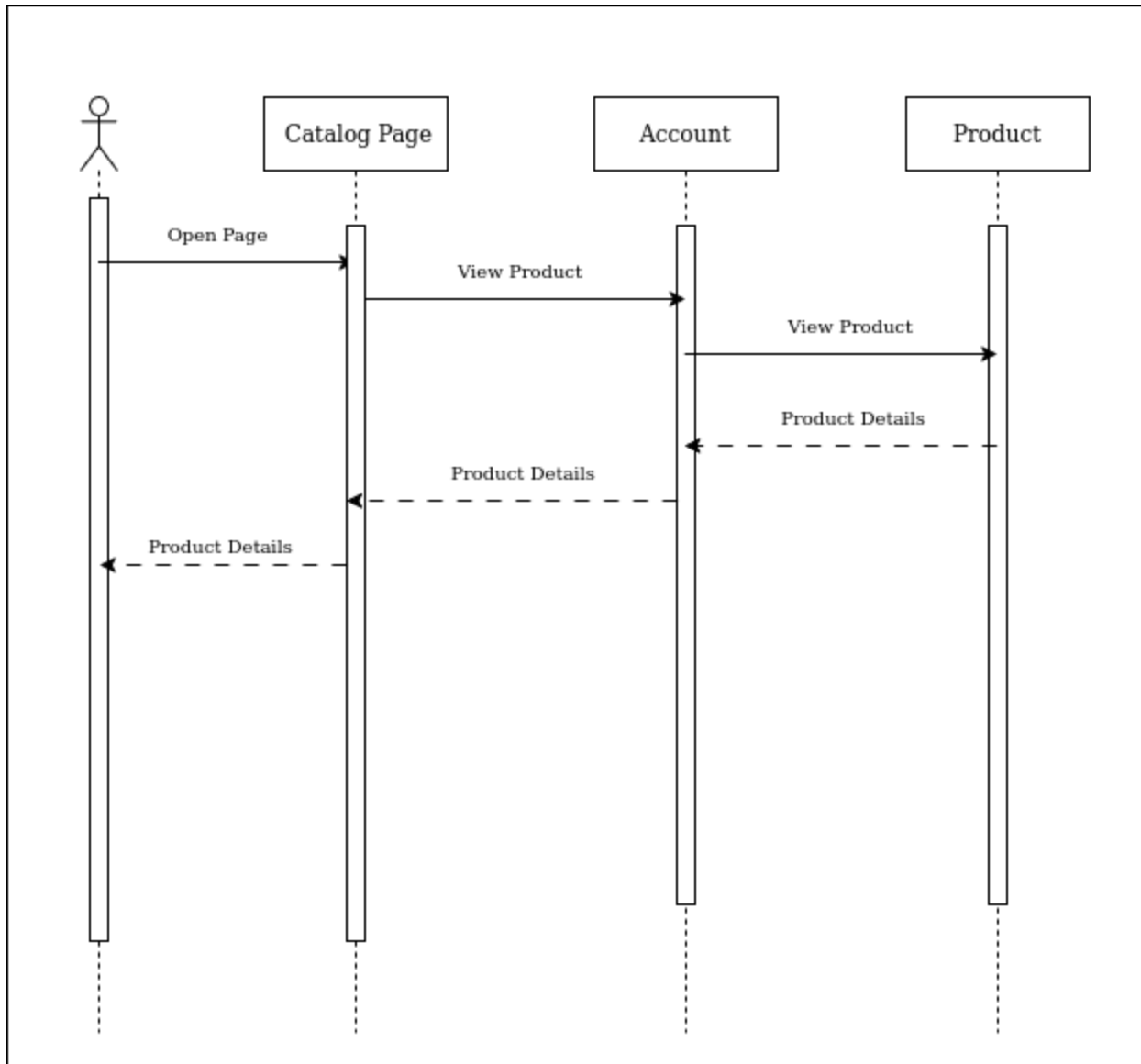
Customer



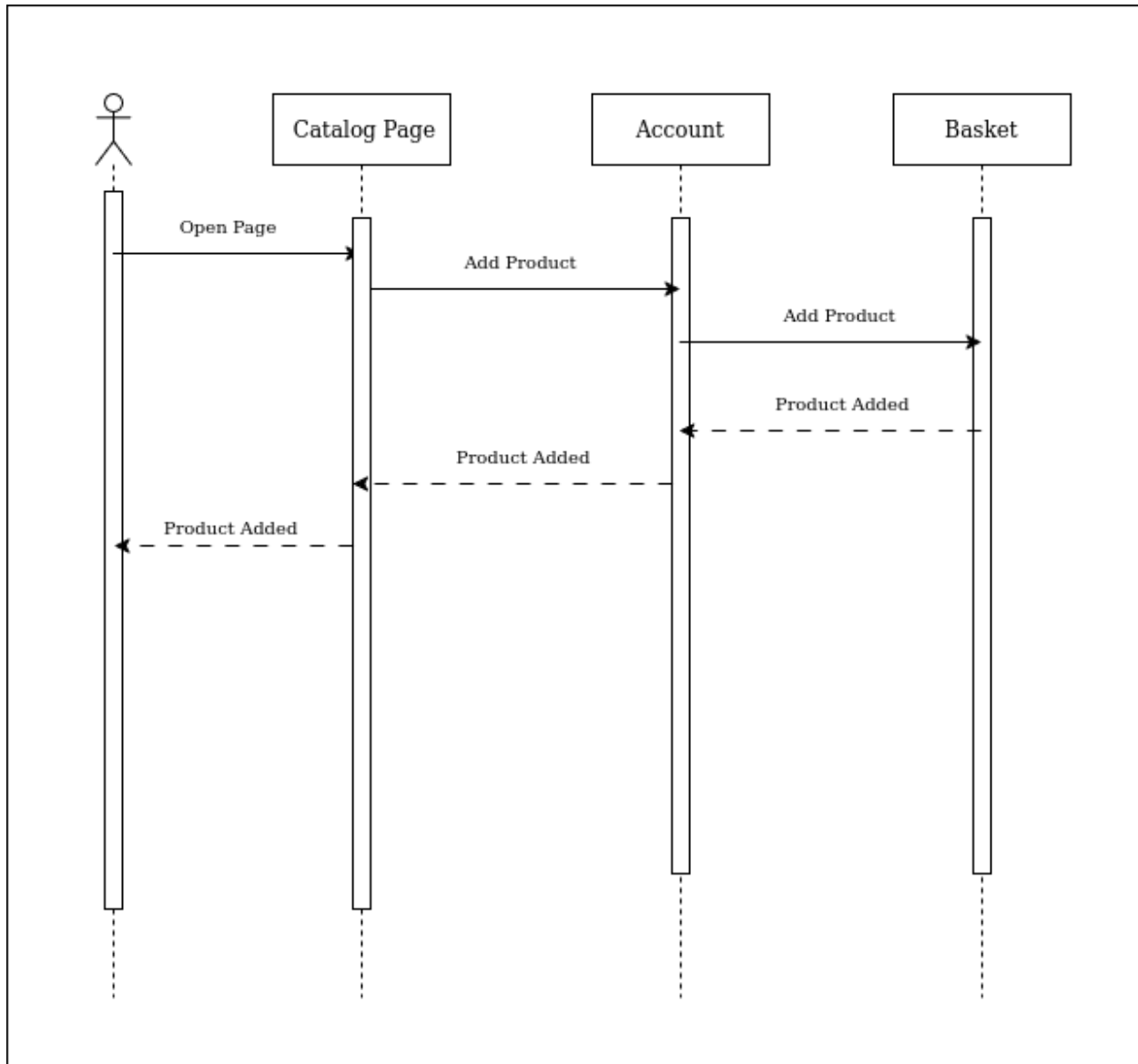


Customer/Guest

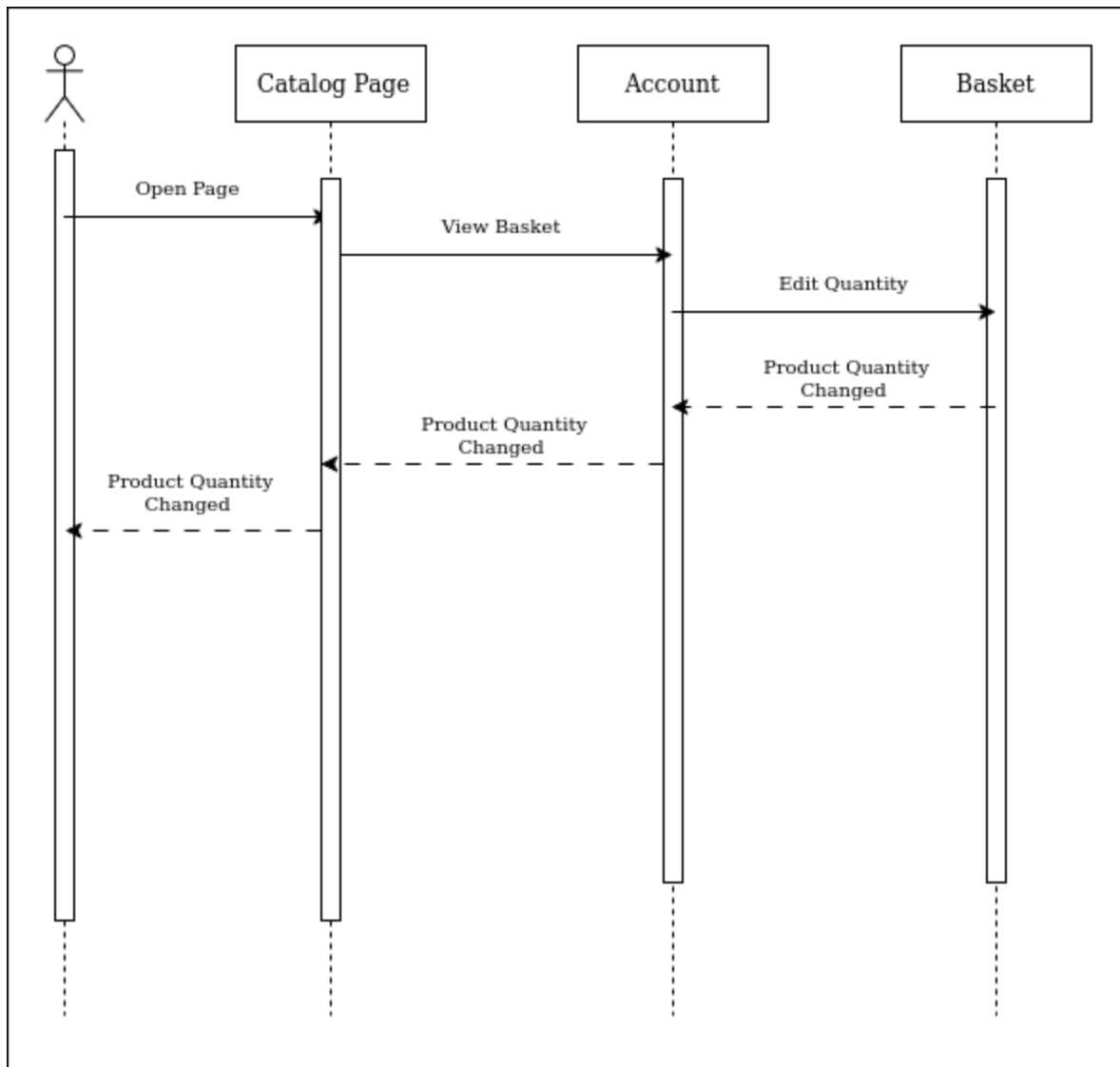
View Product:



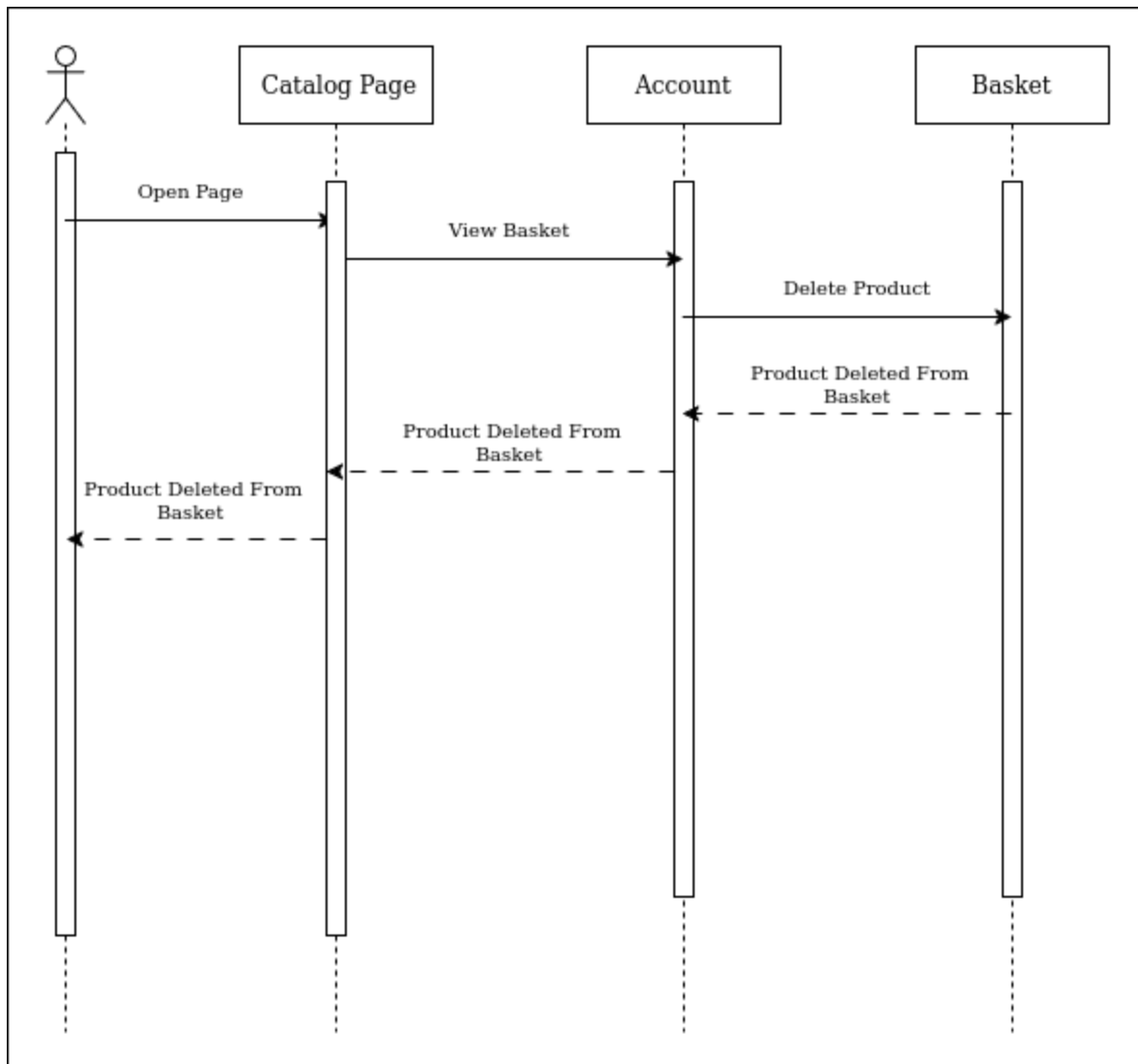
Add Product:



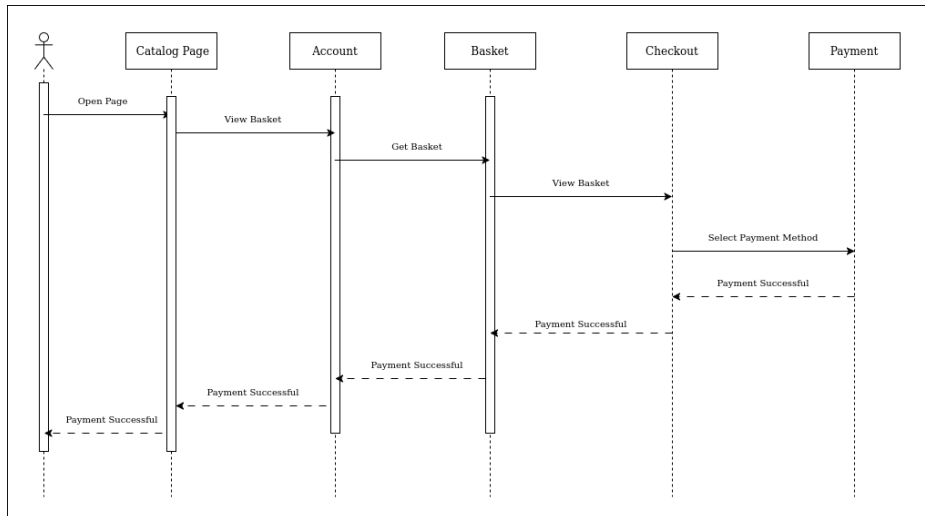
Edit Quantity:



Delete Product:



Checkout:



CRC Cards

CRC Cards will be included in documents under a PDF format.

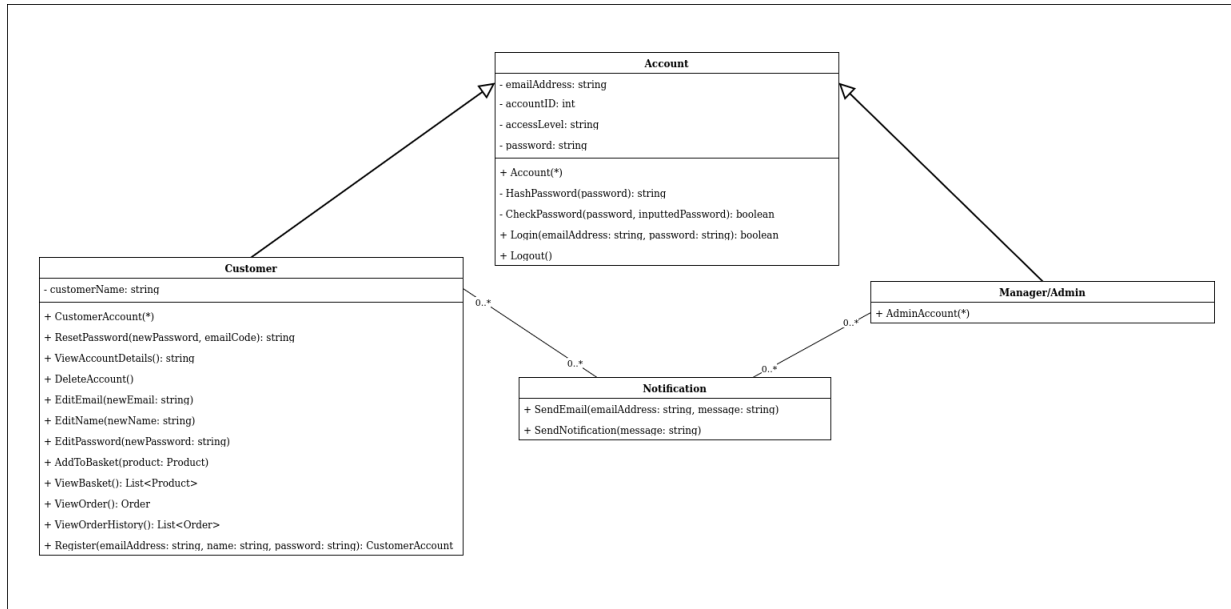
The CRC Cards depict a high-level view of the classes that may be added to the application. The cards capture necessary information about each class, including names, responsibilities, and collaborations with other classes.

Class Diagrams

These diagrams are a more detailed diagram using the necessary from the CRC Cards. These diagrams focus on a lower-level view of each class/model in the application, they cover the attributes, relationships and methods of the classes used within the application.

Account Class Diagram

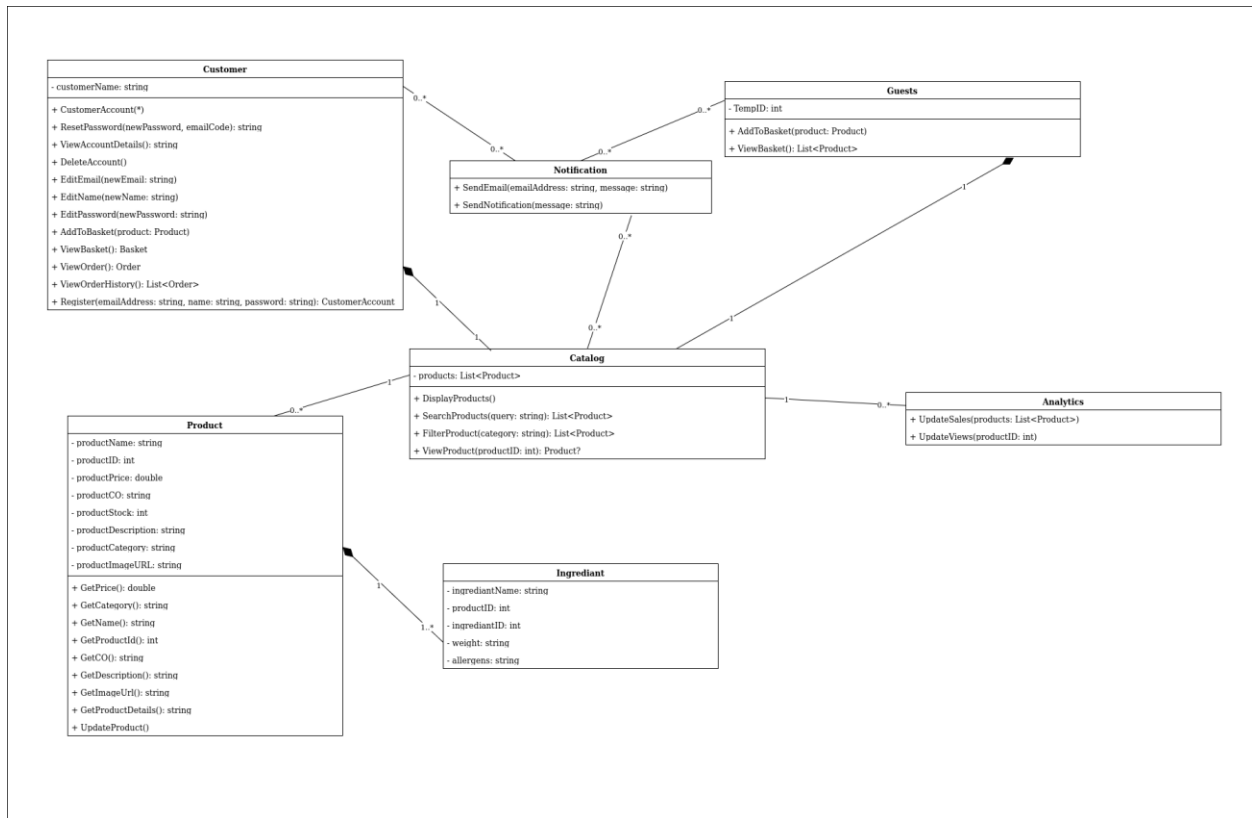
Added an accountId attribute to the account class to differentiate users. I have realized a customer will need both a first name and last name, this will be added in the implementation phase.



Catalog Class Diagram

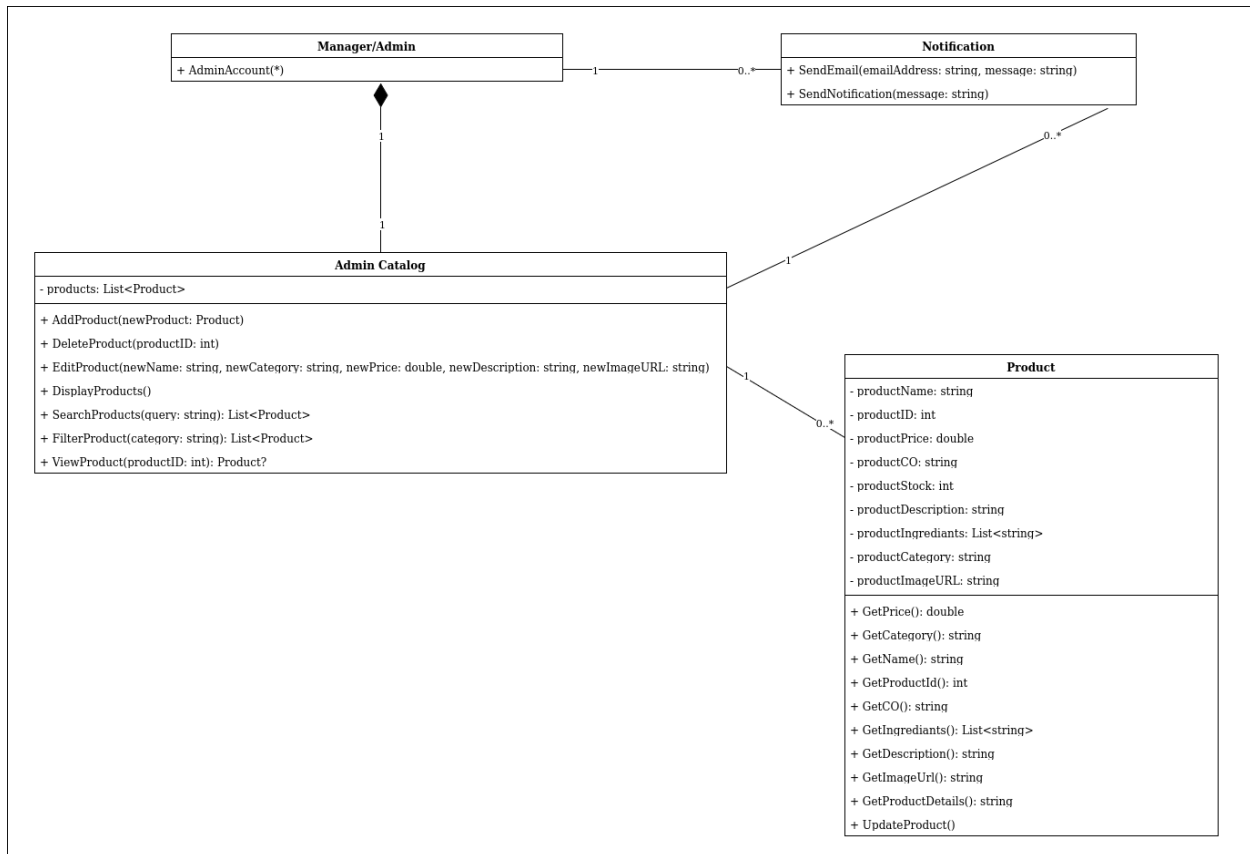
Removed inventory management and replaced with an update function to product as the inventory management class seemed to be unnecessary. Added a temporary id to guest.

Added an ingredients class as its necessary for following the normalisation rule 1NF.



Manager Class Diagram

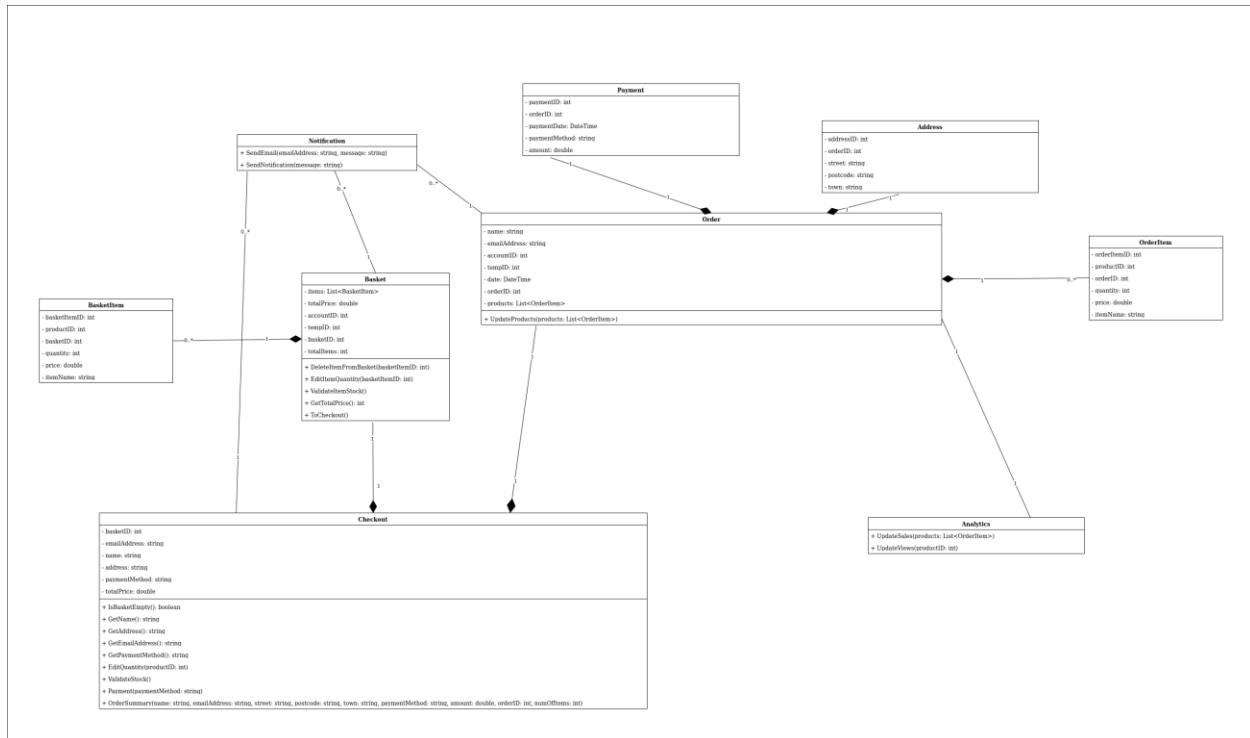
Ingredient class will be added to this in the implementation phase.



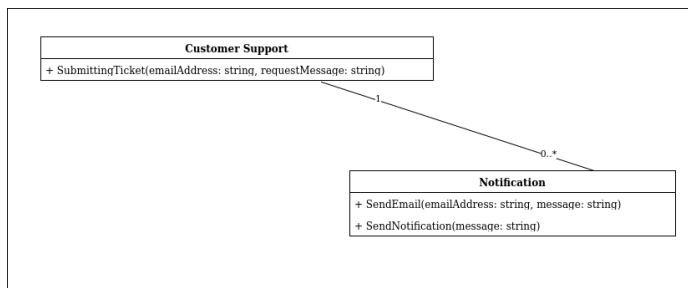
Checkout Class Diagram

Added BasketItem and OrderItem class to add more specific details to products such as quantity without changing the product class. Added date in order class to distinguish orders at different times, it is also needed in the order history wireframes. Added address

class as it needs to follow the 1NF rule. The payment class is added for clarity and separation of responsibility between order and payment.



Customer Support Class Diagram



Normalisation

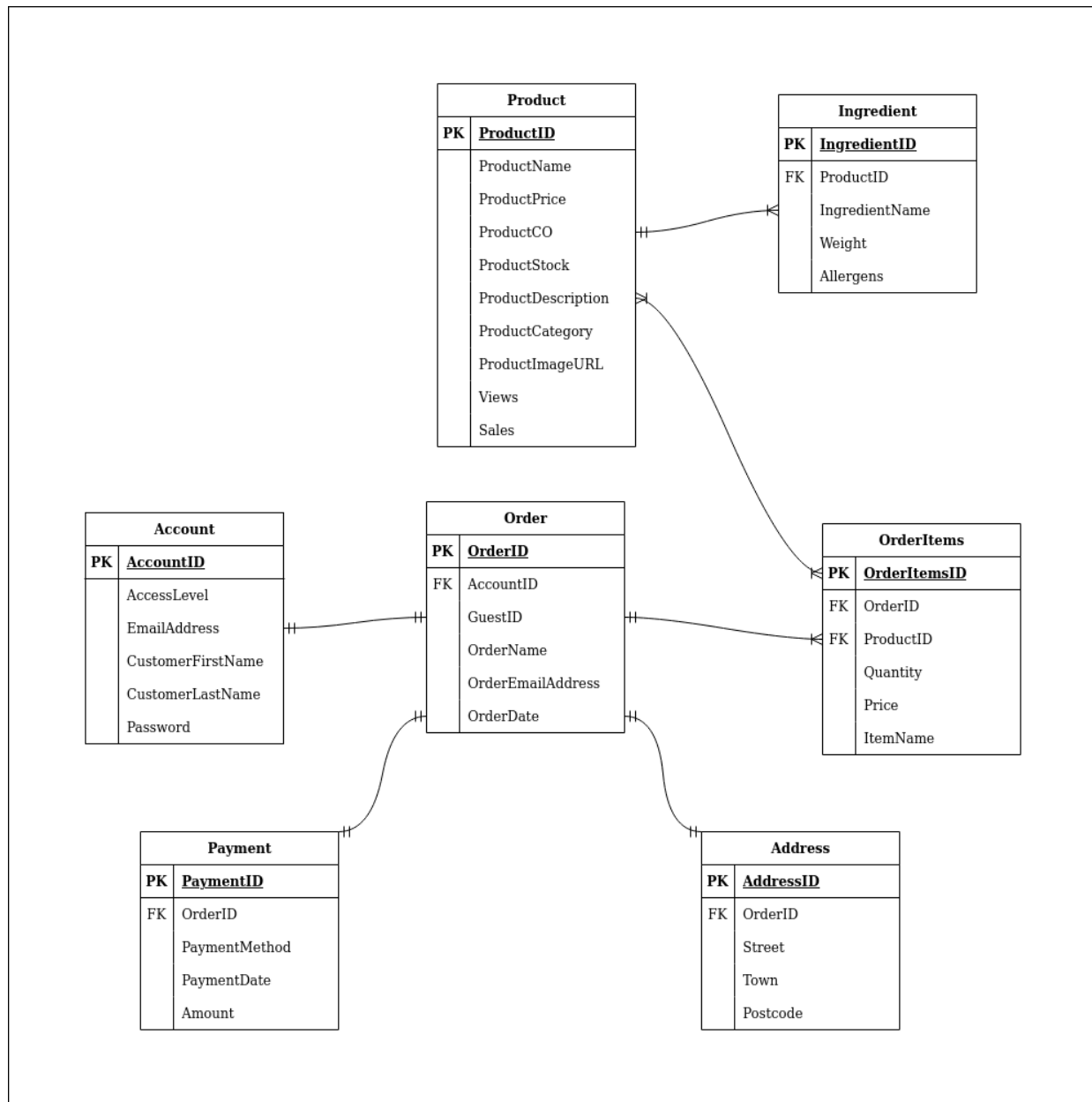
UNF	1NF	2NF	3NF
AccountID AccessLevel Password EmailAddress CustomerName	AccountID AccessLevel Password EmailAddress CustomerFirstName	AccountID AccessLevel Password EmailAddress CustomerFirstName	AccountID AccessLevel Password EmailAddress CustomerFirstName

ProductID	CustomerLastName	CustomerLastName	CustomerLastName
ProductName	ProductID		
ProductPrice	ProductName	<u>ProductID</u>	<u>ProductID</u>
ProductCO	ProductPrice	ProductName	ProductName
ProductStock	ProductCO	ProductPrice	ProductPrice
ProductDescription	ProductStock	ProductCO	ProductCO
ProductCategory	ProductDescription	ProductStock	ProductStock
ProductImageURL	ProductCategory	ProductDescription	ProductDescription
ProductIngredients	ProductImageURL	ProductCategory	ProductCategory
Views	Views	ProductImageURL	ProductImageURL
Sales	Sales	Views	Views
OrderID	IngredientID	Sales	Sales
OrderName	IngredientName		
OrderEmailAddress	Weight	<u>IngredientID</u>	<u>IngredientID</u>
Address	Allergens	ProductID*	ProductID*
GuestID	OrderID	IngredientName	IngredientName
PaymentMethod	OrderName	Weight	Weight
TotalPrice	OrderEmailAddress	Allergens	Allergens
OrderDate	Street		
OrderItemsID	Town	<u>OrderID</u>	<u>OrderID</u>
Quantity	Postcode	AccountID*	AccountID*
Price	PaymentMethod	GuestID	GuestID
ItemName	TotalPrice	OrderName	OrderName
	OrderDate	OrderEmailAddress	OrderEmailAddress
	OrderItemsID	PaymentMethod	OrderDate
	Quantity	TotalPrice	
	Price	OrderDate	<u>PaymentID</u>
	ItemName		OrderID*
		<u>AddressID</u>	PaymentMethod
		OrderID*	PaymentDate
		Street	Amount
		Town	
		Postcode	<u>AddressID</u>
			OrderID*
		<u>OrderItemsID</u>	Street
		OrderID*	Town
		ProductID*	Postcode
		Quantity	
		Price	<u>OrderItemsID</u>
		ItemName	OrderID*
			ProductID*
			Quantity
			Price
			ItemName

Entity Name
Account
Product
Ingredient
Order
Address
Payment
OrderItems

Entity Relation Diagram

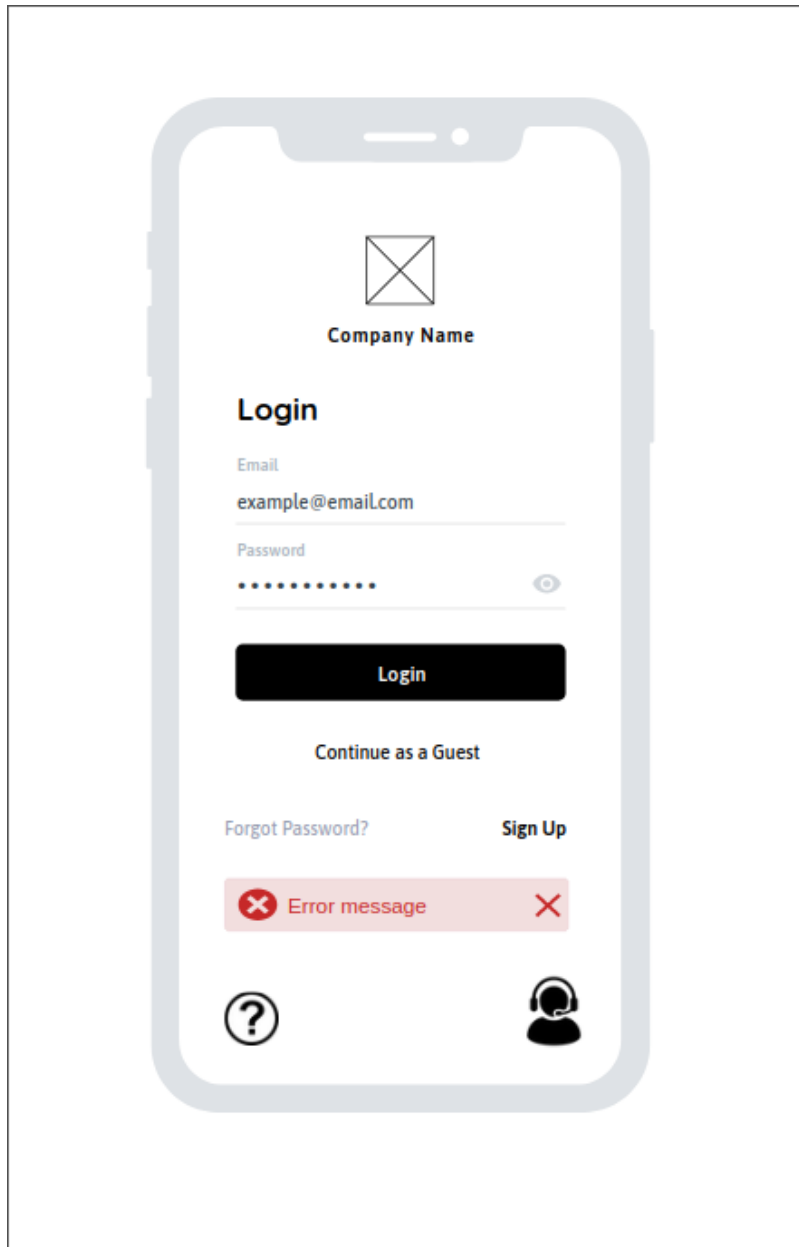
This diagram is a detailed blueprint that illustrates how the applications data is structured and related to the database. It focuses on entities (tables), attributes (columns), and relationships between them.

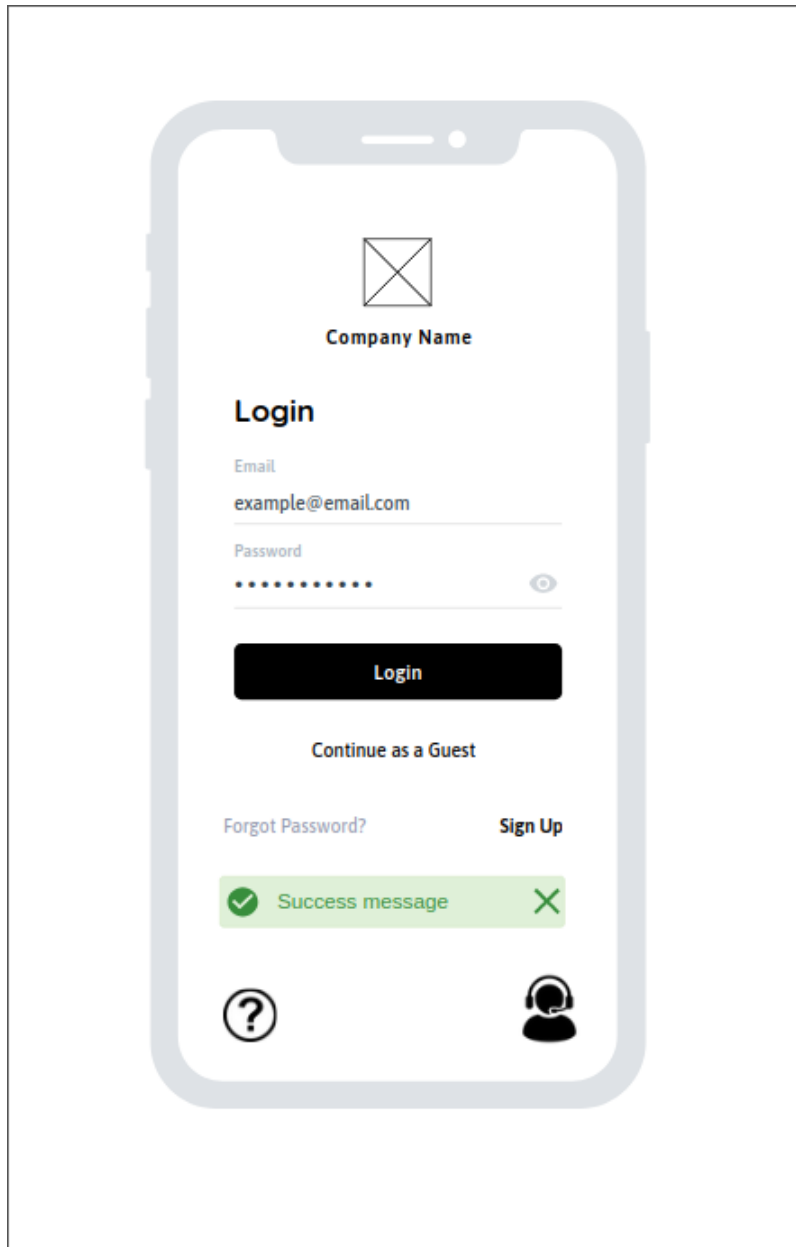


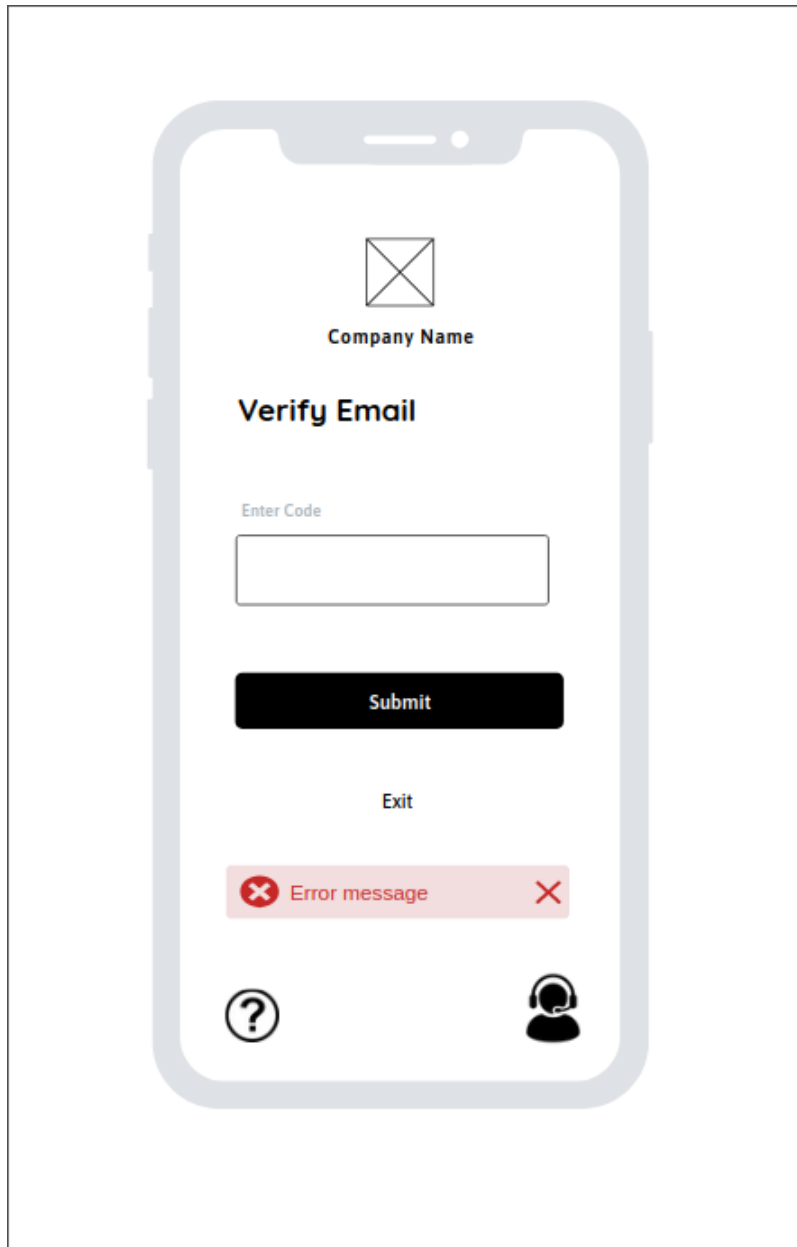
View Model

Wireframes

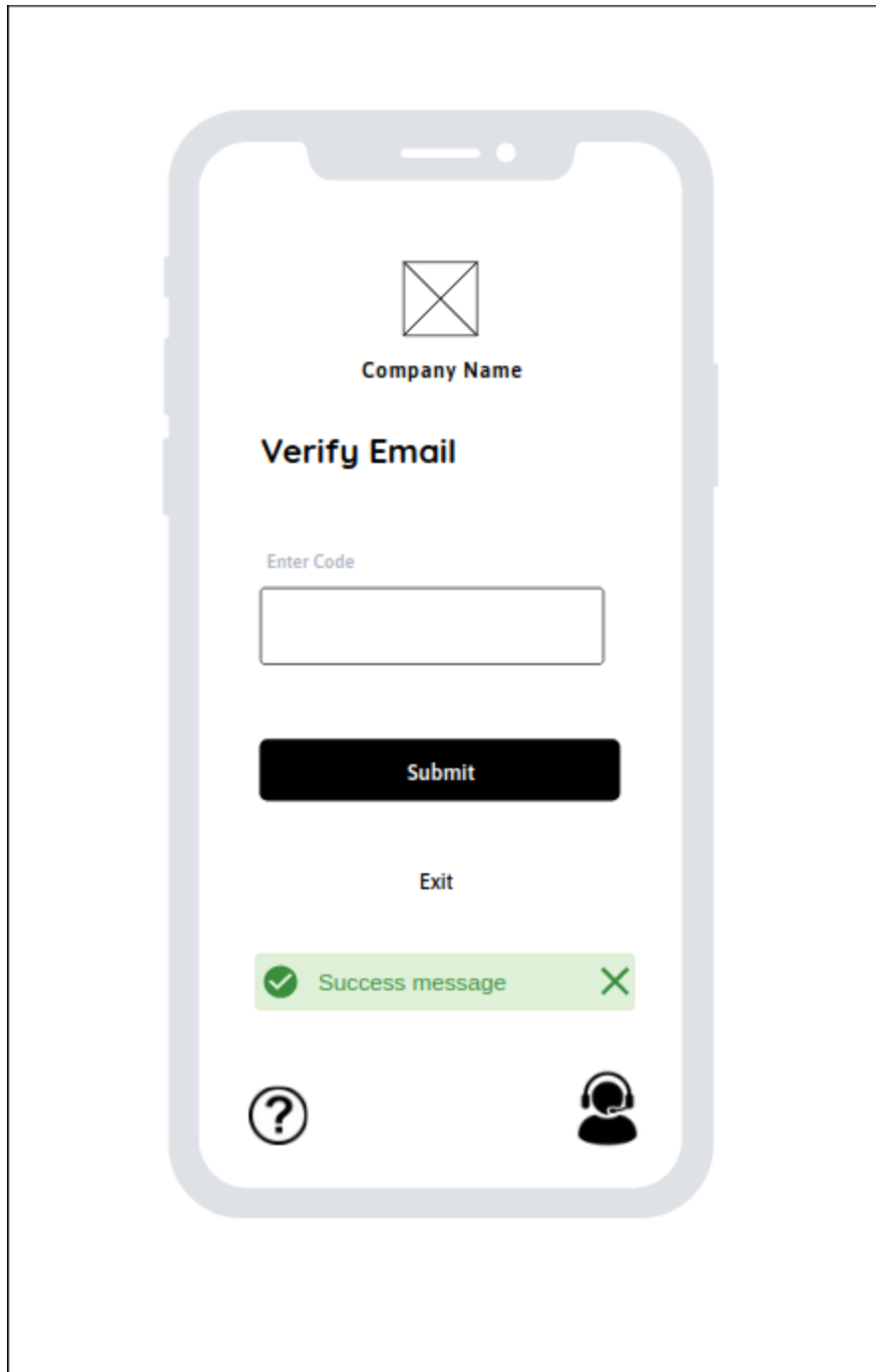
Login Page





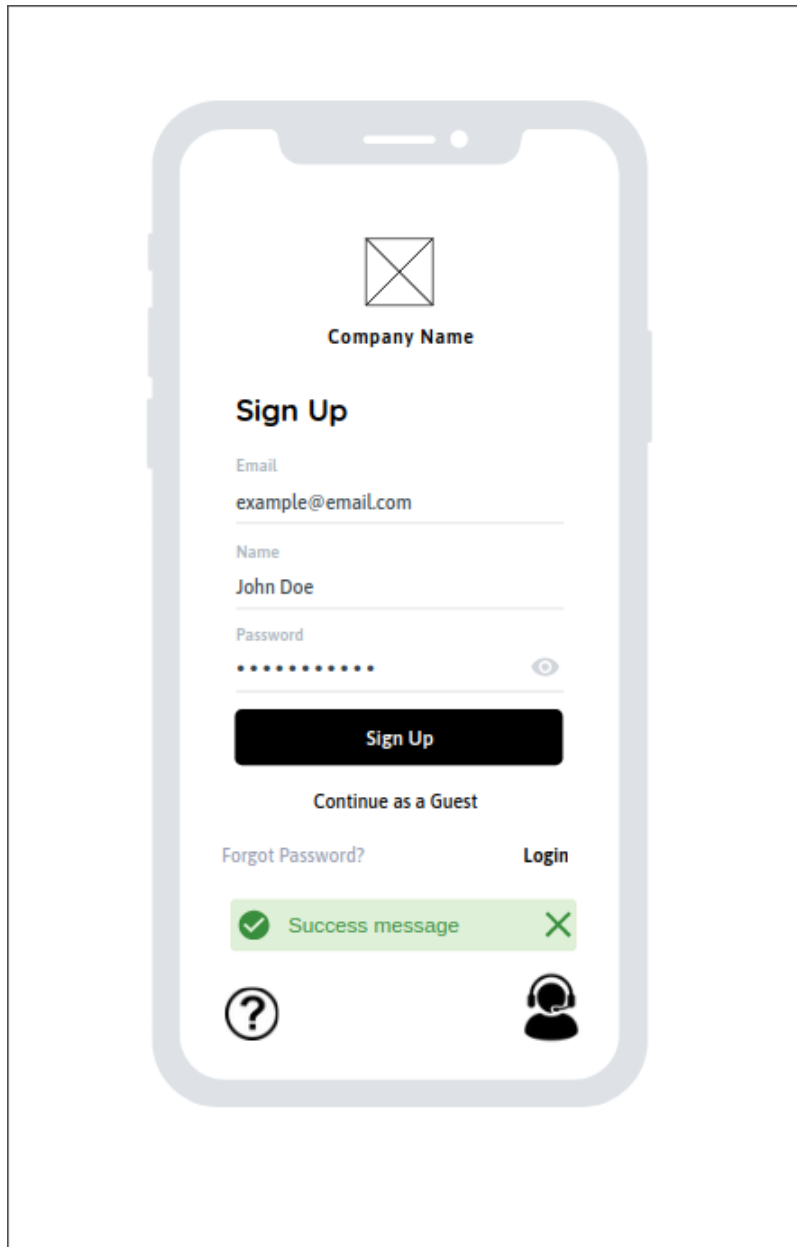


The image shows a mobile app interface for a 'Verify Email' screen. At the top, there is a square icon with an 'X' inside, followed by the text 'Company Name'. Below this is the title 'Verify Email'. A label 'Enter Code' is positioned above a text input field. A black 'Submit' button is located below the input field. Underneath the button is the text 'Exit'. A red error message banner with a red 'X' icon on the left and a red 'X' icon on the right contains the text 'Error message'. At the bottom of the screen, there are two circular icons: a question mark on the left and a headset icon on the right.

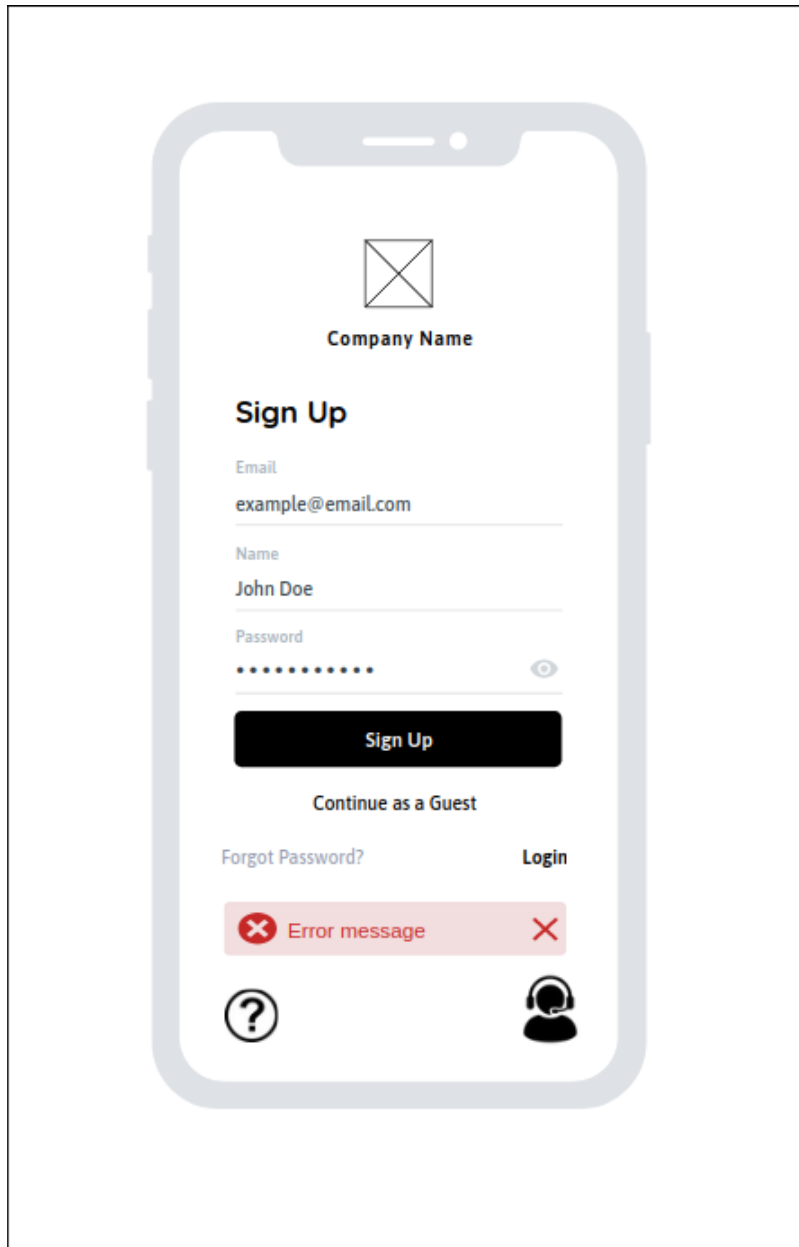


Register Page

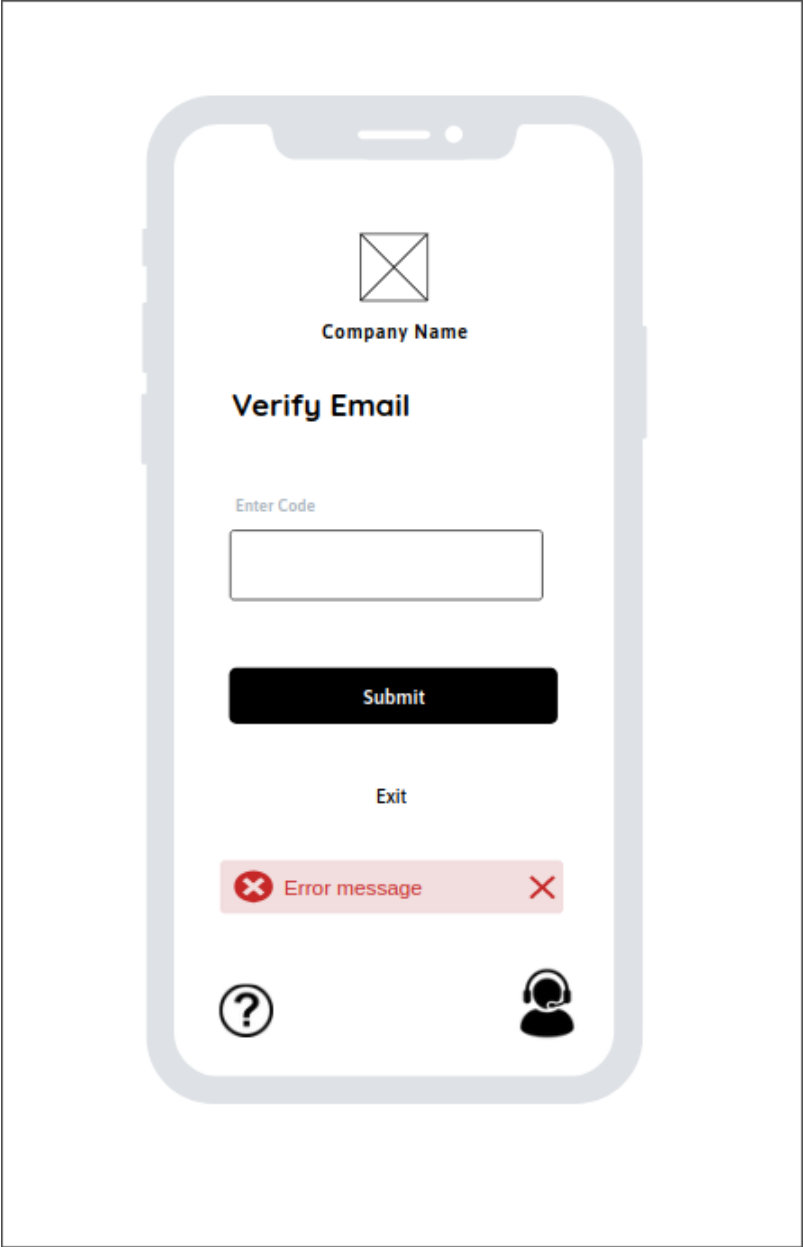
Name will be split into first name and last name.

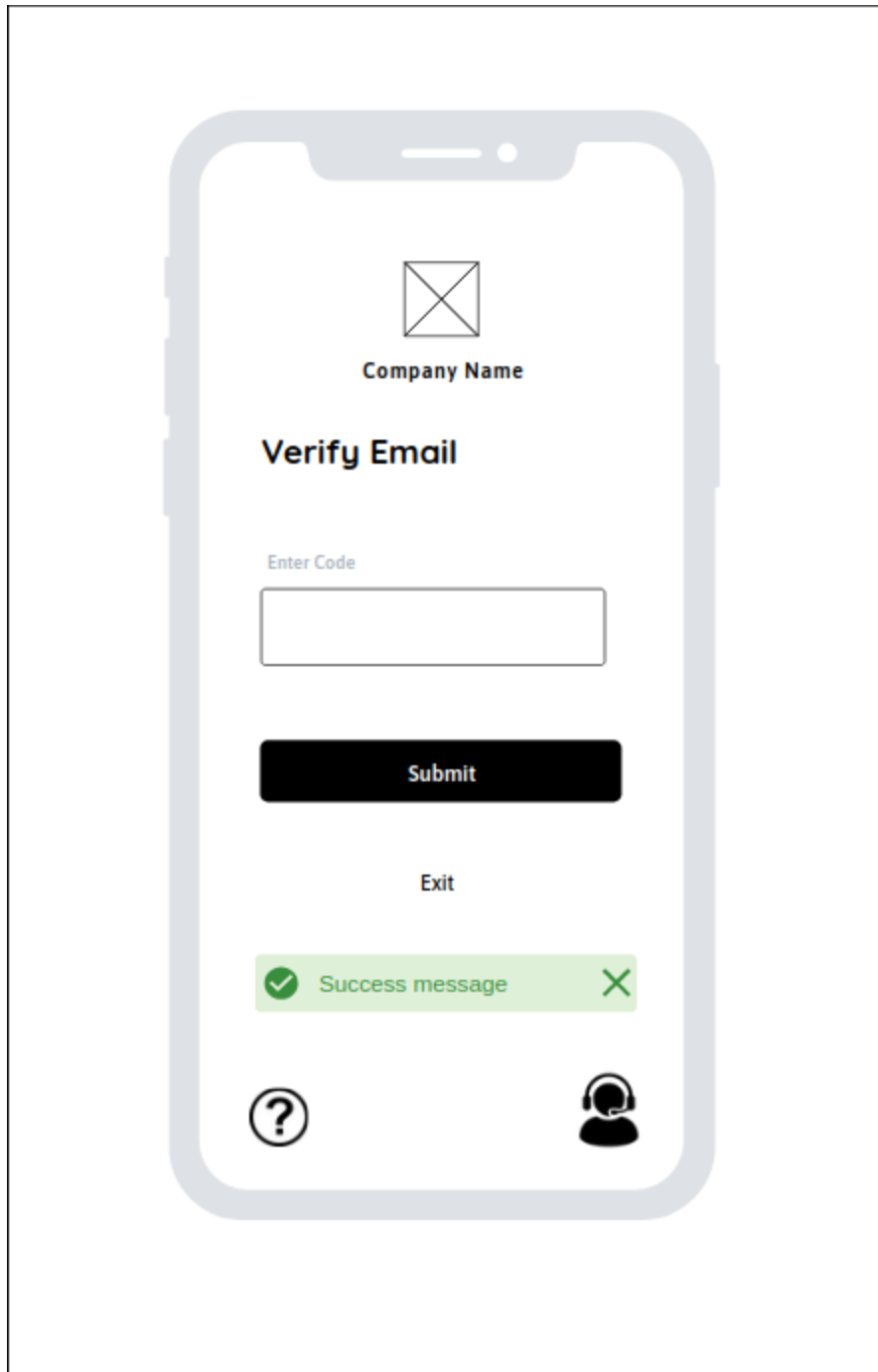


A mobile app sign-up screen design displayed within a smartphone frame. The screen features a placeholder icon (a square with an 'X') and the text "Company Name" below it. The main heading is "Sign Up". Below this are three input fields: "Email" with the value "example@email.com", "Name" with the value "John Doe", and "Password" with masked characters "....." and a toggle eye icon. A black "Sign Up" button is positioned below the password field. Below the button is the text "Continue as a Guest". At the bottom of the form area are two links: "Forgot Password?" and "Login". A green success message banner with a checkmark icon, the text "Success message", and a close 'X' icon is shown below the links. At the very bottom of the screen are two circular icons: a question mark on the left and a headset on the right.

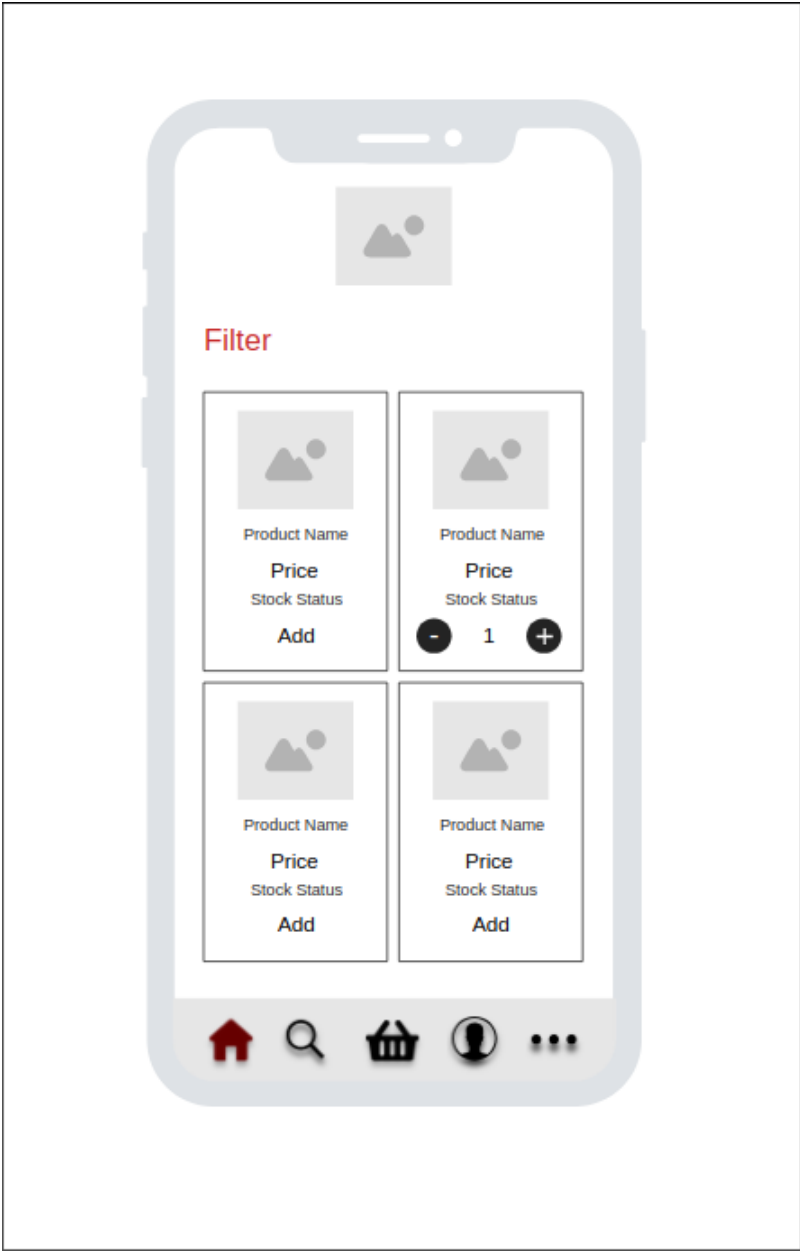


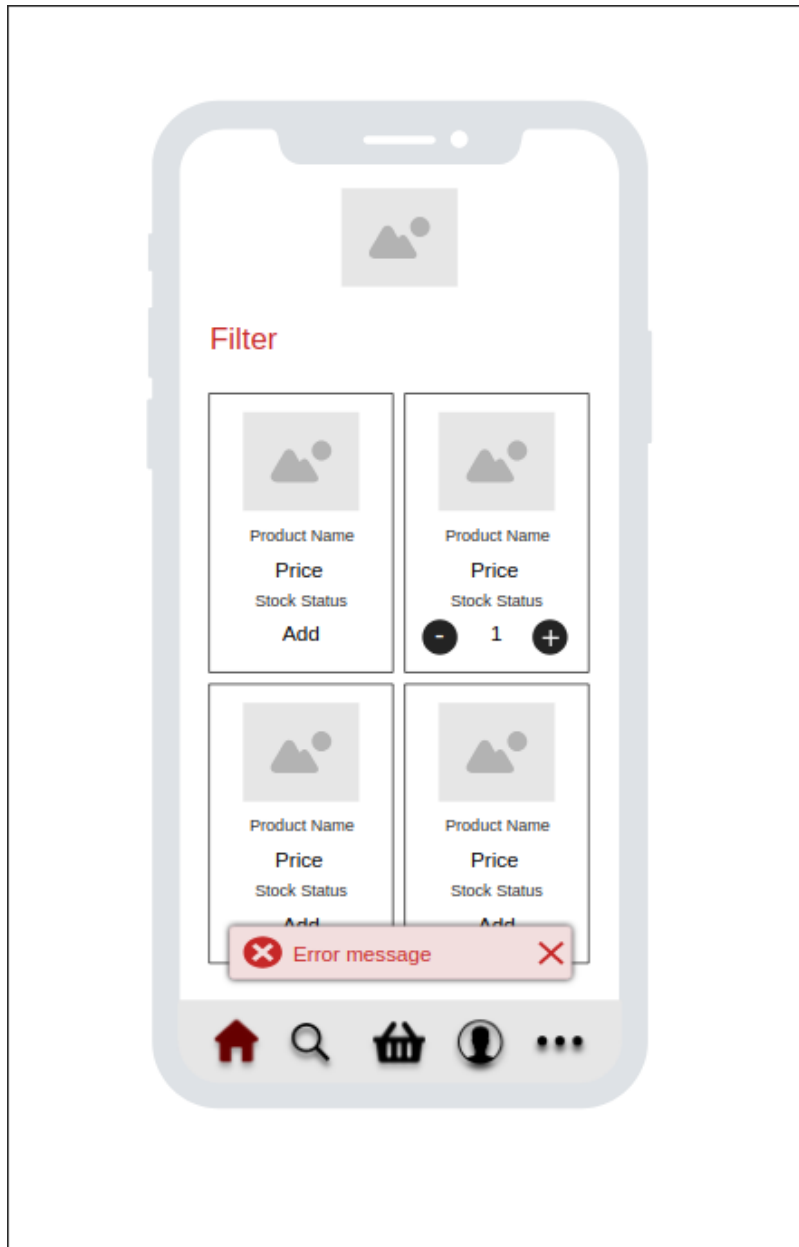
A mobile app sign-up screen design. At the top, there is a placeholder icon for a company logo (a square with an 'X') and the text "Company Name". Below this is the "Sign Up" title. The form includes three input fields: "Email" with the value "example@email.com", "Name" with the value "John Doe", and "Password" with masked characters ".....". A black "Sign Up" button is positioned below the password field. Below the button is the text "Continue as a Guest". At the bottom of the form area, there are two links: "Forgot Password?" and "Login". Below these links is a red error message box containing a red 'X' icon, the text "Error message", and another red 'X' icon. At the very bottom of the screen, there are two icons: a question mark in a circle on the left and a headset icon on the right.

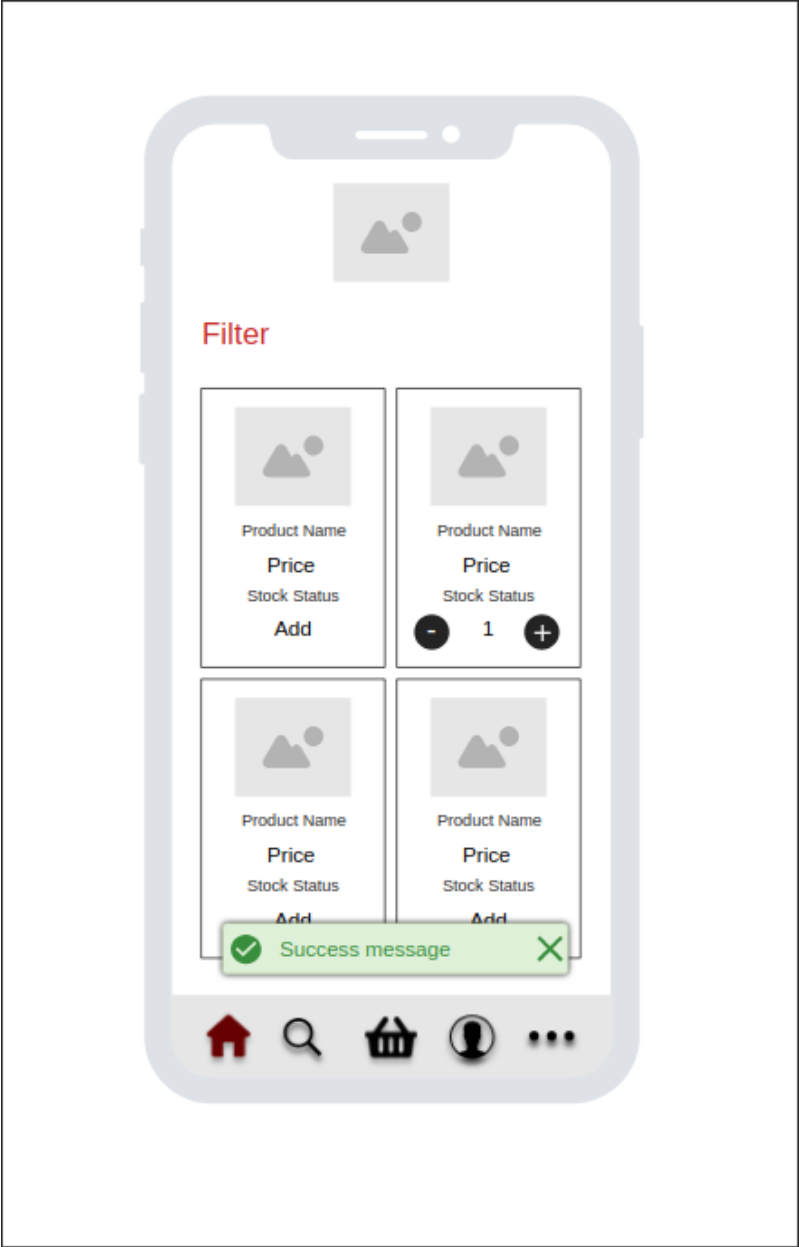


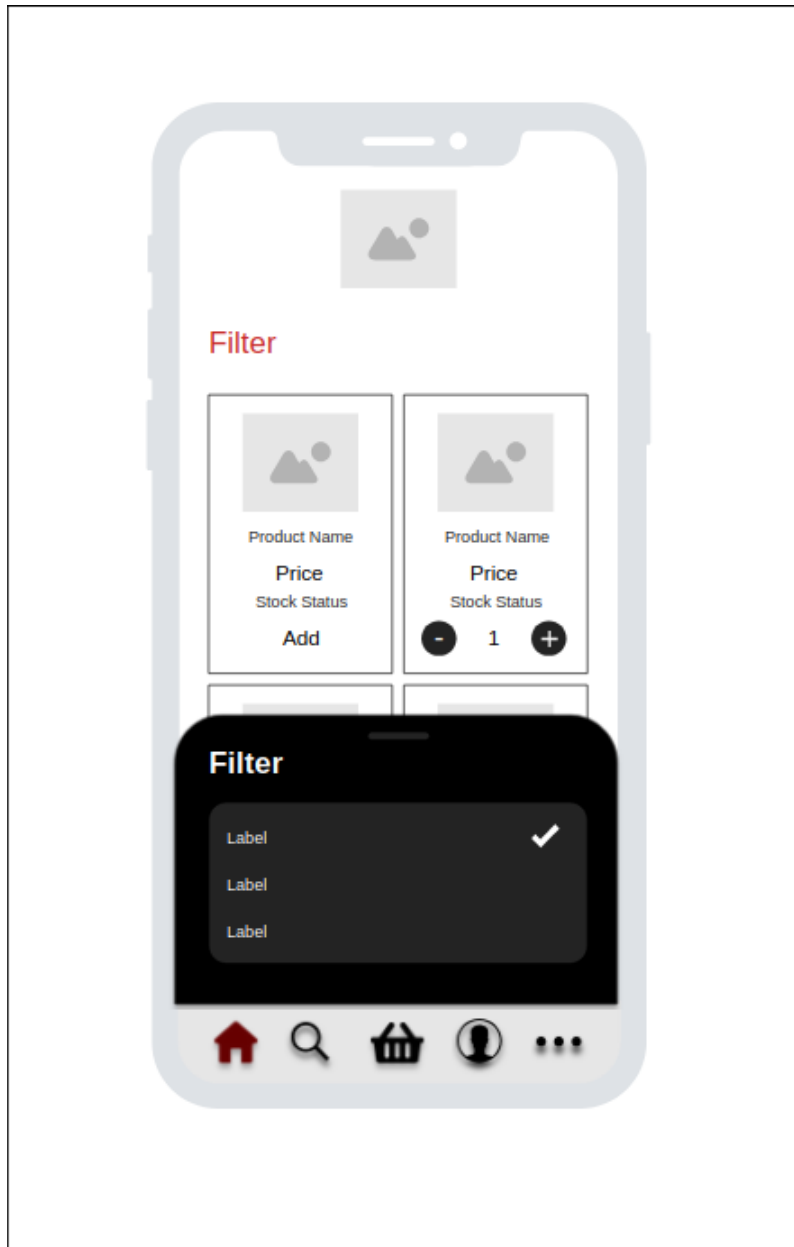


Catalog Page

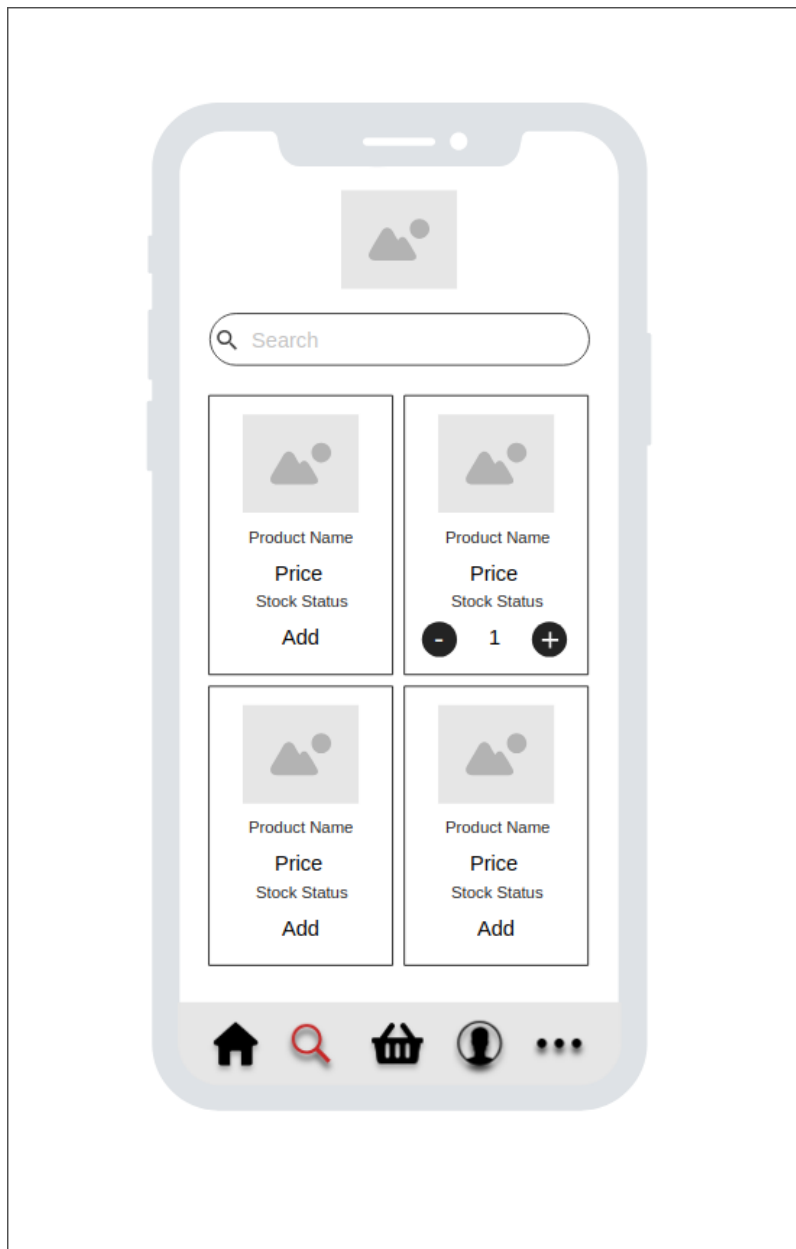


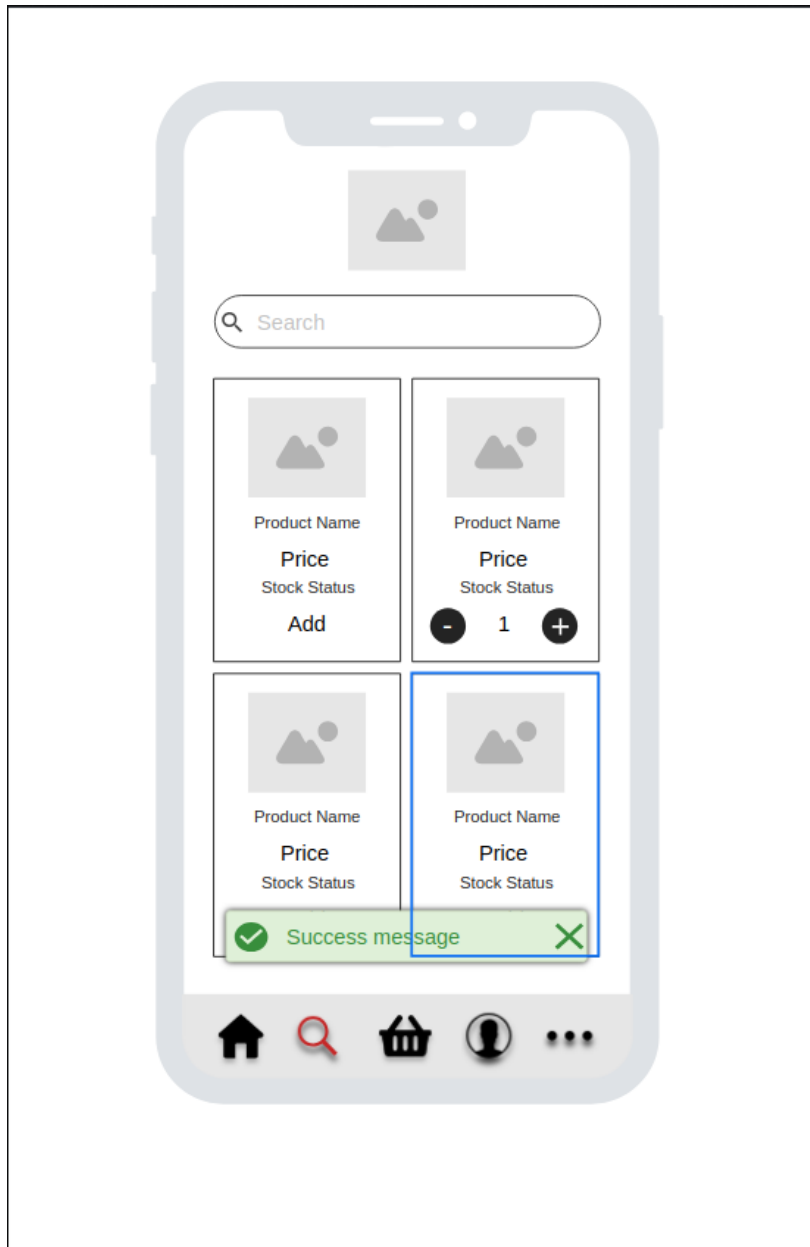


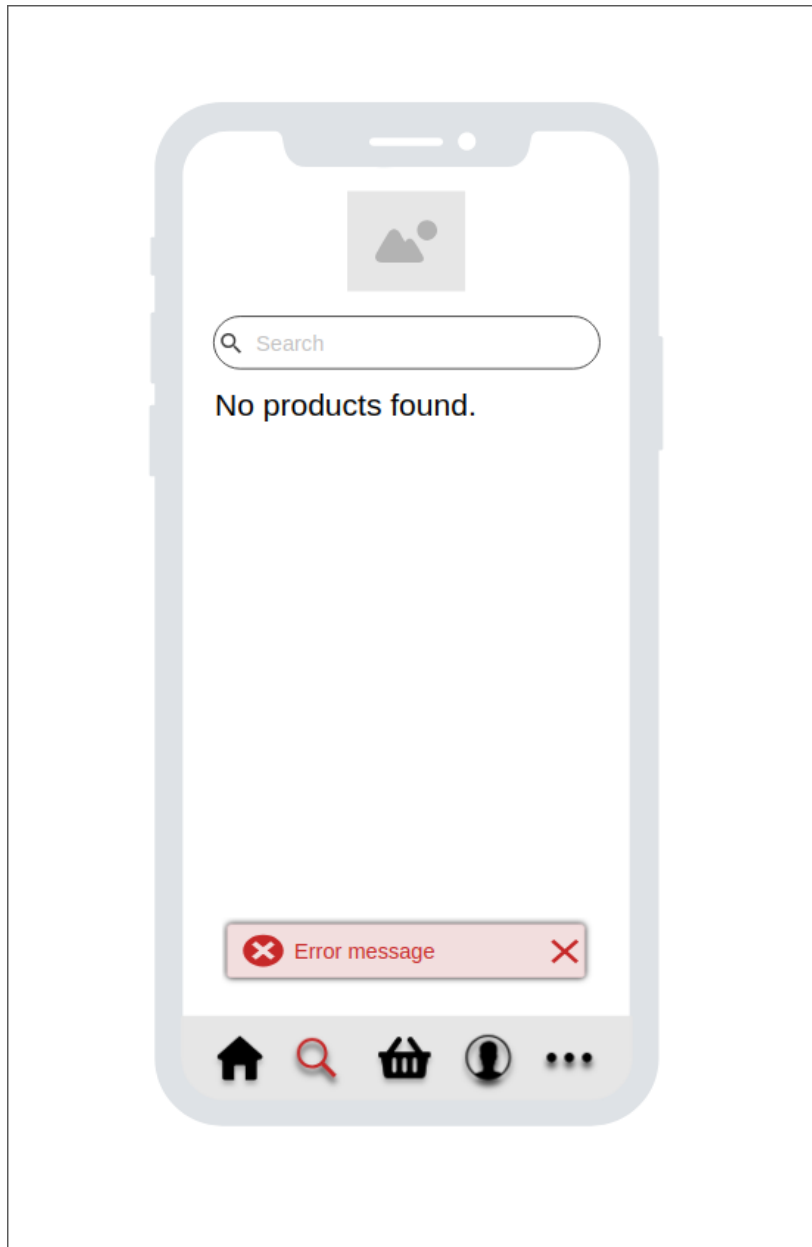




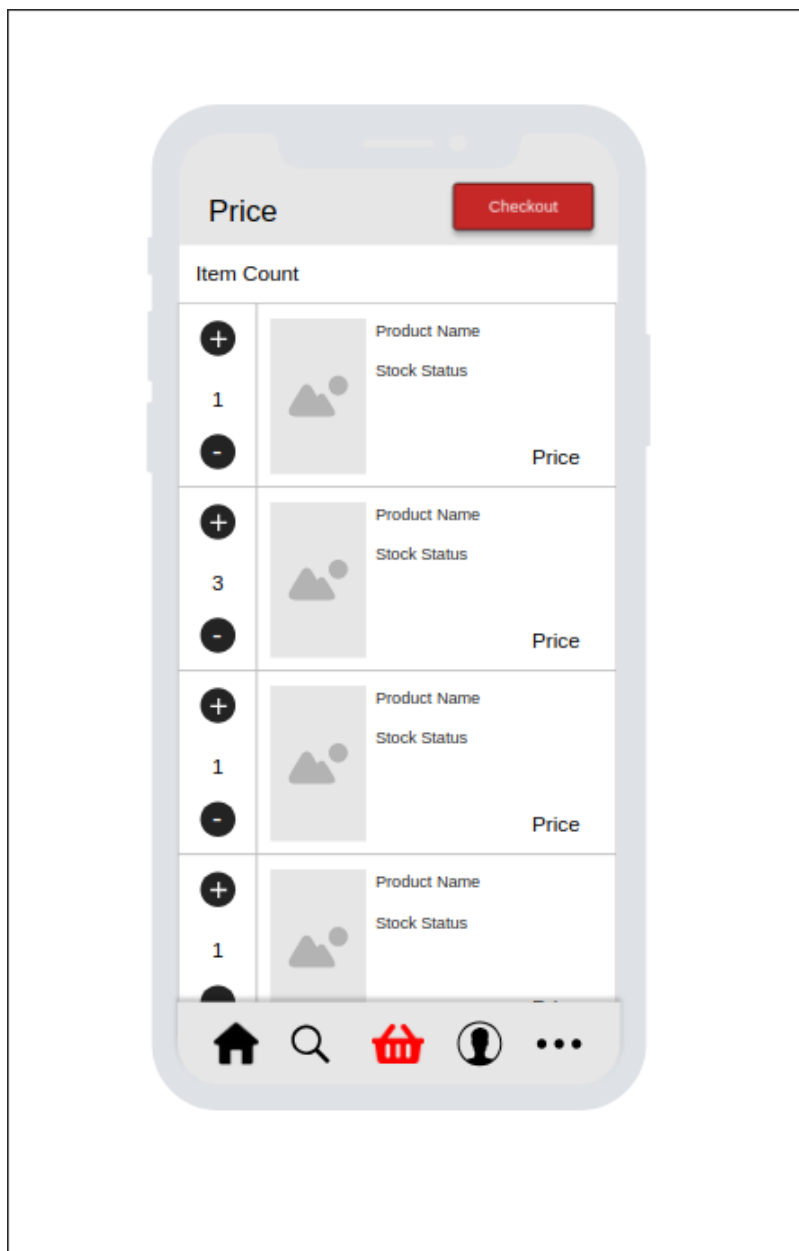
Search Page

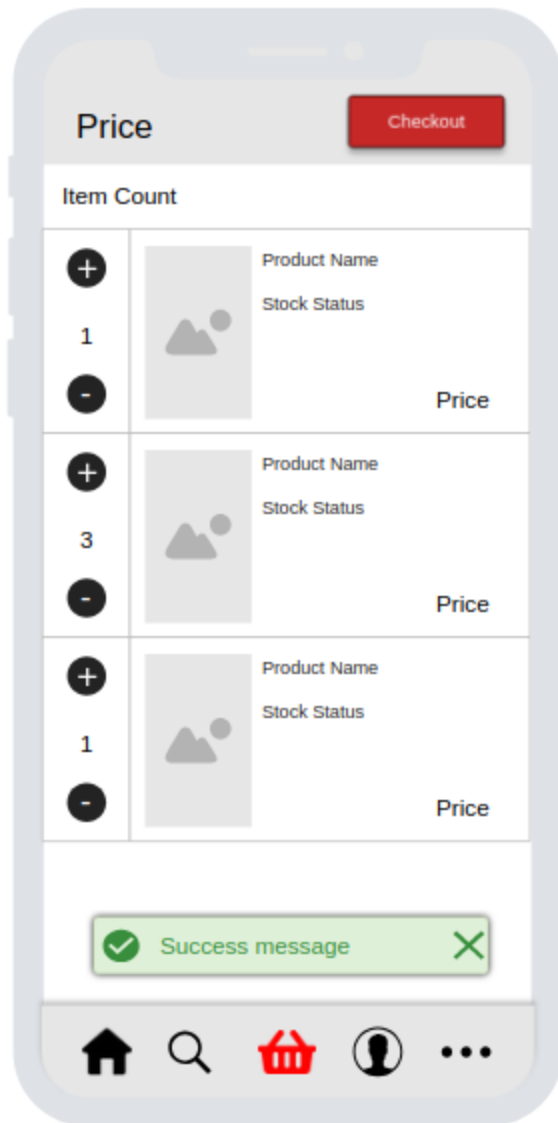


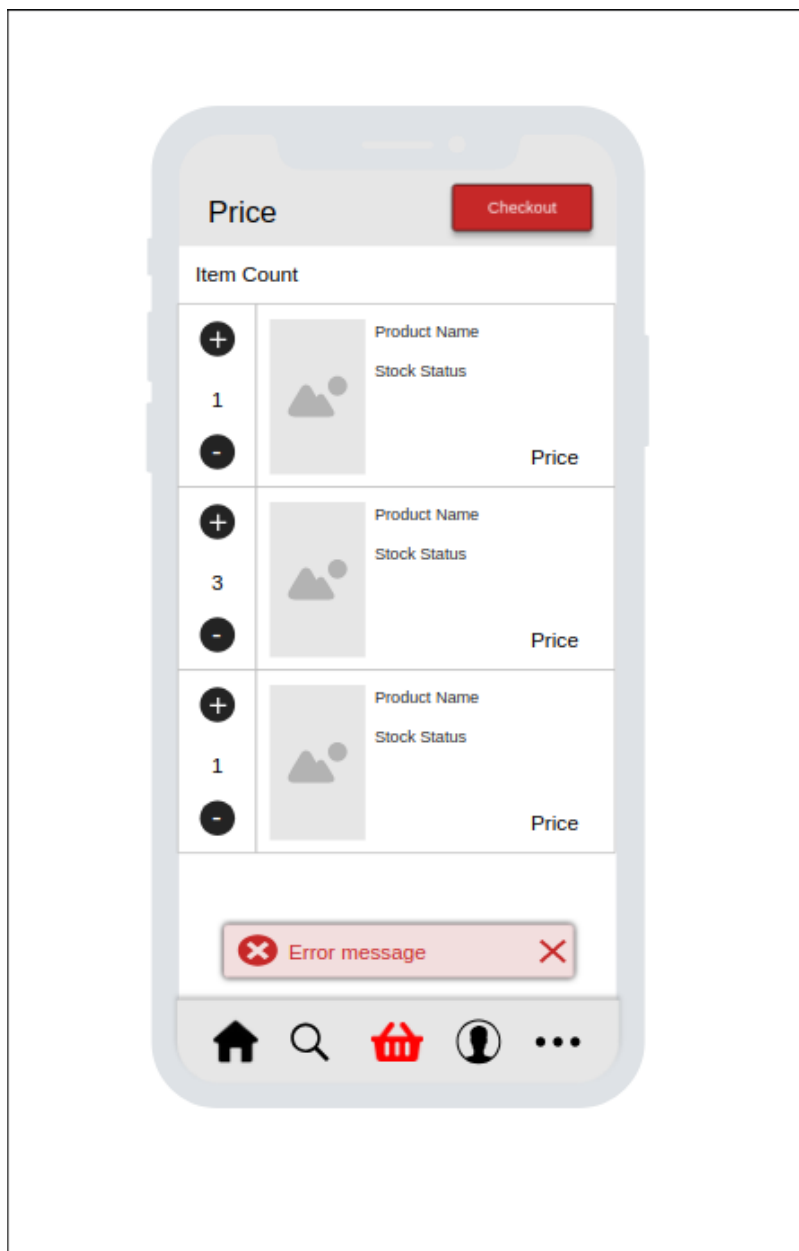




Basket Page



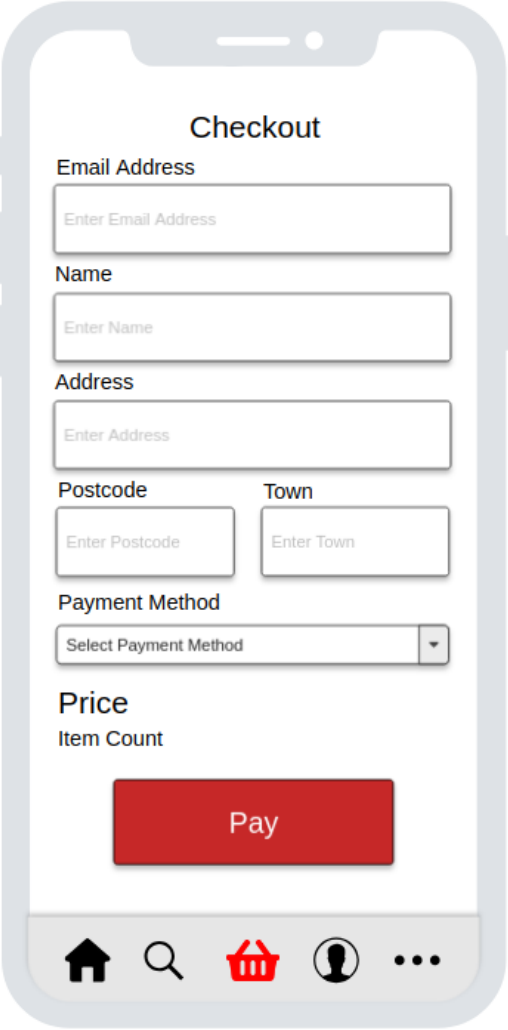




Checkout Page

Replace address with street.

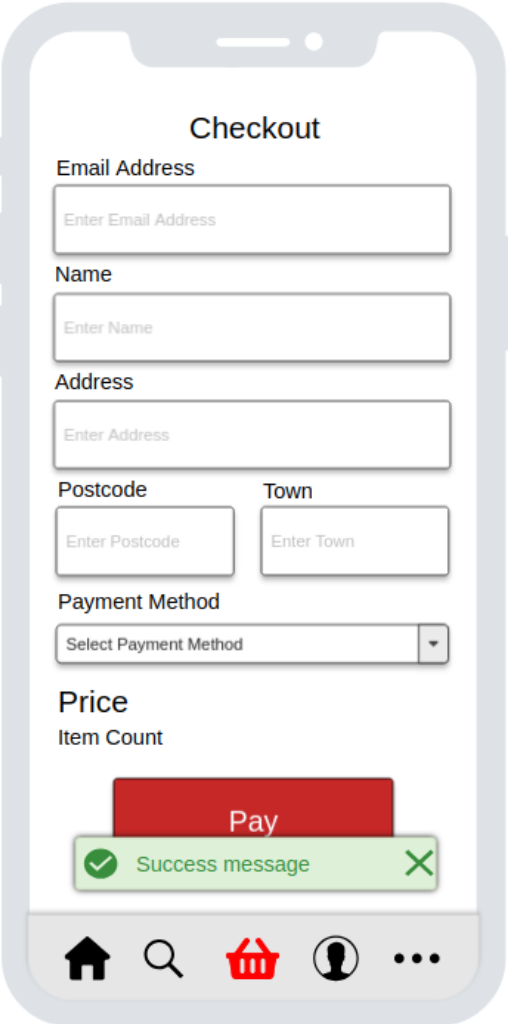
Name will be split into first name and last name.



The image shows a mobile application interface for a checkout process. The screen is titled "Checkout" and contains several input fields and a payment button. The layout is as follows:

- Email Address:** A text input field with the placeholder "Enter Email Address".
- Name:** A text input field with the placeholder "Enter Name".
- Address:** A text input field with the placeholder "Enter Address".
- Postcode:** A text input field with the placeholder "Enter Postcode".
- Town:** A text input field with the placeholder "Enter Town".
- Payment Method:** A dropdown menu with the placeholder "Select Payment Method".
- Price:** A section header.
- Item Count:** A label below the Price header.
- Pay Button:** A large red button with the text "Pay".

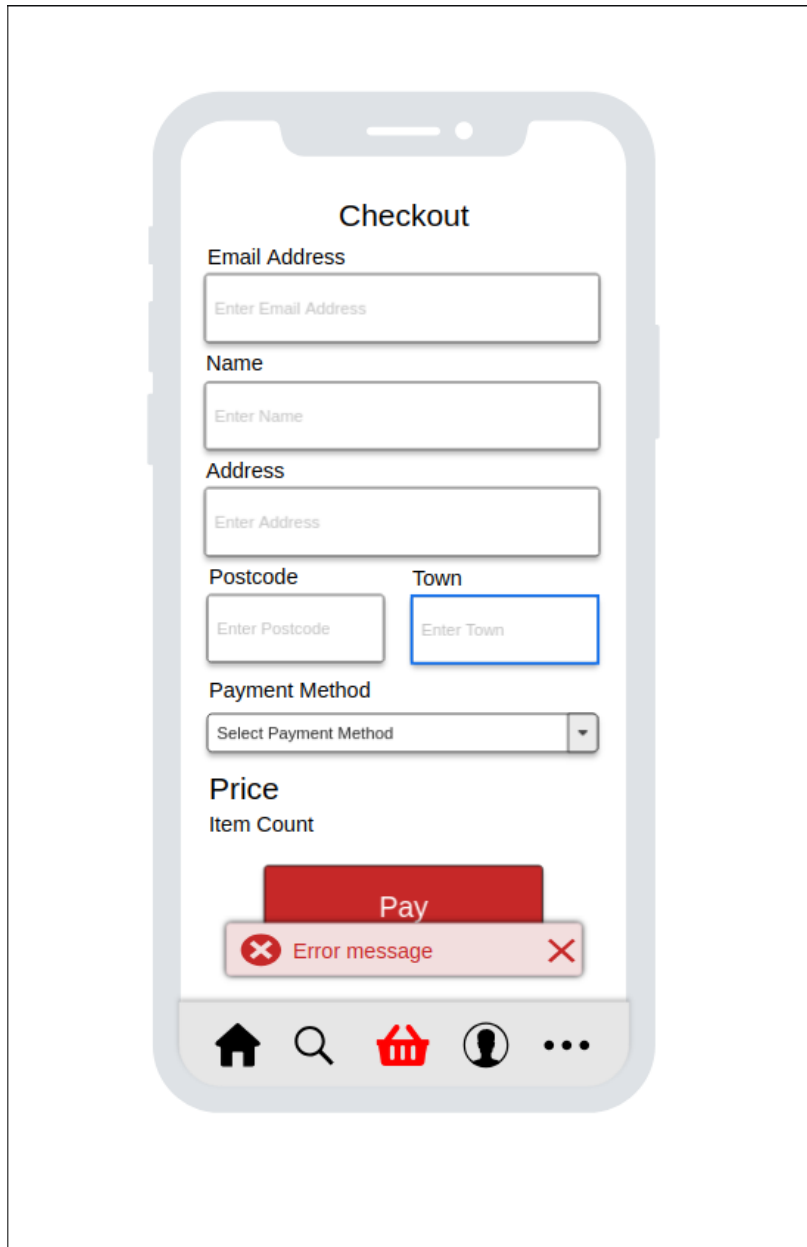
At the bottom of the screen is a navigation bar with five icons: a home icon, a magnifying glass (search), a shopping cart, a user profile, and a menu (three dots).



The image shows a mobile application interface for a checkout process. The screen is titled "Checkout" and contains several input fields and a payment button. The layout is as follows:

- Checkout** (Section Header)
- Email Address** (Label)
- Name** (Label)
- Address** (Label)
- Postcode** (Label)
- Town** (Label)
- Payment Method** (Label)
- Price** (Section Header)
- Item Count** (Label)
- Pay** (Red button)
- Success message** (Green message box with a checkmark and an 'X' icon)

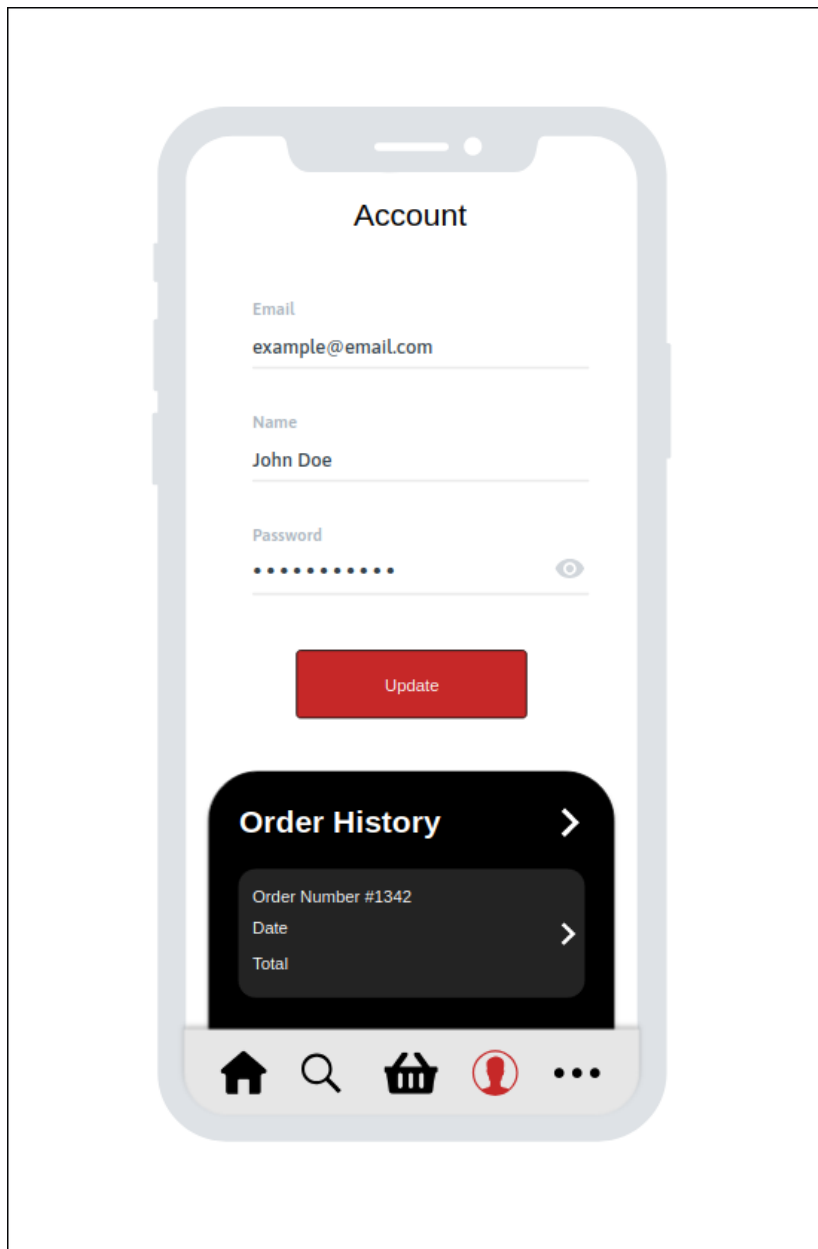
The bottom of the screen features a navigation bar with five icons: a home icon, a magnifying glass (search), a shopping cart, a user profile, and a menu (three dots).

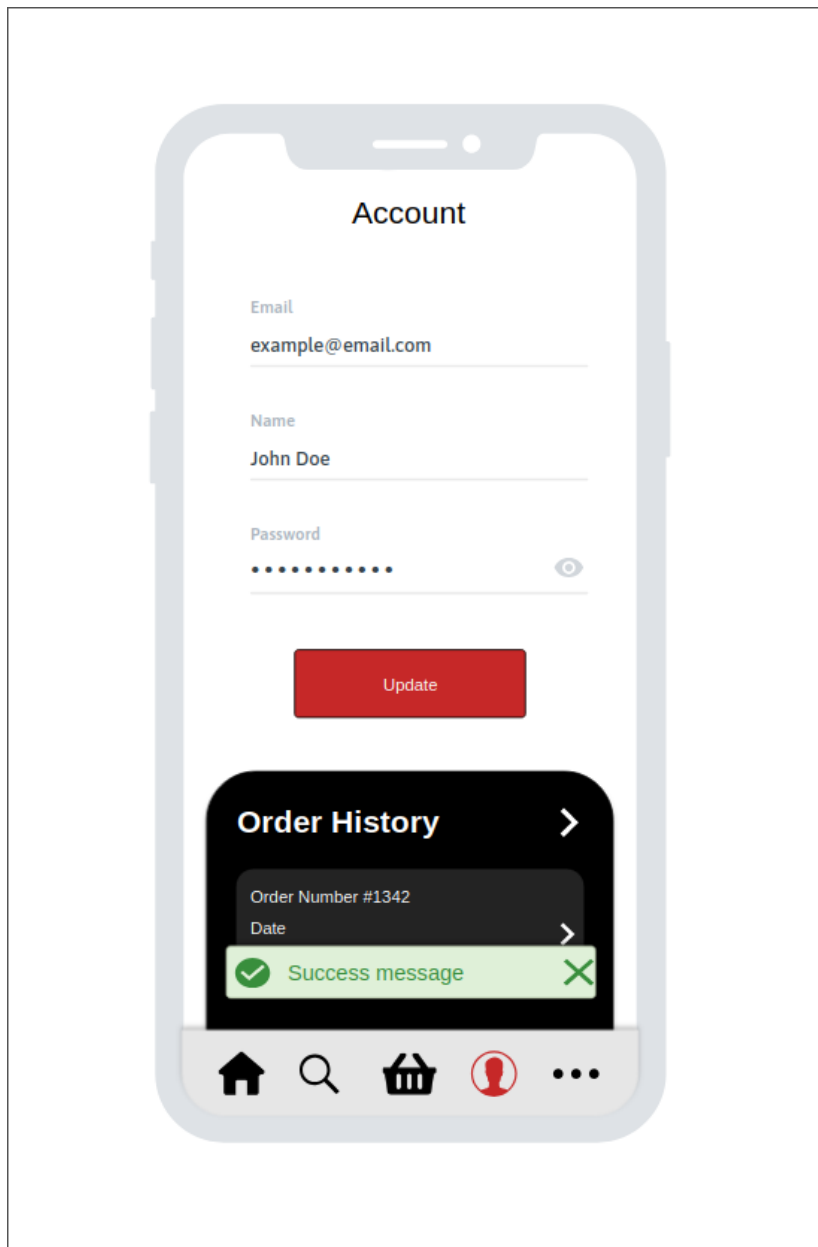


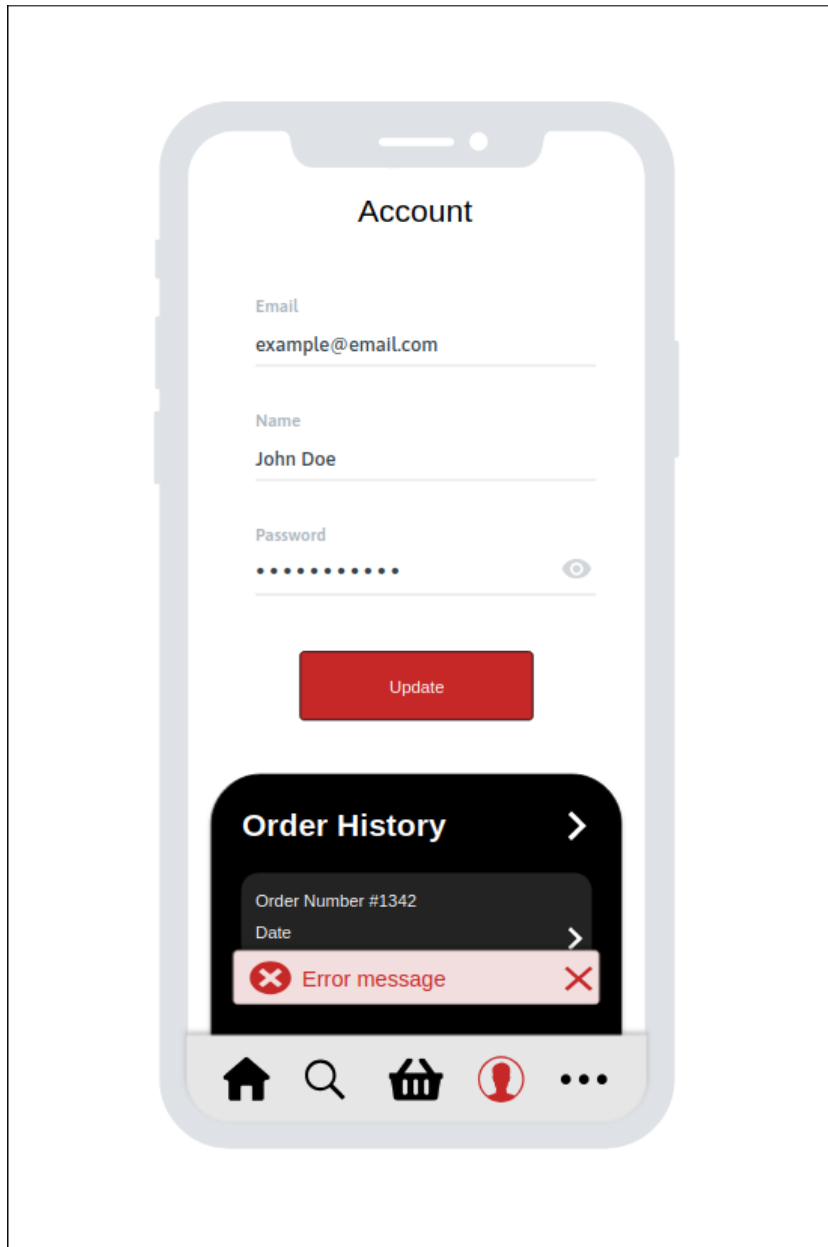
The image shows a mobile app checkout screen. At the top, the title "Checkout" is centered. Below it are several form fields: "Email Address" with a placeholder "Enter Email Address", "Name" with a placeholder "Enter Name", "Address" with a placeholder "Enter Address", "Postcode" with a placeholder "Enter Postcode", and "Town" with a placeholder "Enter Town". The "Town" field is highlighted with a blue border. Below these is a "Payment Method" dropdown menu with the text "Select Payment Method". Underneath is a "Price" section with the label "Item Count". At the bottom of the form is a red "Pay" button. Below the button is a red error message box with the text "Error message" and two red 'X' icons. At the very bottom is a navigation bar with five icons: a home icon, a magnifying glass, a shopping cart, a user profile, and a menu icon.

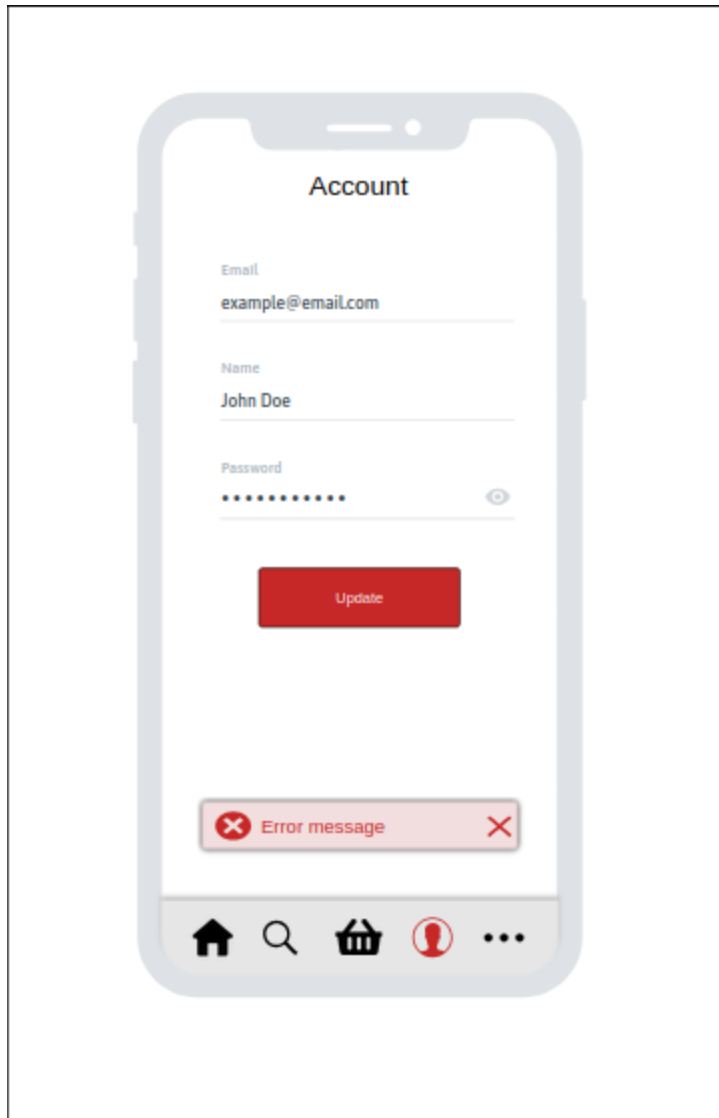
Account Page

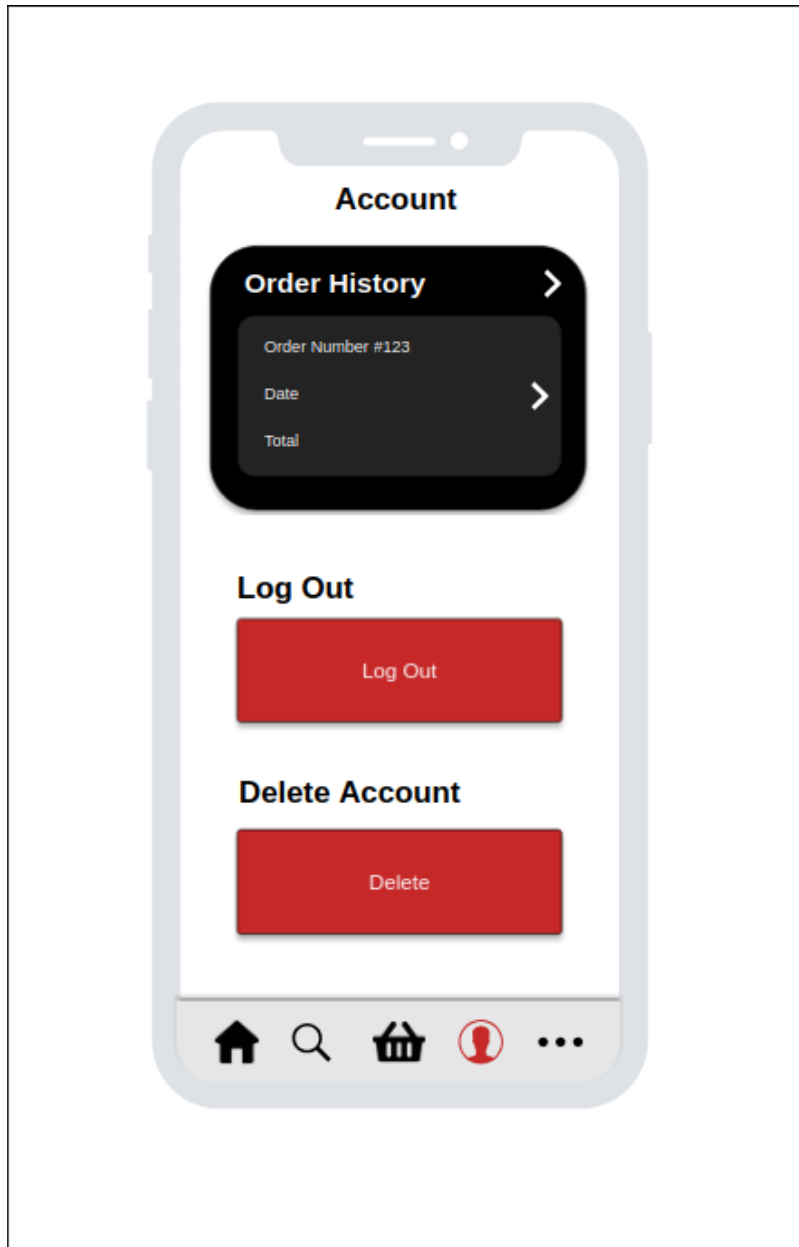
Name will be split into first name and last name.

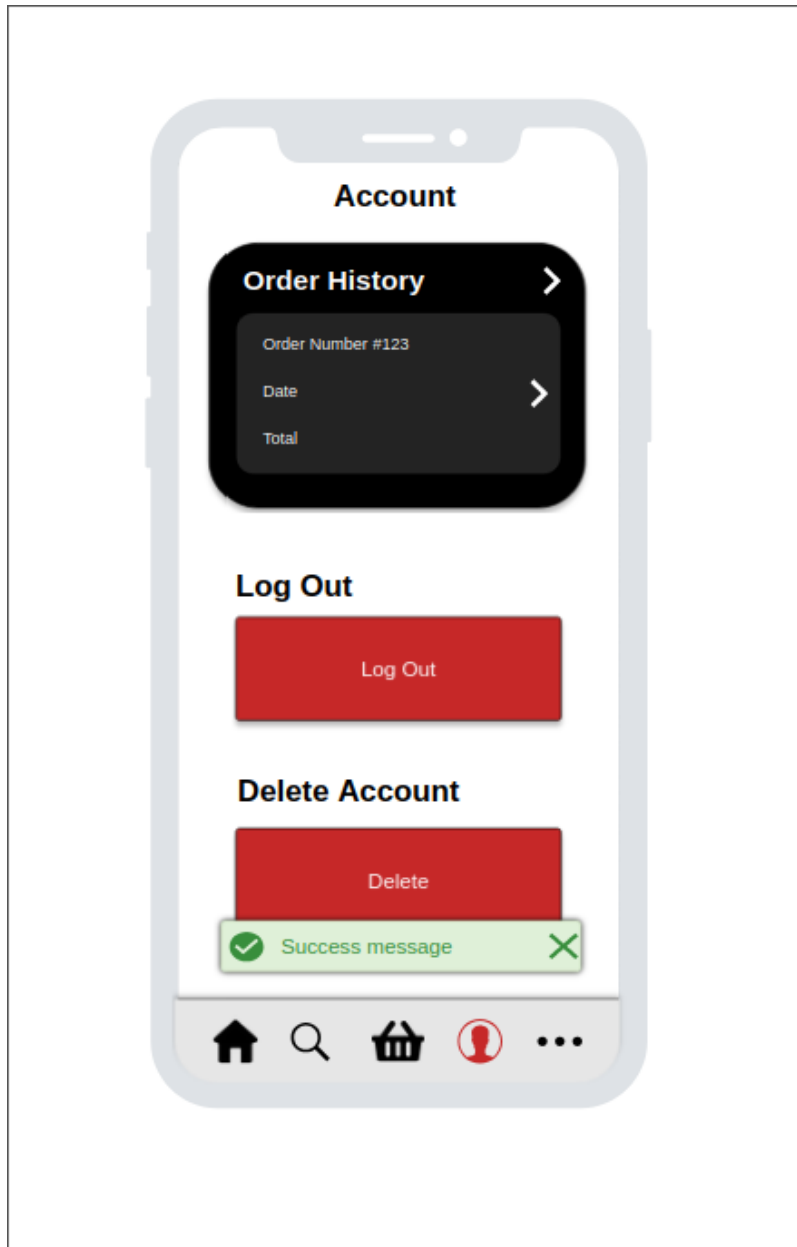


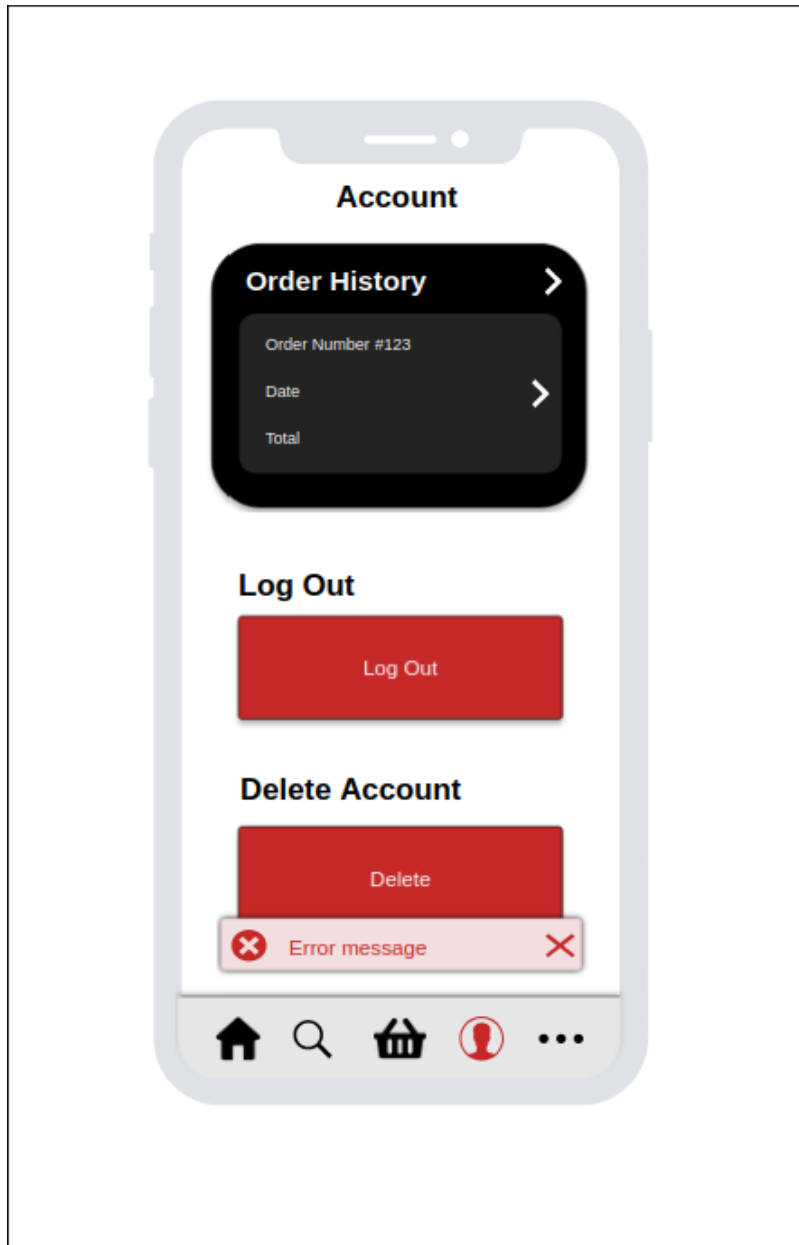




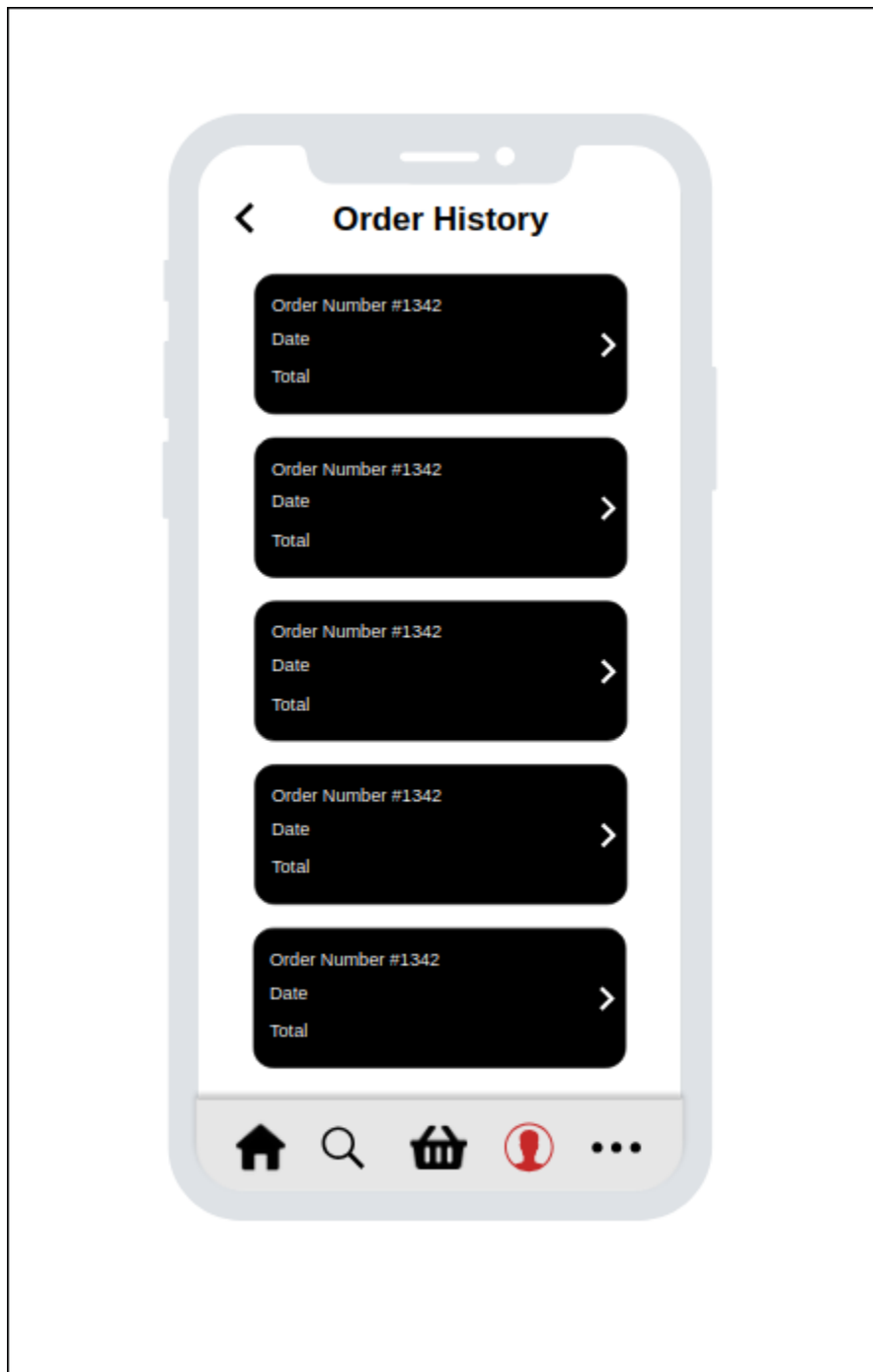


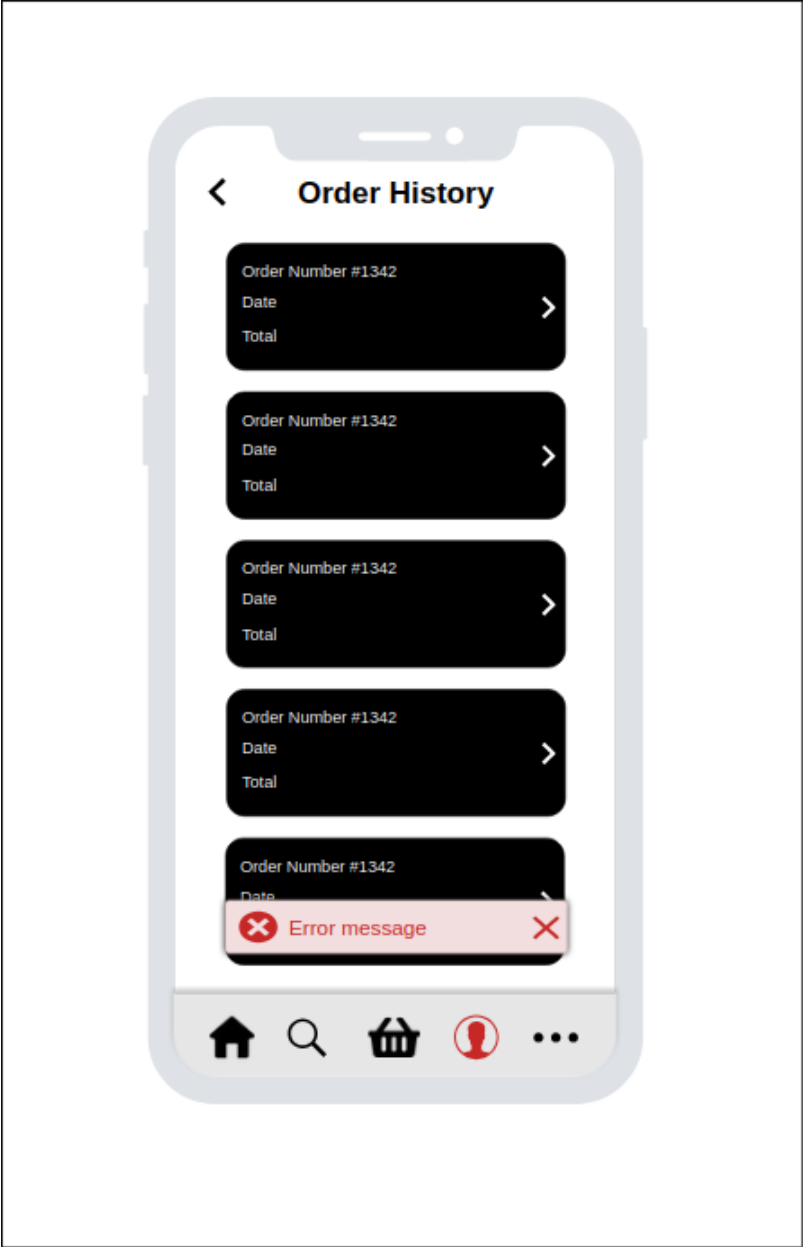


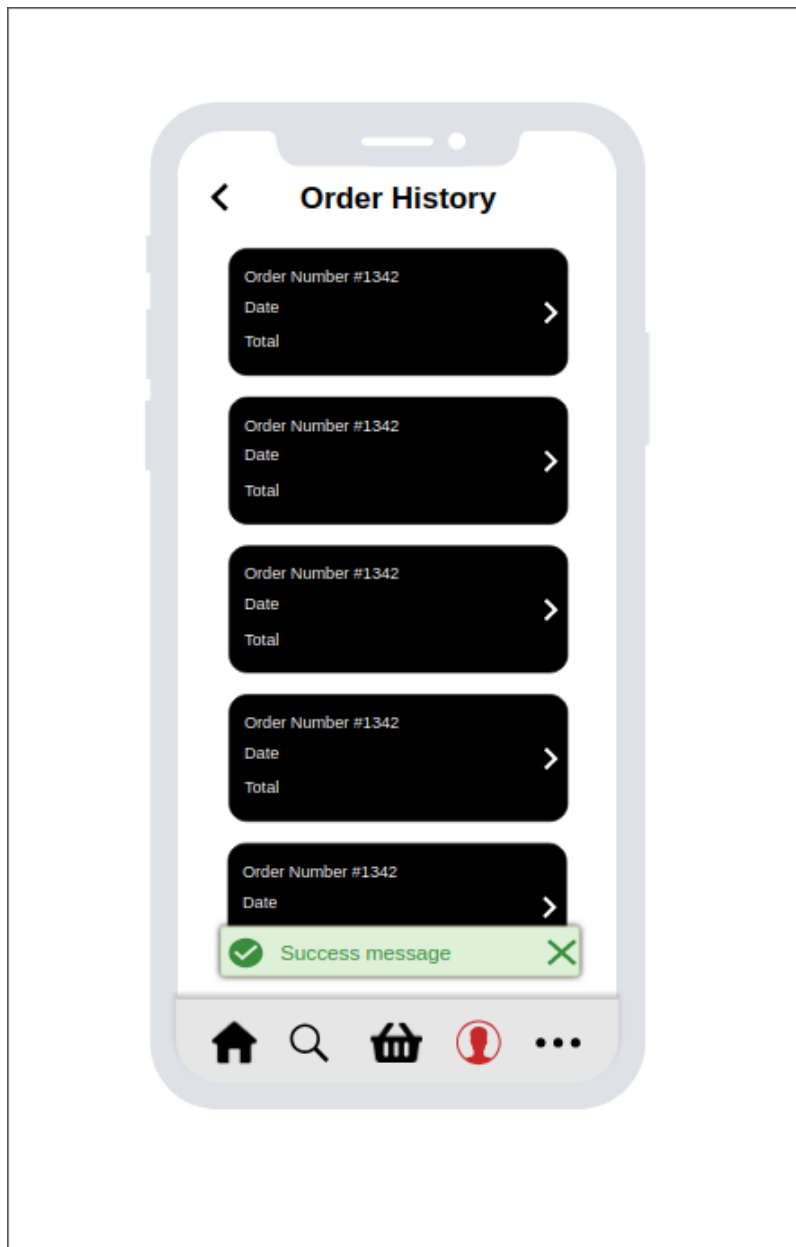




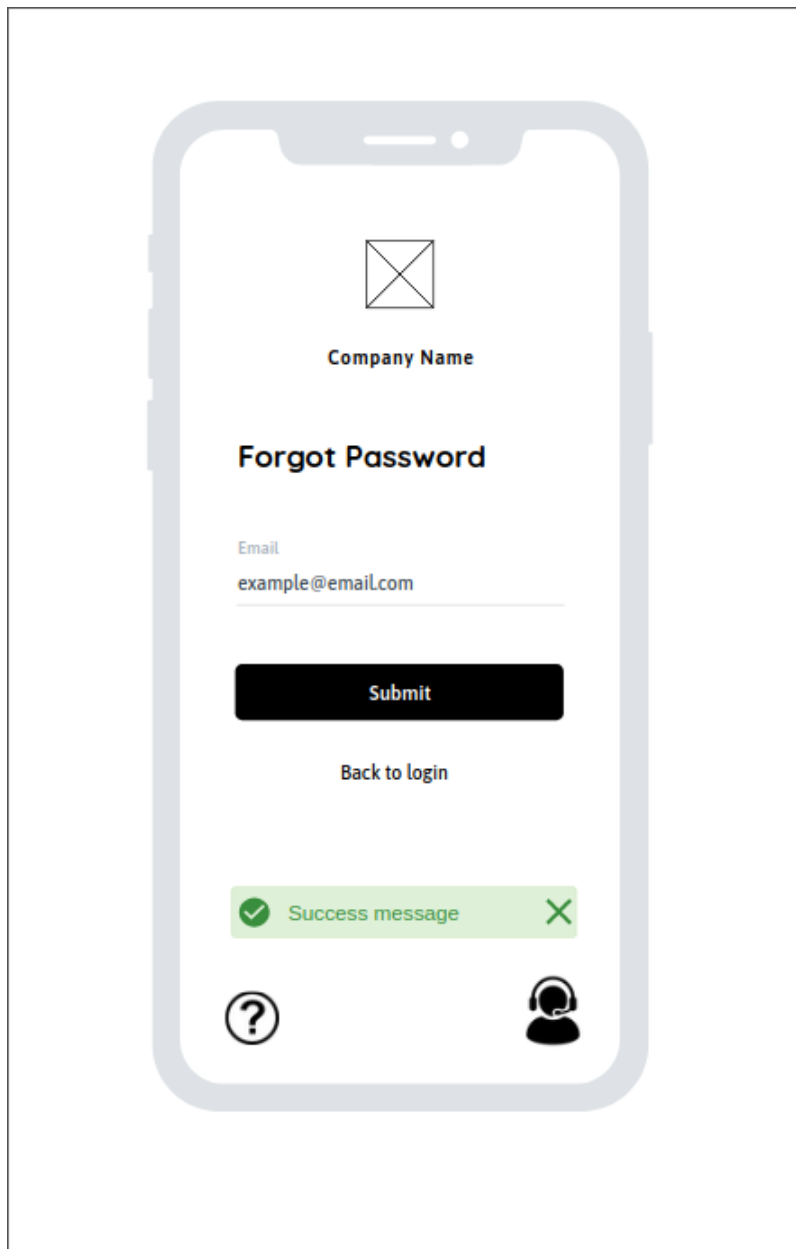
Order History Page



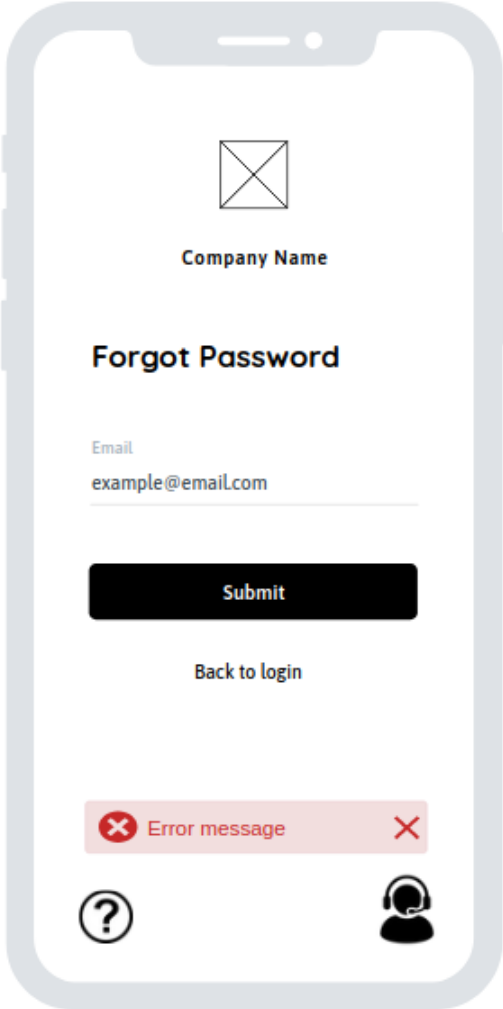




Forgot Password Page



A mobile app mockup of a 'Forgot Password' page. The page is displayed on a smartphone frame. At the top, there is a square icon with an 'X' inside, followed by the text 'Company Name'. Below this is the title 'Forgot Password'. A text input field is labeled 'Email' and contains the text 'example@email.com'. A black 'Submit' button is positioned below the input field. Underneath the button is a link that says 'Back to login'. A green success message banner at the bottom of the form area contains a checkmark icon, the text 'Success message', and a close 'X' icon. At the very bottom of the screen, there are two circular icons: one with a question mark and another with a headset icon representing customer support.





Company Name

Forgot Password

Email

example@email.com

Submit

Back to login

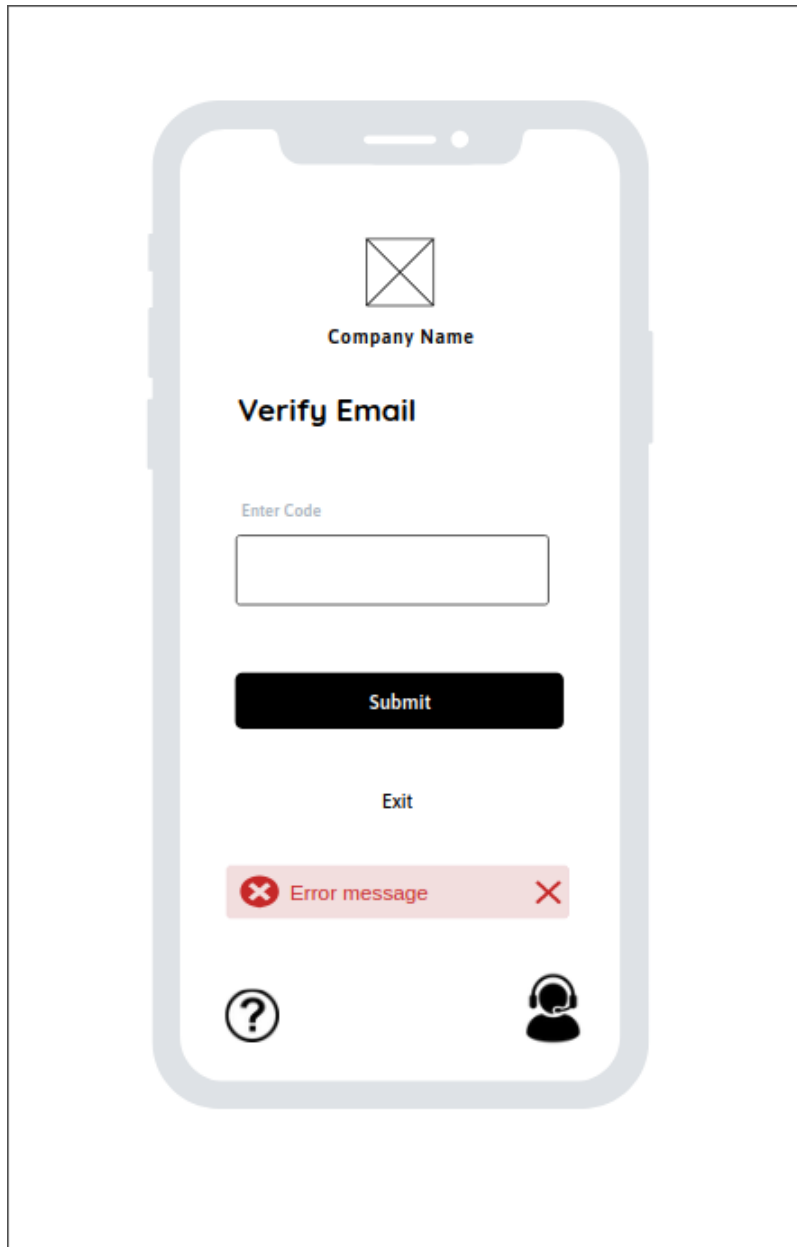


Error message

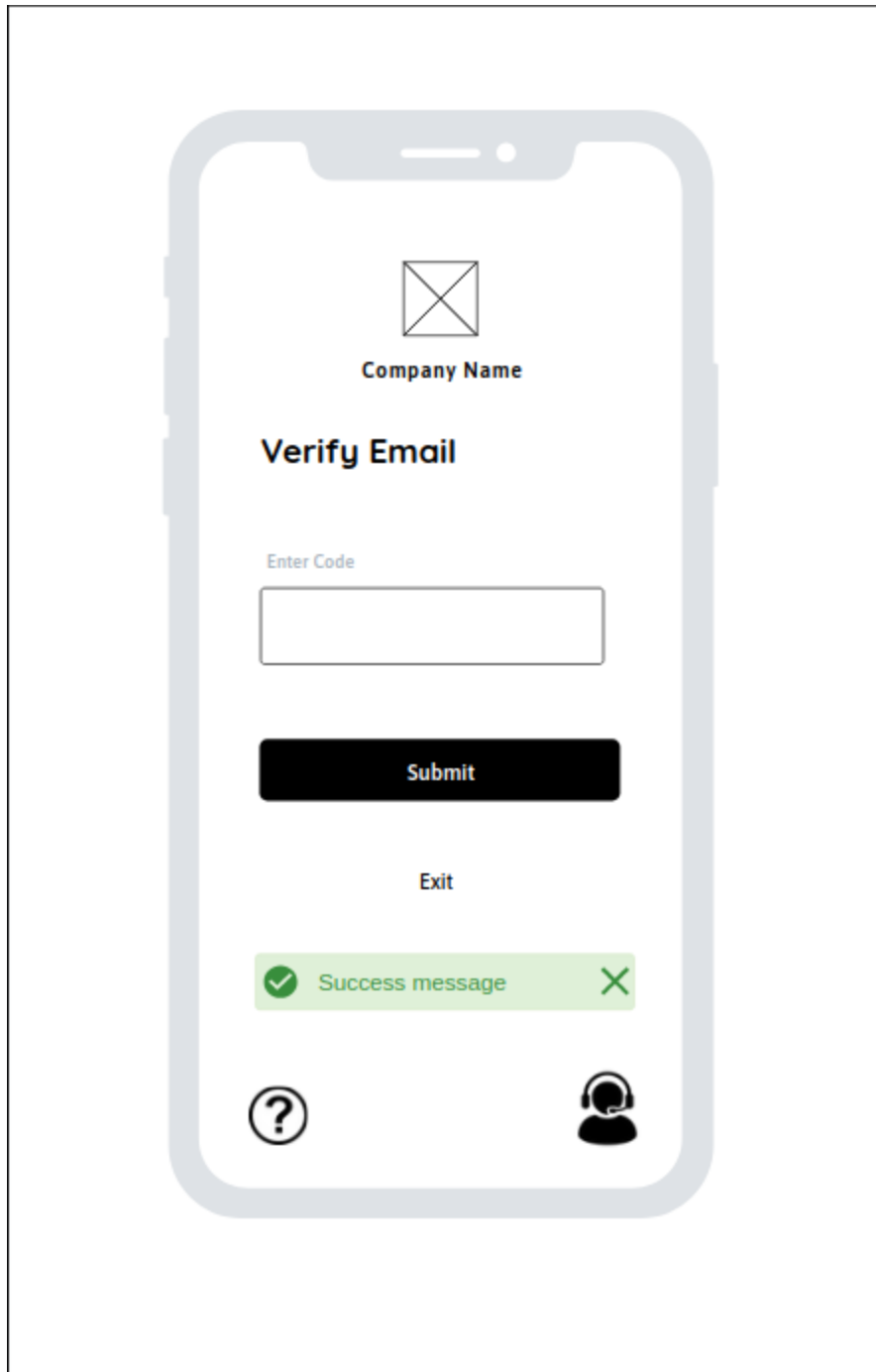


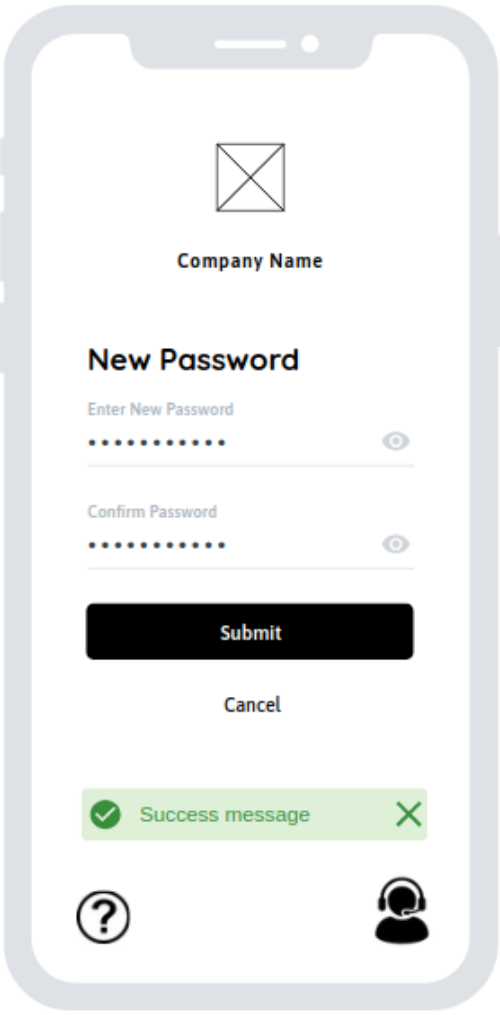




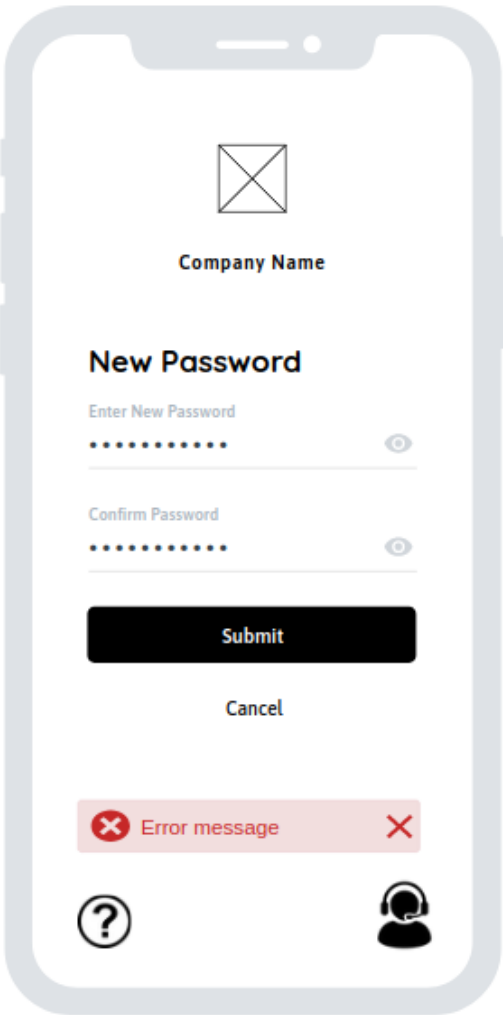


The image shows a mobile app interface for a 'Verify Email' screen. At the top, there is a square icon with an 'X' inside, followed by the text 'Company Name'. Below this is the title 'Verify Email'. A label 'Enter Code' is positioned above a text input field. A black 'Submit' button is located below the input field. Underneath the button is the text 'Exit'. A red error message banner with a red 'X' icon on the left and a red 'X' icon on the right contains the text 'Error message'. At the bottom of the screen, there are two circular icons: a question mark on the left and a headset icon on the right.



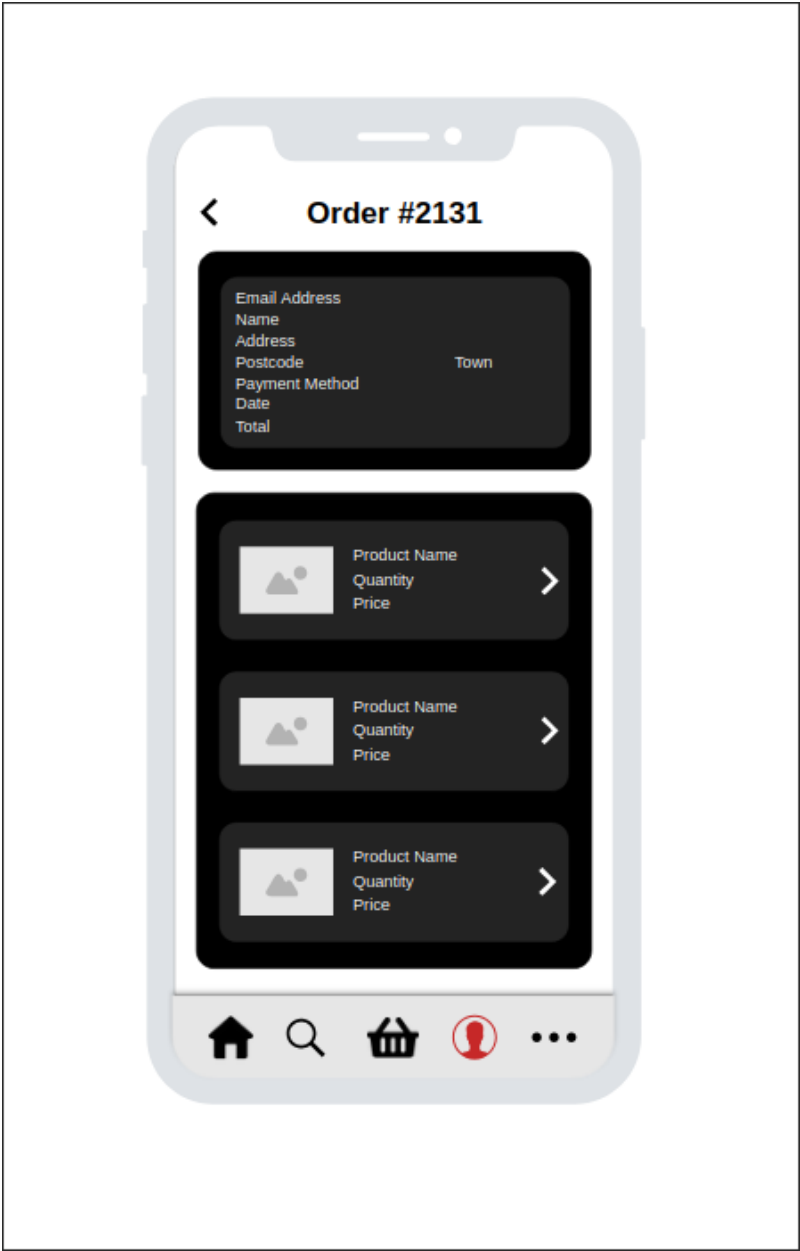


The image shows a mobile application interface for creating a new password. At the top, there is a placeholder icon for a company logo, represented by a square with an 'X' inside. Below this is the label "Company Name". The main section is titled "New Password". It contains two input fields: "Enter New Password" and "Confirm Password", both masked with dots and featuring an eye icon to toggle visibility. Below the input fields are two buttons: a black "Submit" button and a "Cancel" button. A green success message banner at the bottom reads "Success message" with a checkmark icon on the left and a close 'X' icon on the right. At the very bottom, there are two circular icons: a question mark on the left and a headset on the right, representing help and support options.

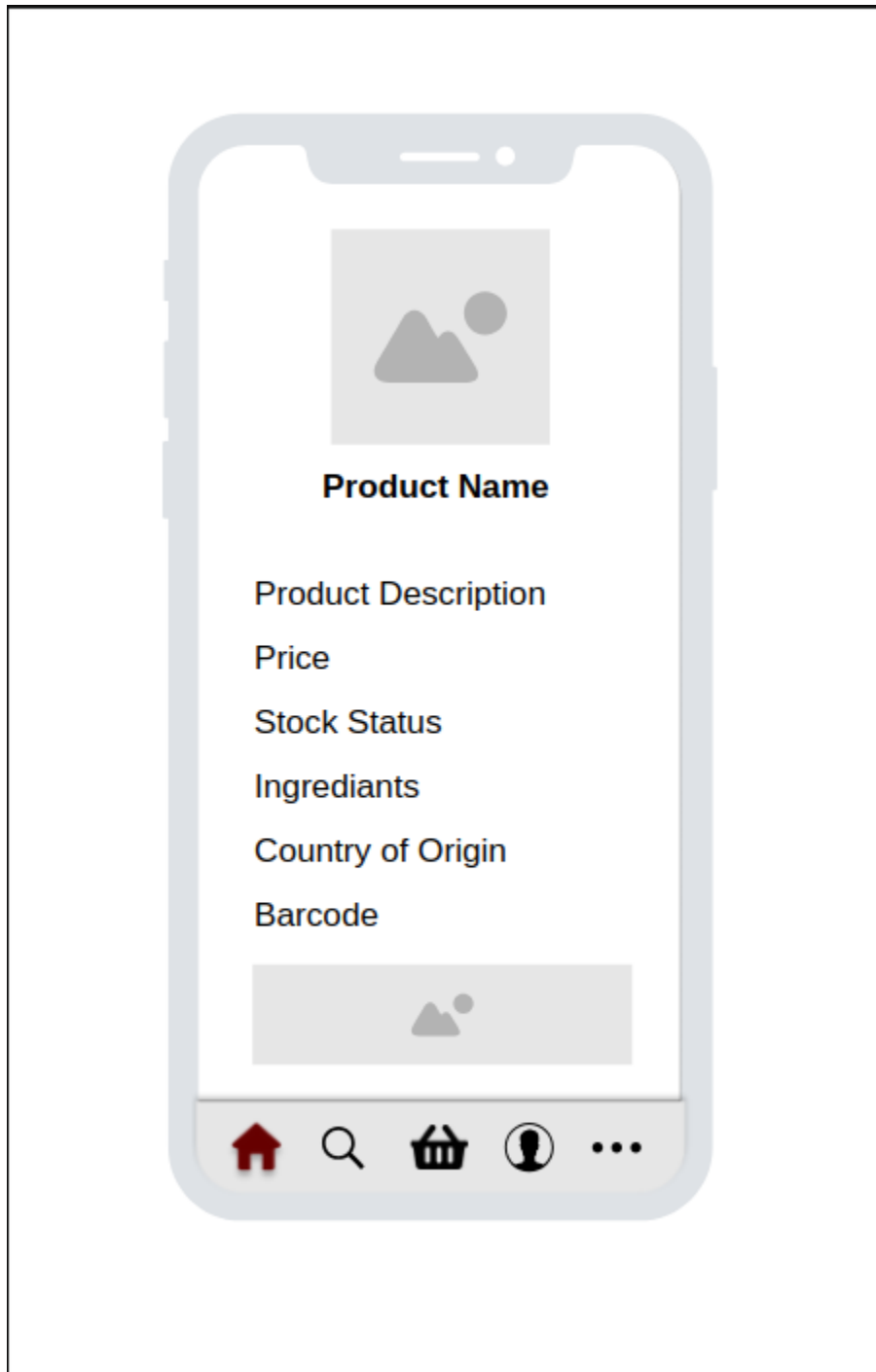


A mobile application interface for a password reset screen. At the top, there is a placeholder icon for a company logo, represented by a square with an 'X' inside. Below this is the label "Company Name". The main heading is "New Password". Underneath, there are two input fields: "Enter New Password" and "Confirm Password", both masked with dots and featuring an eye icon for toggling visibility. A black "Submit" button is positioned below the input fields, followed by a "Cancel" link. At the bottom of the form, there is a red error message banner that reads "Error message" with red 'X' icons at both ends. The bottom of the screen features two circular icons: a question mark on the left and a headset icon on the right, representing help and support options.

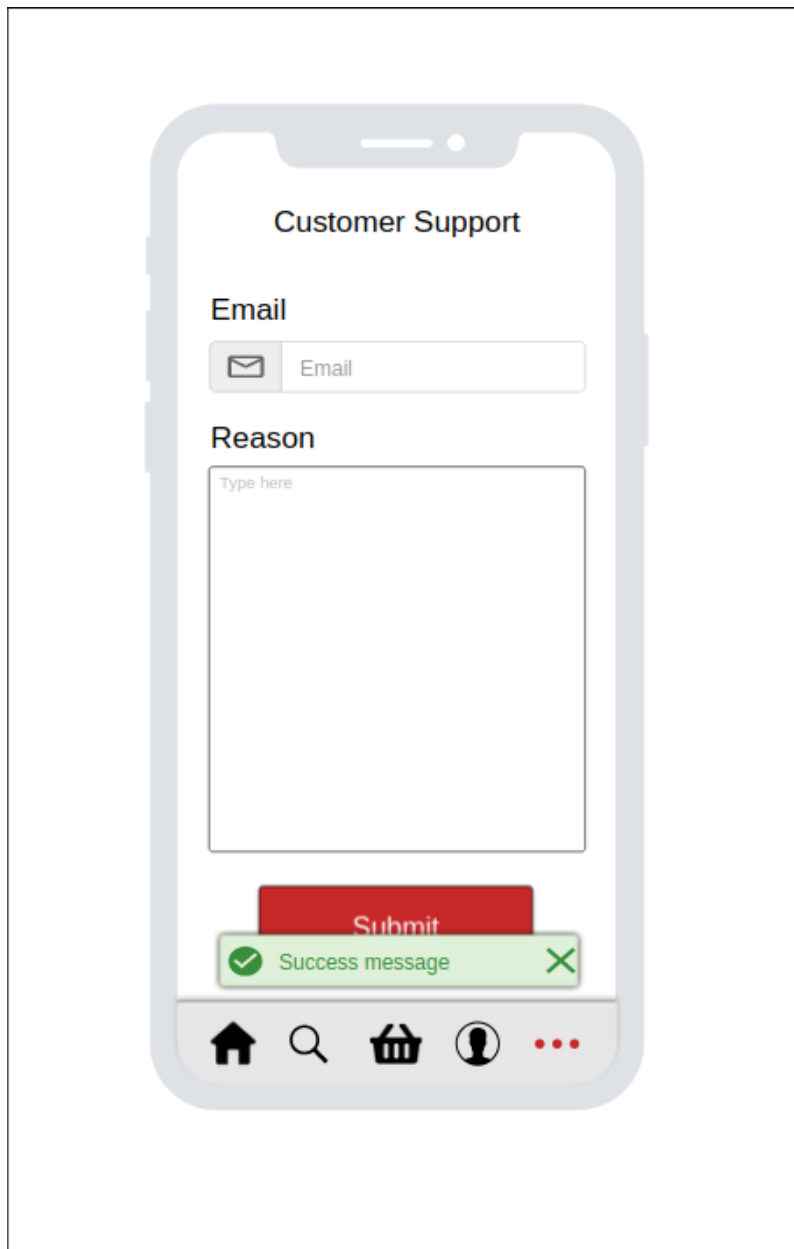
Order Page



Product Page



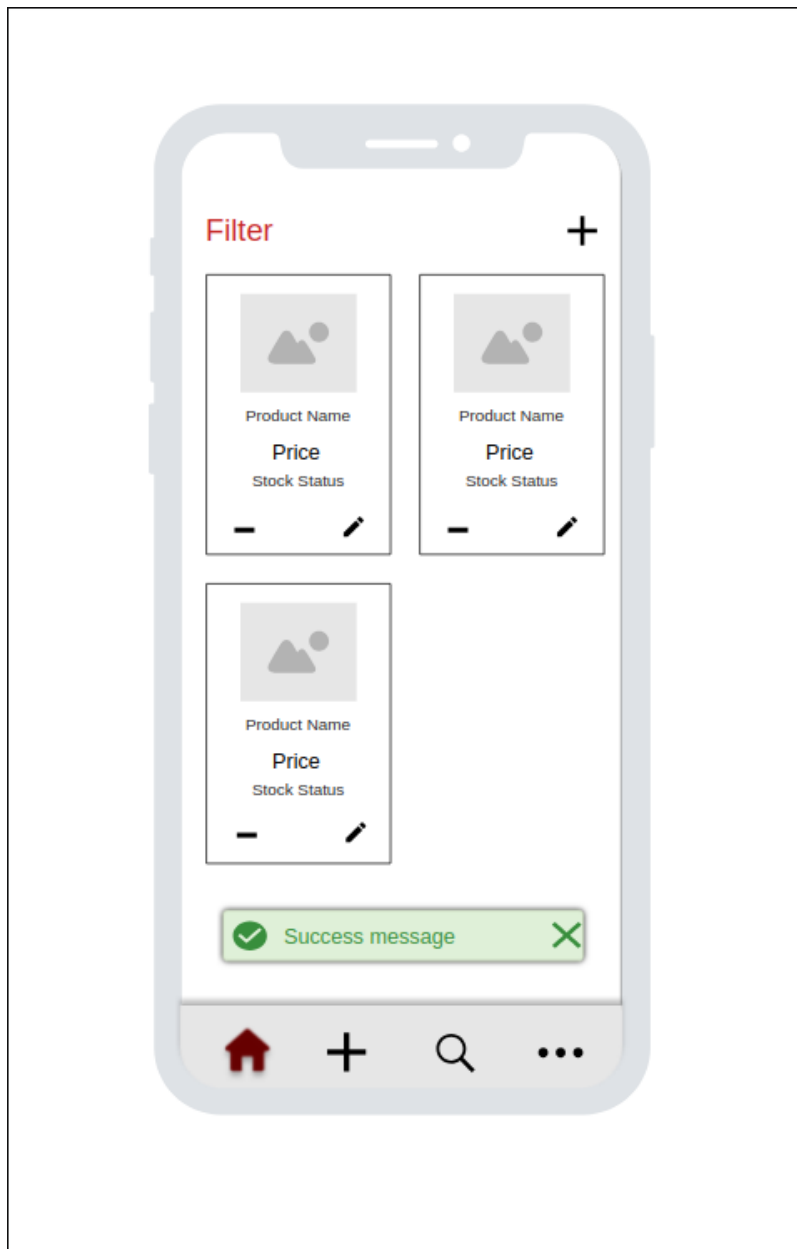
Customer Support Page

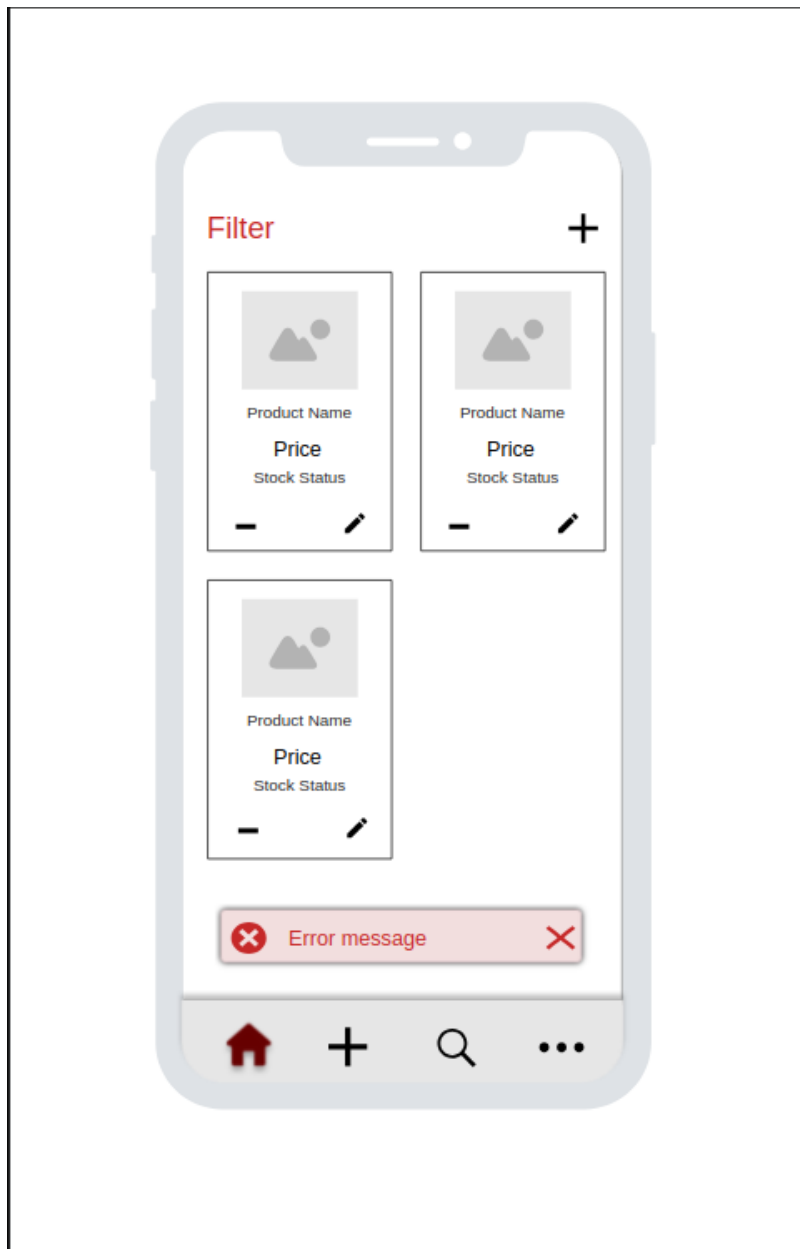


The image shows a mobile application interface for a 'Customer Support' page. The page is displayed within a grey smartphone frame. At the top, the title 'Customer Support' is centered. Below the title, there is an 'Email' section with a text input field containing the placeholder 'Email' and an envelope icon to its left. Underneath the email field is a 'Reason' section with a large text area containing the placeholder 'Type here'. A red 'Submit' button is positioned below the text area. At the bottom of the form, there is a green success message bar with a checkmark icon, the text 'Success message', and a close 'X' icon. The bottom of the screen features a navigation bar with five icons: a home icon, a magnifying glass (search), a shopping cart, a user profile, and a menu (three dots).

A mobile application mockup for a 'Customer Support' form. The screen is titled 'Customer Support'. Below the title is an 'Email' section with a text input field containing the placeholder 'Email'. Underneath is a 'Reason' section with a large text area containing the placeholder 'Type here'. A red 'Submit' button is positioned below the text area. A red error message banner is displayed below the button, containing a red 'X' icon, the text 'Error message', and another red 'X' icon. At the bottom of the screen is a navigation bar with five icons: a home icon, a magnifying glass (search), a shopping cart, a user profile, and a menu (three dots).

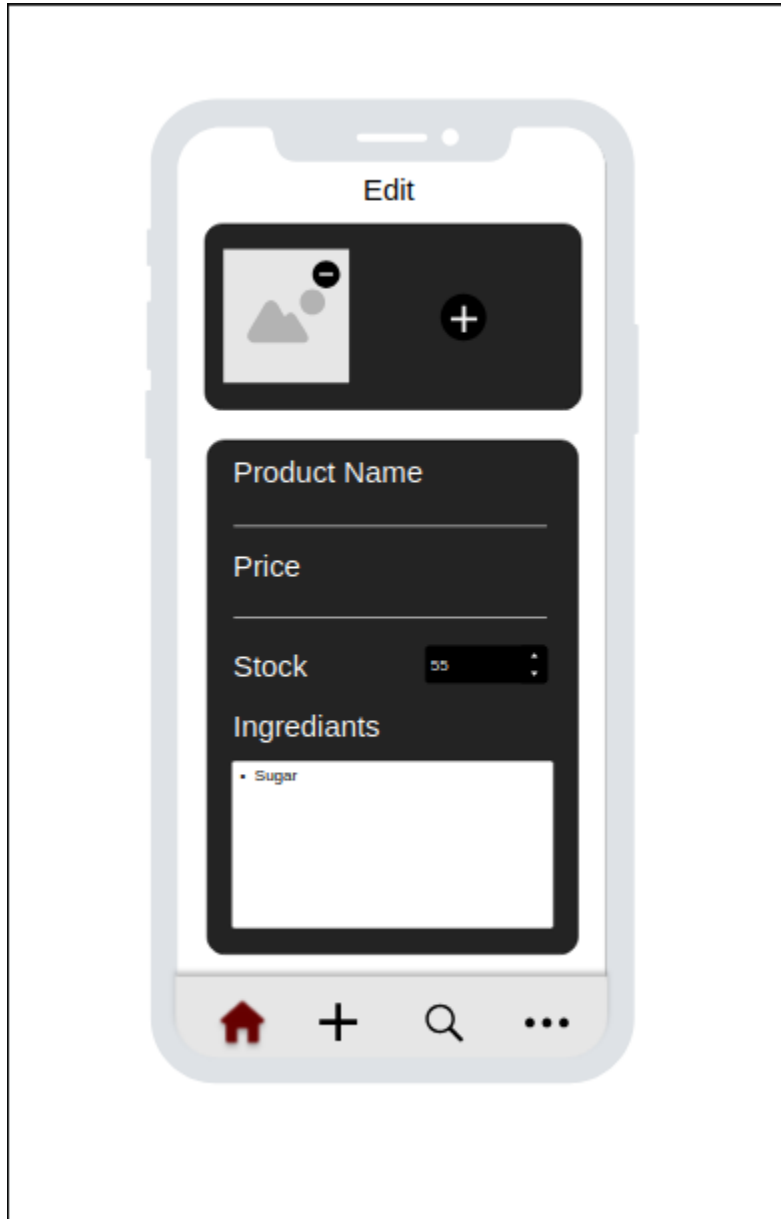
Admin Catalog Page

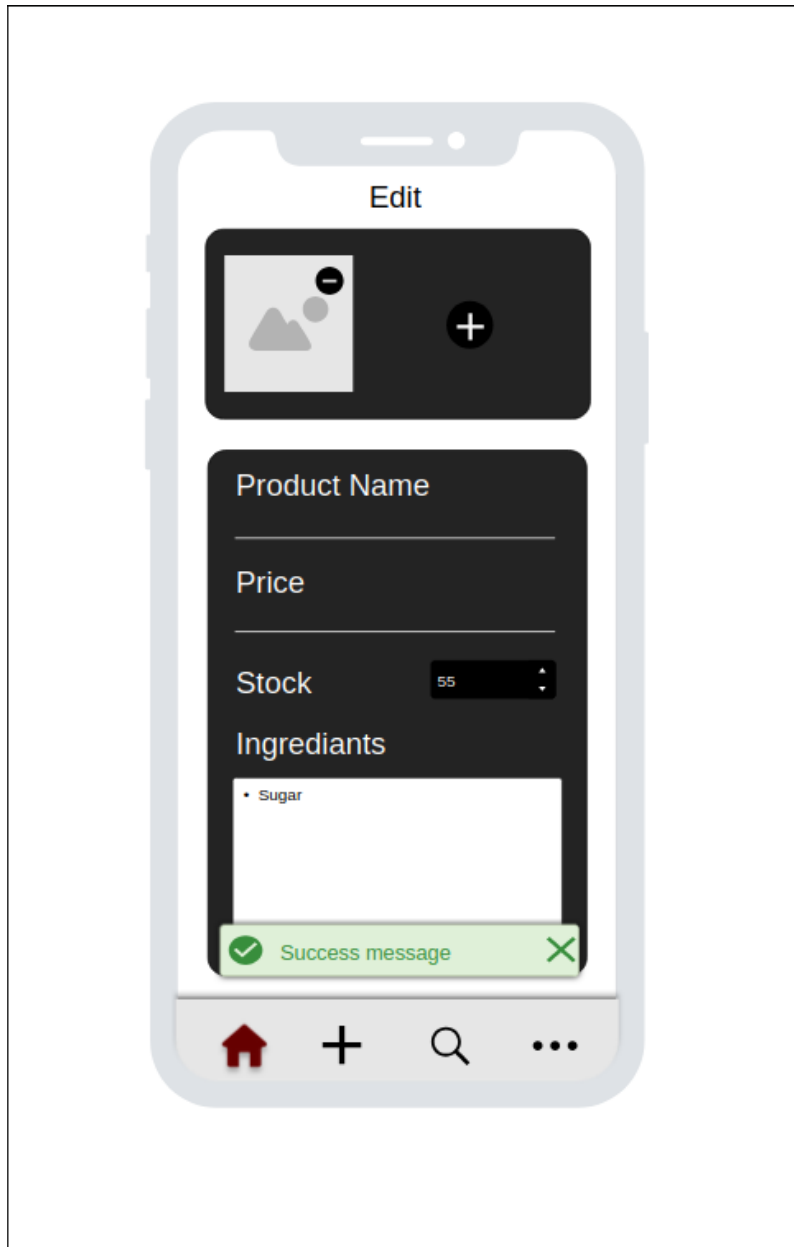


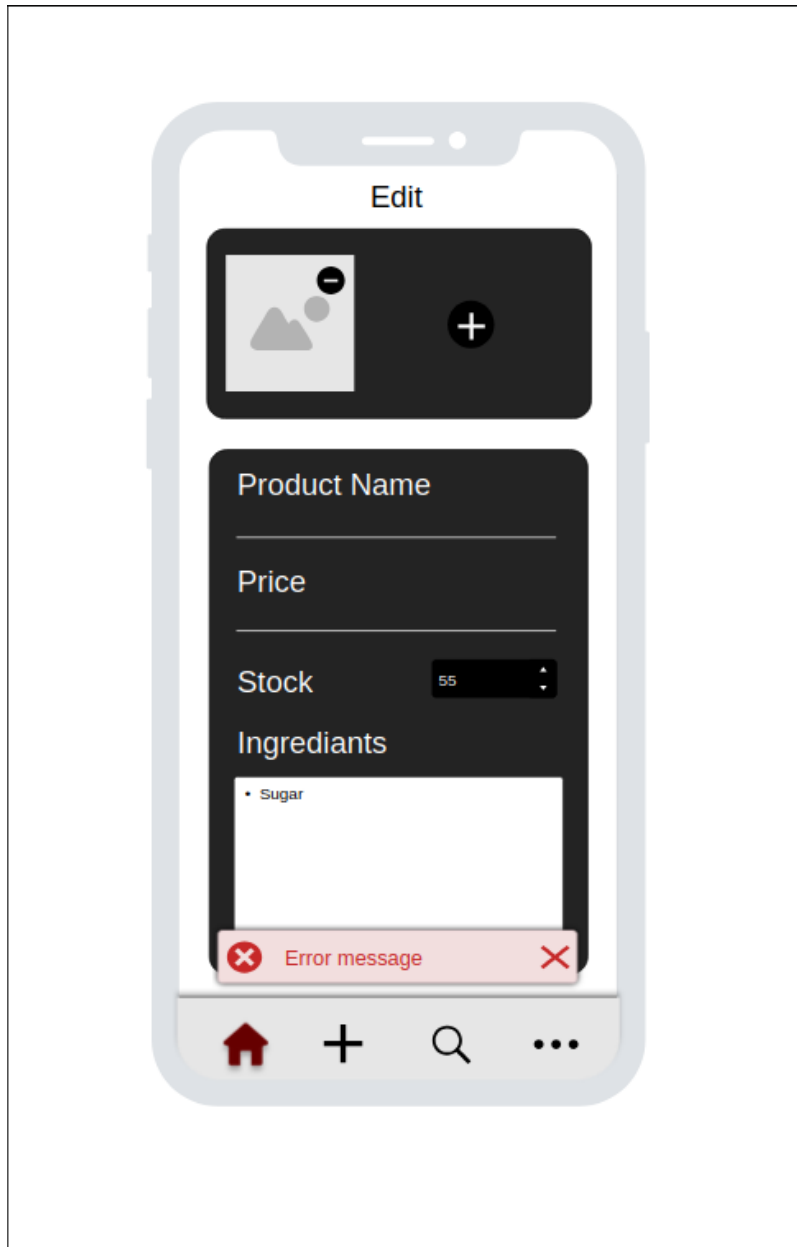


Edit Product Page

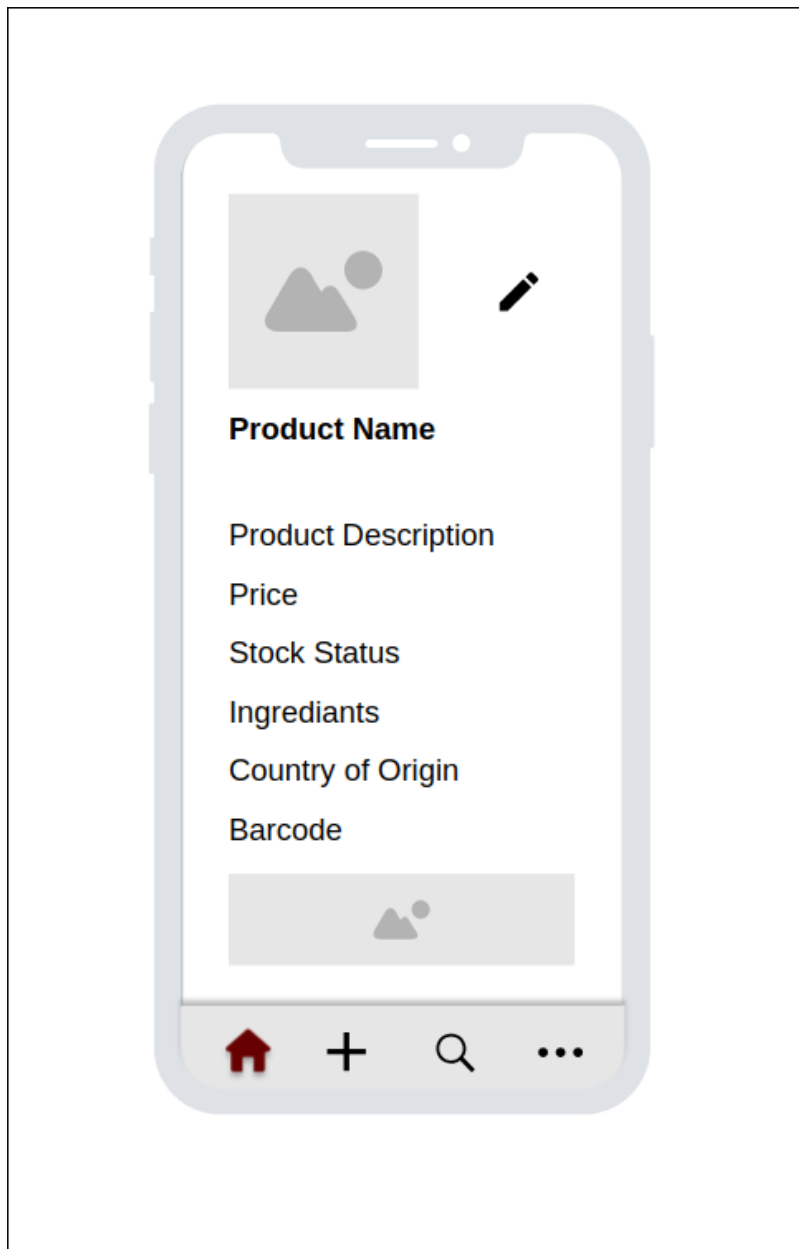
Ingredients section may need to be changed due to a new class being made for it with extra details such as weight and allergens.

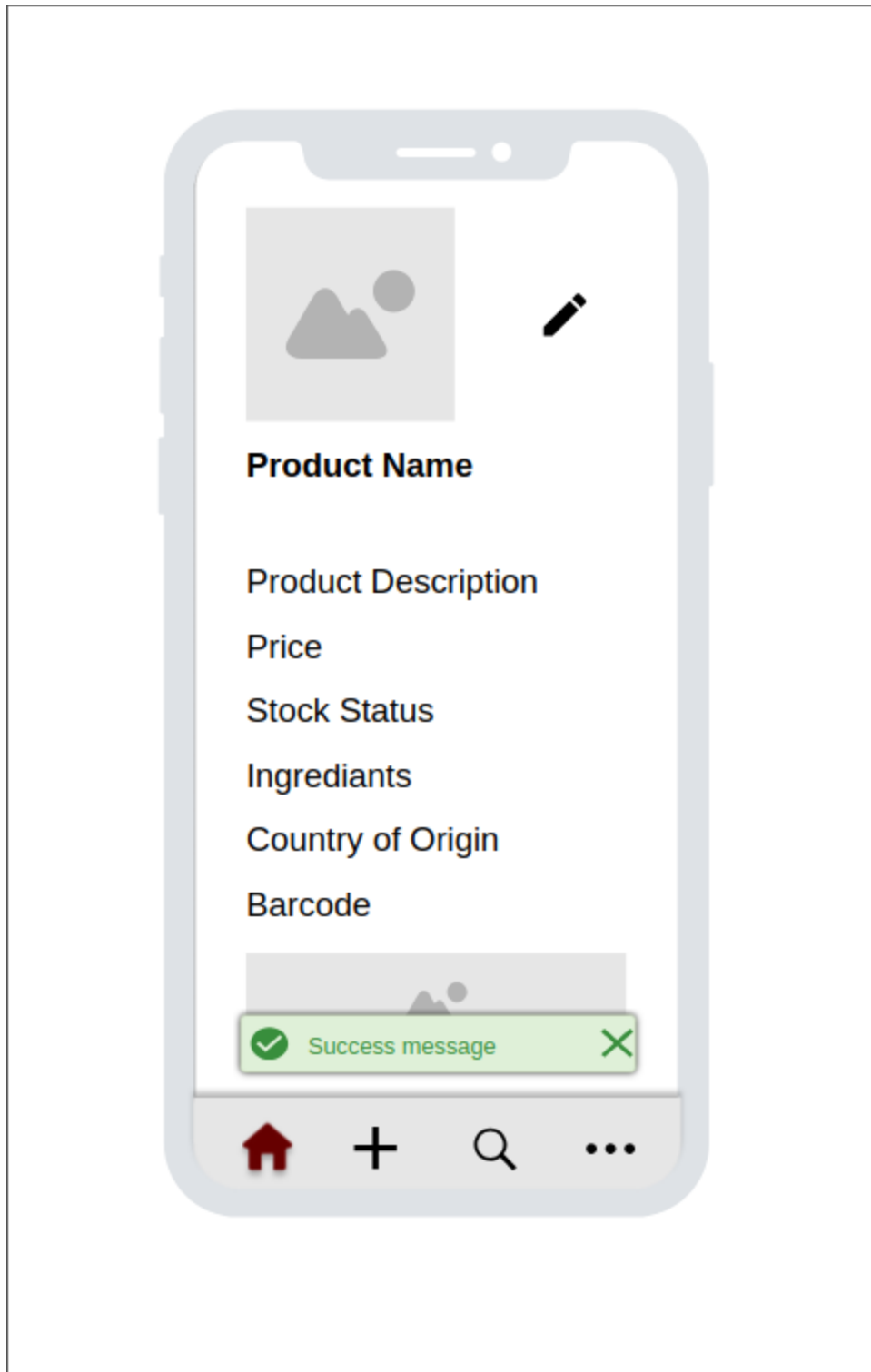


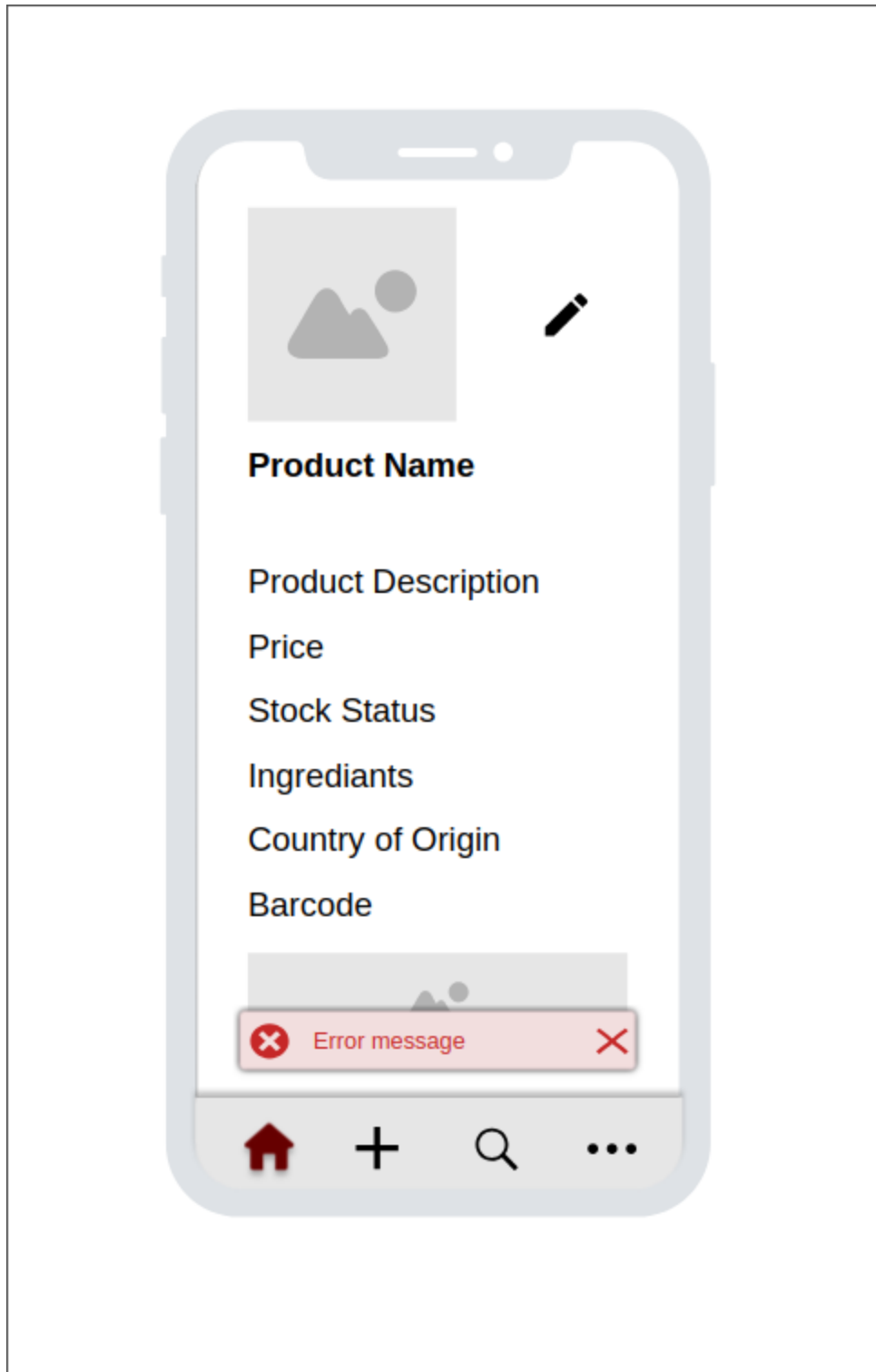




Admin Product Page

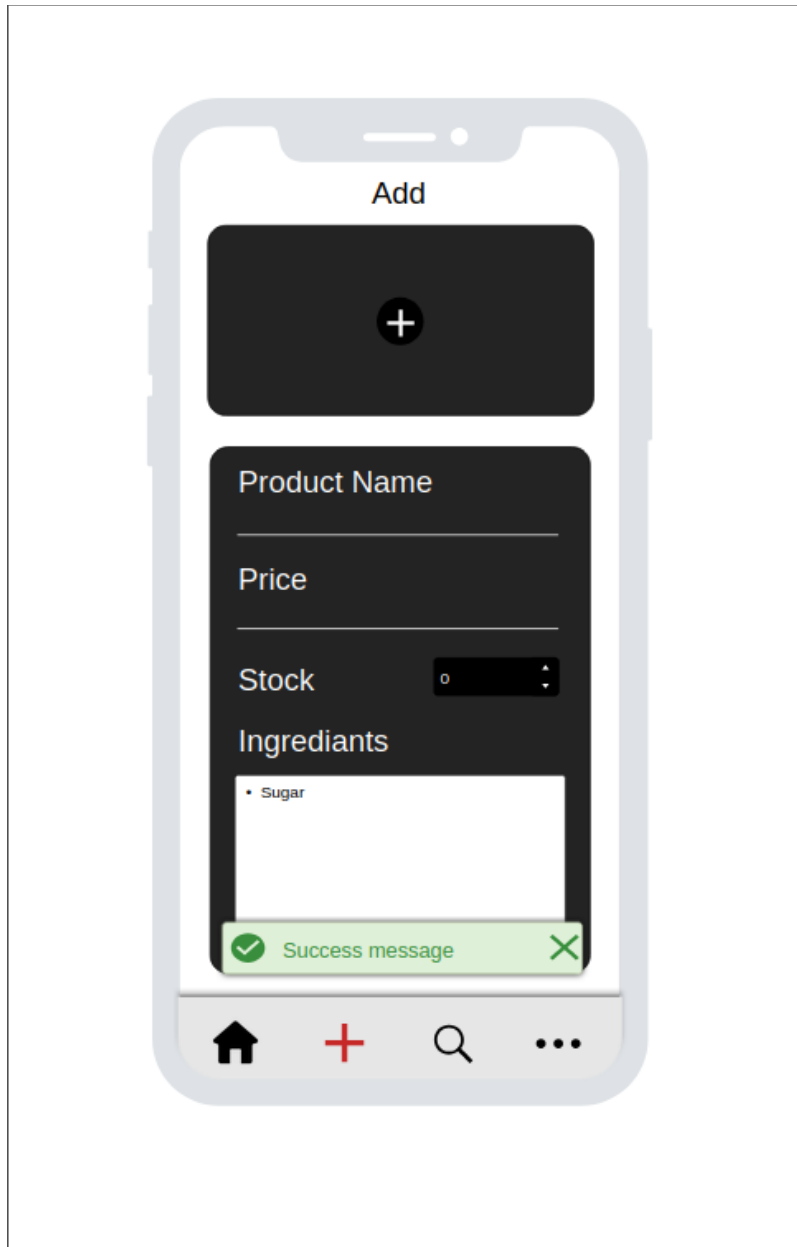


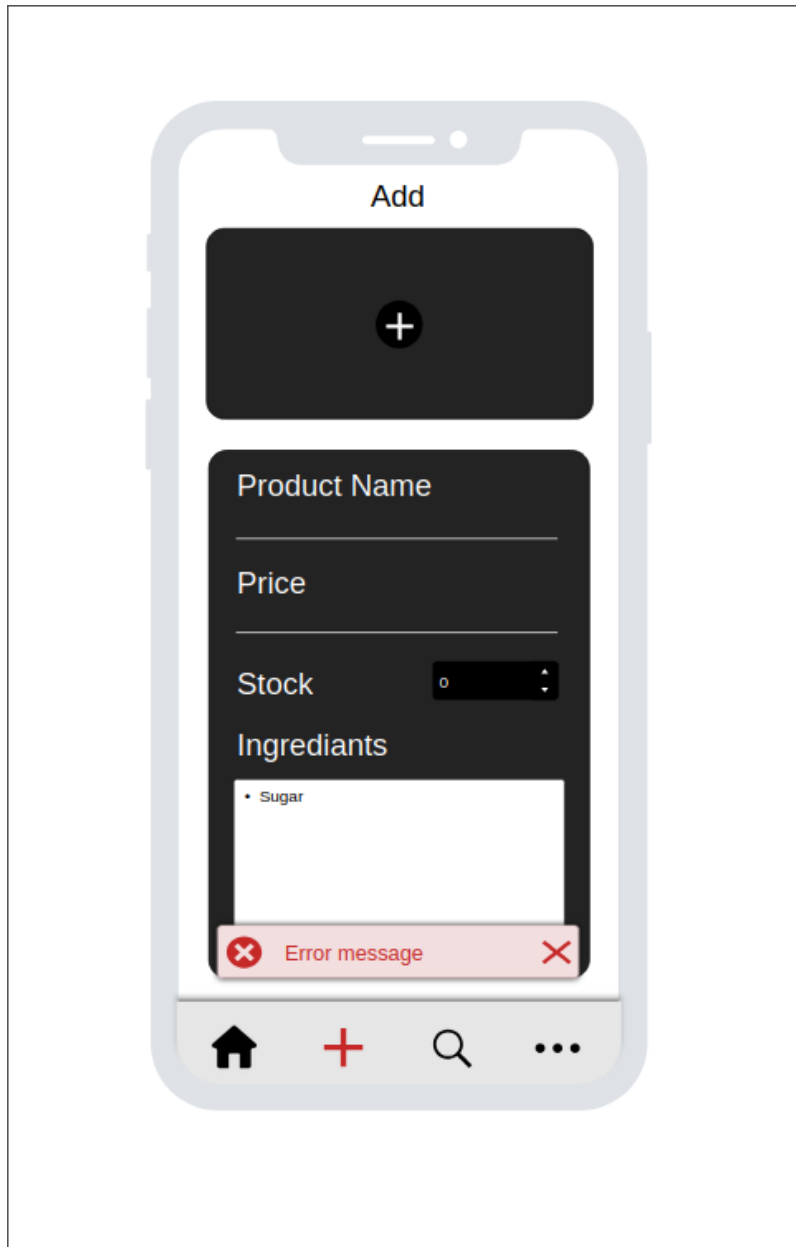




Add Product Page

Ingredients section may need to be changed due to a new class being made for it with extra details such as weight and allergens.





HCI

Visibility of System Status.

“The design should always keep users informed about what is going on through appropriate feedback within a reasonable amount of time” (Nielsen, 2024). Notifications achieve this as it informs customers and managers about whether a system has successfully completed a task, or an error had occurred within the system. Examples of this are when a user successfully logs in a notification in green will notify the user saying “Successfully logged in” or when a user fails to log in a message in red will appear “Failed to log in, please try a different email or password, or make sure your log in is spelt right.”

Match Between the System and the Real World.

“The design should speak the users' language. Use words, phrases, and concepts familiar to the user rather than internal jargon. Follow real-world conventions, making information appear in a natural and logical order” (Nielsen, 2024). Nearly all interact-able elements of the application use little to no jargon and allowing users to naturally understand what the elements mean such as using familiar icons to give a clear impression of the interact-able element like a basket icon to make the user understand that the element is a basket.

User Control and Freedom

“Users often perform actions by mistake. They need a clearly marked ‘emergency exit’ to leave the unwanted action without having to go through an extended process” (Nielsen, 2024). Most actions will be added with a “Are you sure you want to proceed with {task}” notification to make sure the user would like to proceed with the action they are making thus helping prevent any misinputs/errors the user can make.

Consistency and Standards

“Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform and industry conventions” (Nielsen, 2024). Nearly all pages follow a similar design to other applications in the market and follow a strict colour scheme helping users quickly adjust and understand the layout of the application.

Error Prevention

“Good error messages are important, but the best designs carefully prevent problems from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action” (Nielsen, 2024). Notifications will appear on nearly all tasks asking if the user would like to proceed with the action, they are making preventing any accidental actions that could cause errors.

Another prevention is used in forms, inputs are validated and cleaned in prevention of backend errors that could occur.

Recognition Rather than Recall

“Minimise the user's memory load by making elements, actions, and options visible. The user should not have to remember information from one part of the interface to another. Information required to use the design (e.g. field labels or menu items) should be visible or easily retrievable when needed” (Nielsen, 2024). Recognisable icons and clear headings in forms achieve this and allow the user to seamlessly spot the element they are looking for.

Flexibility and Efficiency of Use

“Shortcuts — hidden from novice users — may speed up the interaction for the expert user so that the design can cater to inexperienced and experienced users. Allow users to tailor frequent actions” (Nielsen, 2024). There was no need for this as most tasks can be completed within a short length of time both by novice and experienced users of the application due to the simplicity of the design and application.

Aesthetic and Minimalist Design

“Interfaces should not contain information that is irrelevant or rarely needed. Every extra unit of information in an interface competes with the relevant units of information and diminishes their relative visibility” (Nielsen, 2024). This is achieved with pages showing only necessary information to the user for them to understand and complete tasks. Extra information such as country of origin in the products class can be accessed in a more detailed page when a product is tapped on.

Recognise, Diagnose and Recover from Errors

“Error messages should be expressed in plain language (no error codes), precisely indicate the problem, and constructively suggest a solution” (Nielsen, 2024). All error messages will include solutions to assist users in fixing their problem on their own. These solutions and error messages will avoid any hard to understand/jargon contained within them for users to easily understand.

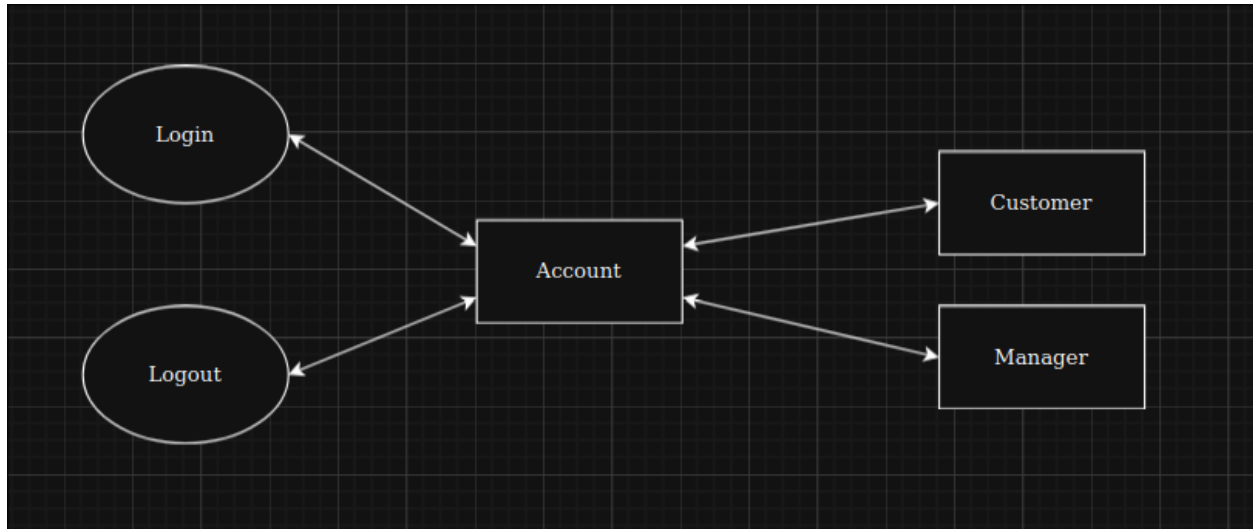
Help and Documentation

“It's best if the system doesn't need any additional explanation. However, it may be necessary to provide documentation to help users understand how to complete their tasks” (Nielsen, 2024). A user guide will be included within the application and design under a question mark icon for users to easily access information they may need to gain and understanding of a certain element in the design.

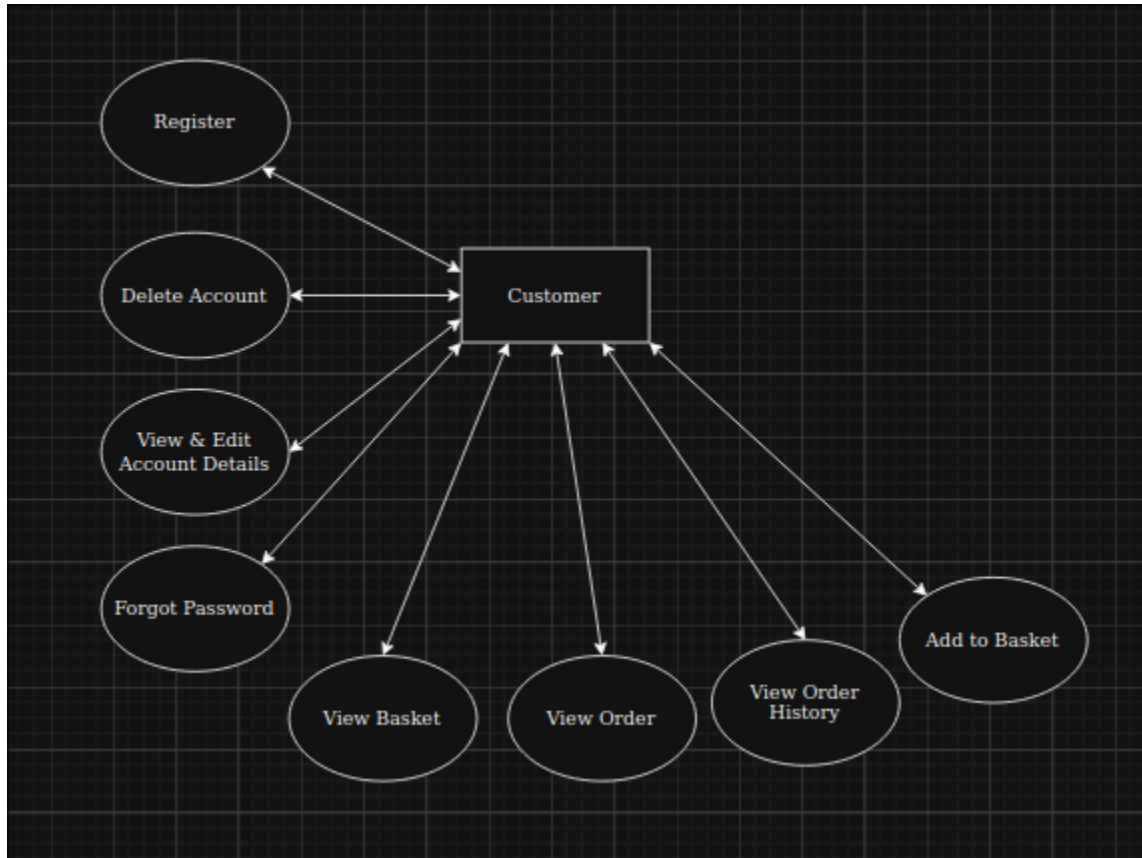
Data Binding

This diagram shows how the transfer of data over, view/pages and models/classes interact with one another.

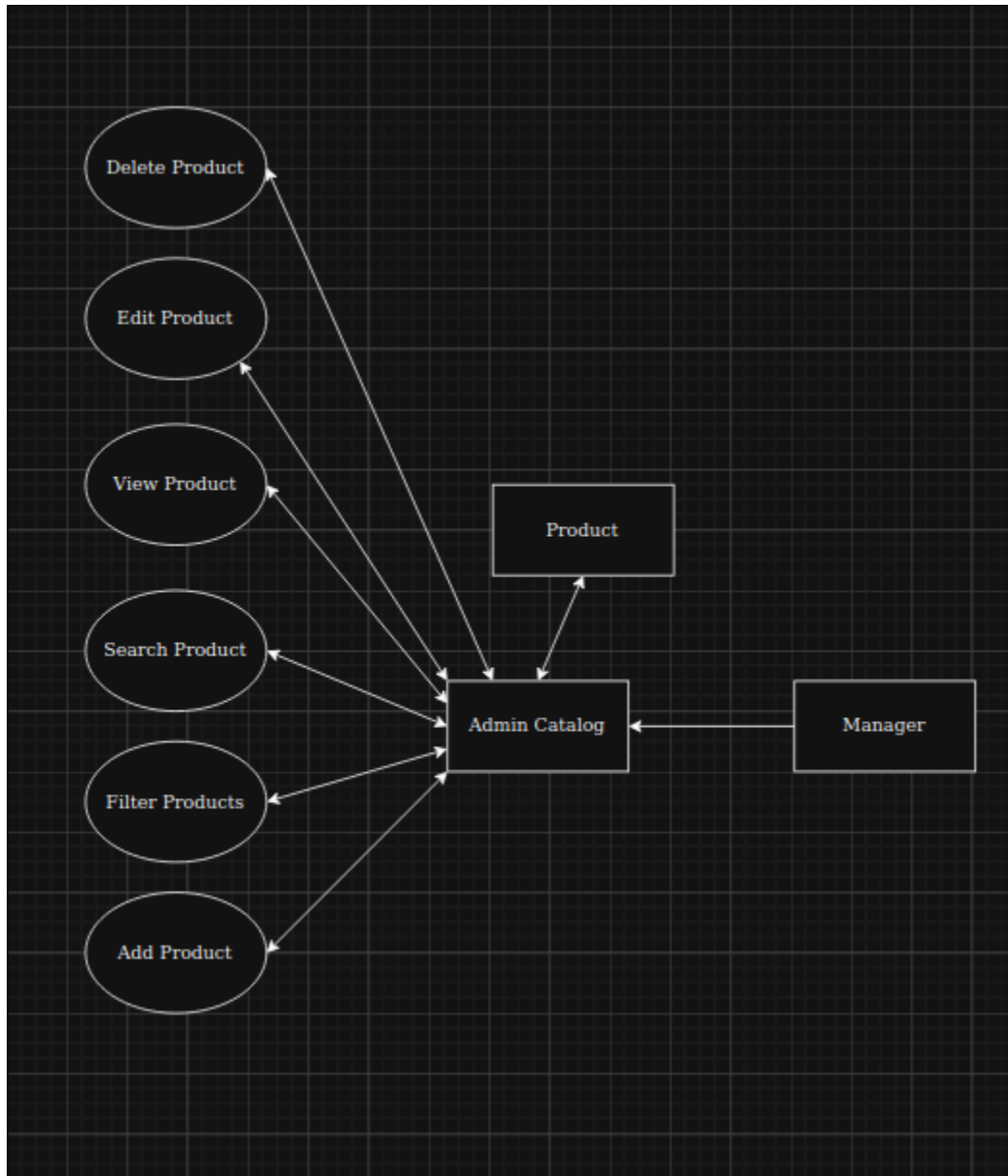
Customer/Manager



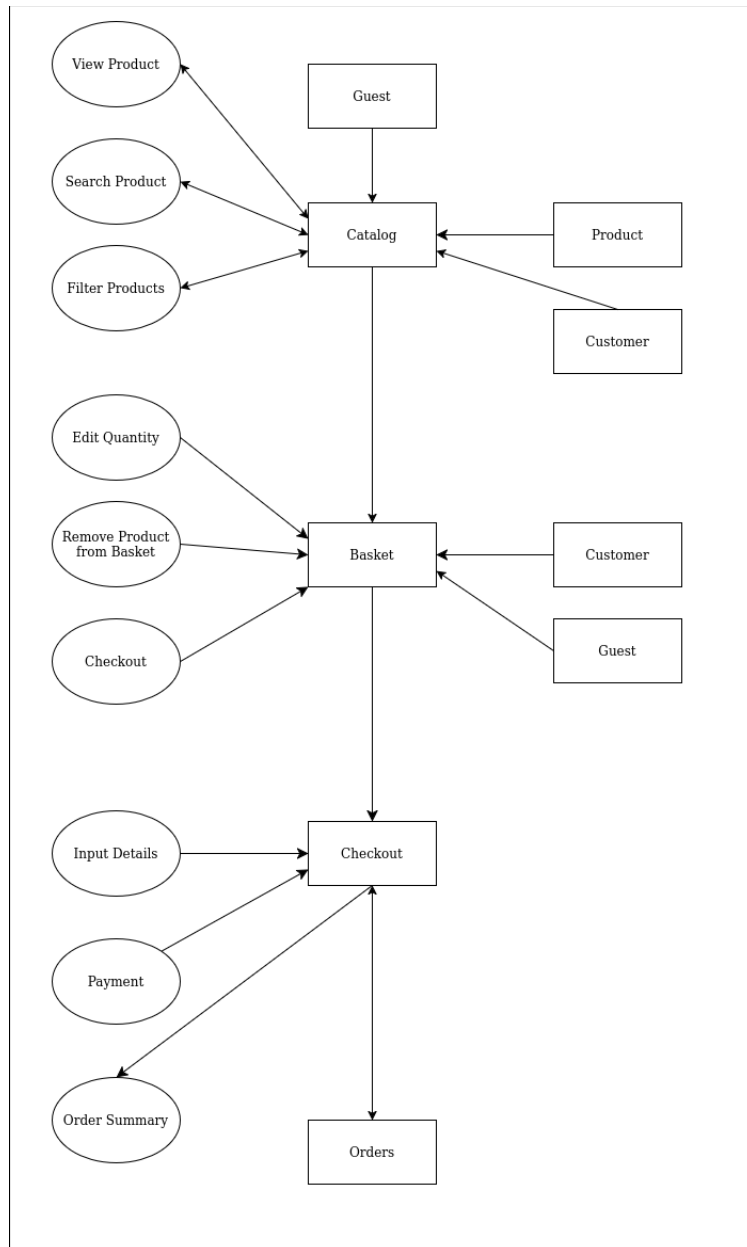
Customer



Manager



Guest/Customer



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